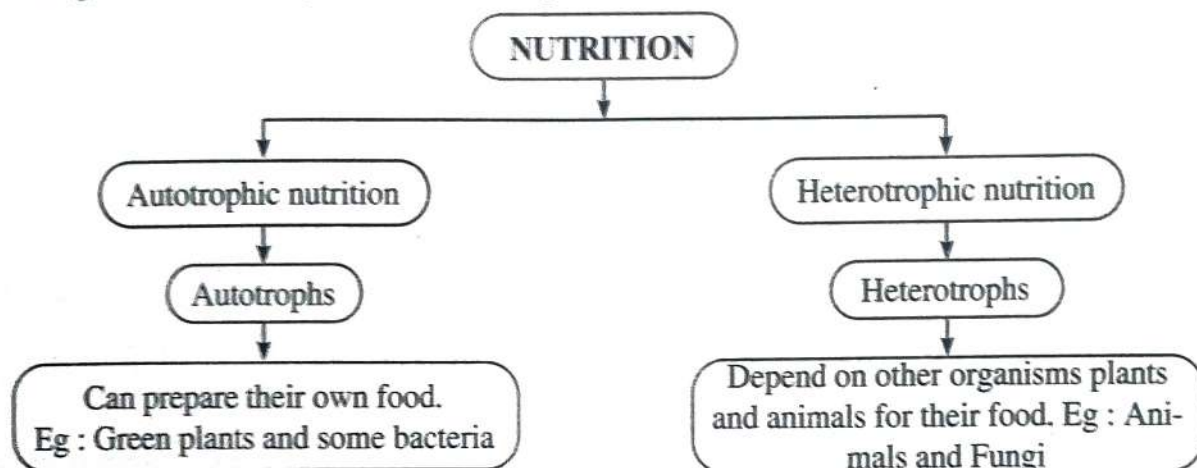


1. NUTRITION

CHAPTER AT A GLANCE

Nutrition : Nutrition is a process in which the food is consumed by the organisms and utilising the nutrients from the food to obtain energy required for life.

Depending on the mode of obtaining nutrition, organisms are classified into two major categories as autotrophs and heterotrophs.

**Autotrophic nutrition :**

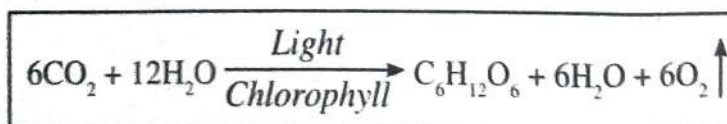
Autotrophs are the organisms capable of using light energy to synthesize chemical compounds. They acquire nutrients like some minerals and water from the soil as well as some gases from the air. They are capable of producing complex compounds like carbohydrates, proteins, lipids etc from these very simple substances. These compounds produced by them are utilized for providing energy to most of the living organisms and all animals including human beings. Eg : Plants

Photosynthesis :

Photosynthesis is a process by which plants containing the green pigment 'chlorophyll' build up complex organic molecules from relatively simple inorganic ones, using sun light as an energy source. The process of photosynthesis is very complex . The simple equation for photosynthesis proposed by C.B. Van Niel in the year 1931 was -



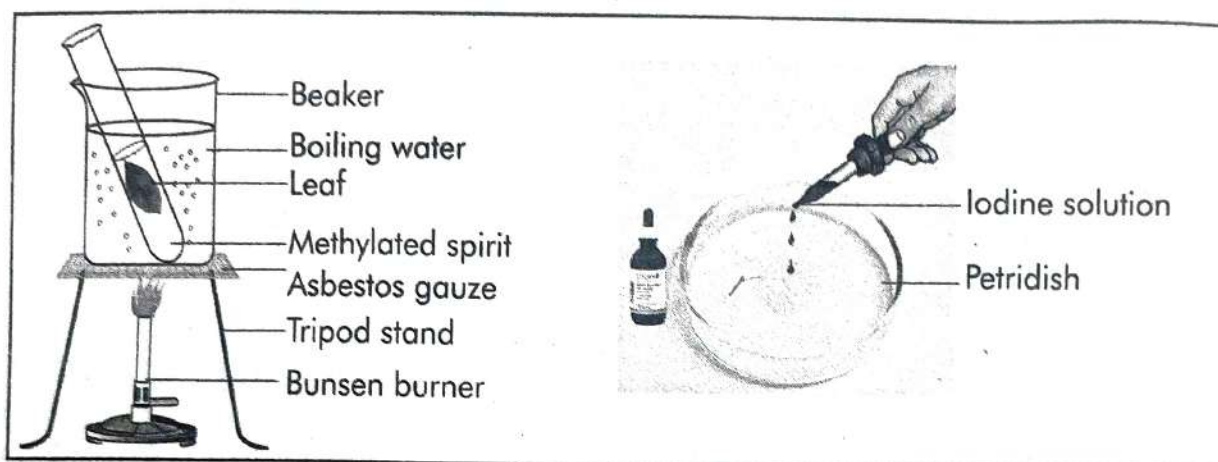
The modified equation of photosynthesis is



Experiments on photosynthesis :**i. Presence of starch (a type of carbohydrate) in leaves :**

To test the presence of starch in leaves take a leaf from a plant (with soft thin leaves) well exposed to sun light. Boil the leaf in methylated spirit over a water bath. It becomes pale white due to the removal of chlorophyll. Observe the leaf. Take the leaf carefully from test tube by using a brush. Spread the leaf in a petridish and add a few drops of tincture iodine/ betadine solution on it. Again observe the leaf.

The presence of starch will be indicated by a blue-black colour.

**Factors (Materials) essential for the process of photosynthesis :**

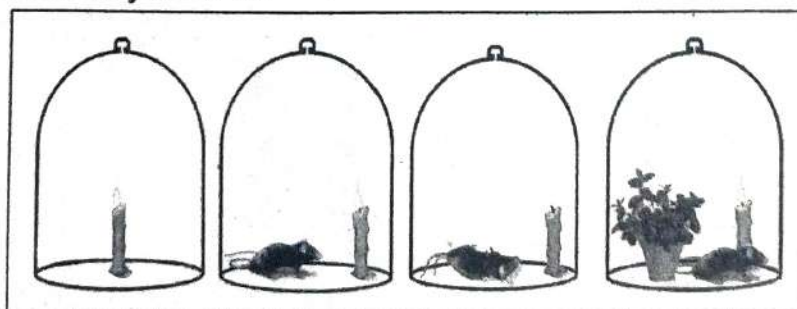
1. Water
2. Air
3. Light

Water and photosynthesis :

Von Helmont found that the water plays an important role in the increase of the plant mass. Von Helmont did not know about photosynthesis then. Later, it was found that increase in plant body mass or material occurred due to the process of photosynthesis.

Air and Photosynthesis :

- i) Joseph Priestley (1733-1804) in 1770 performed a series of experiments that revealed the essential role of air on the growth of green plants. Photosynthesis was still not known to scientists at that time.
- ii) Oxygen was discovered by Priestley in 1774. The name oxygen was coined later by Lavoisier in the year 1775.

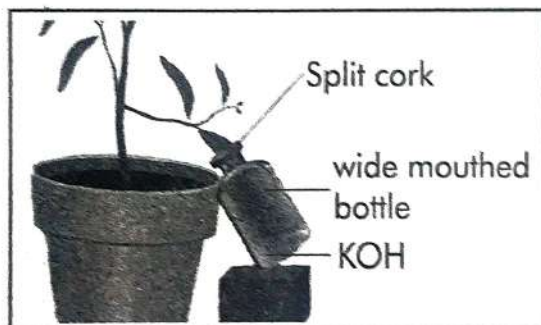


- iii) Priestley observed that a candle burning in a closed bell jar, soon gets extinguished. Similarly, a mouse would soon suffocate in a closed space of the bell jar.

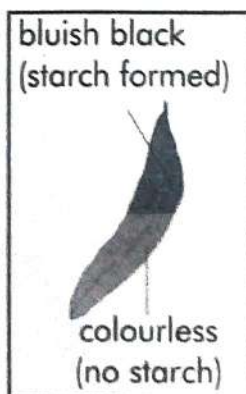
- iv) He concluded that a burning candle or an animal, both somehow, damage air.
- v) When he placed a mint plant in the same bell jars, he found that the mouse stayed alive and the candle when lighted from outside continued burning.
- vi) Priestley hypothesized as follows. Plants restore the air what breathing animals and burning candles remove.
- vii) Priestley's experiment confirmed that gaseous exchange was going on and plants were giving out a gas that supported burning and was essential for the survival of animals.

ii. **Carbon dioxide is necessary for photosynthesis (Mohl's half leaf experiment) :**

- i) Destarch a plant by keeping the plant in the dark for nearly a week to remove the starch (destarching) from the leaves.
- ii) Take a wide mouthed transparent bottle.
- iii) Put potassium hydroxide pellets or potassium hydroxide solution in the bottle. Potassium hydroxide absorbs carbon dioxide.



- iv) Insert splitted cork in the mouth of the bottle.
- v) Insert one of the leaf of destarched plant (through a split cork) into transparent bottle containing potassium hydroxide pellets / potassium hydroxide solution.
- vi) Leave the plant in sunlight for 4 to 5 hours.
- vii) After a few hours, test this leaf and any other leaf of this plant for starch.

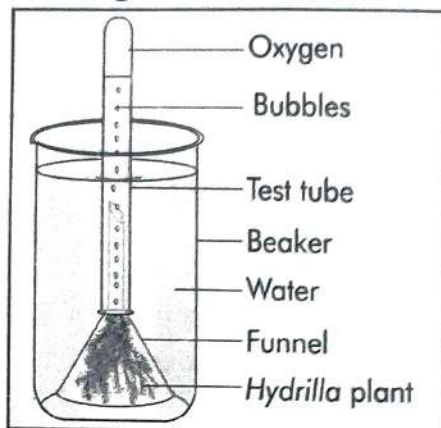


The leaf which was exposed to the atmospheric air becomes bluish black, and the one inside the flask containing potassium hydroxide which absorbs carbon dioxide in the bottle becomes brown instead of blue-black, showing that carbon dioxide is necessary for photosynthesis.

iii. Oxygen is evolved during photosynthesis in the presence of light :

To prove that oxygen is evolved during photosynthesis we need beaker with water, long cylindrical test tube, wide mouthed funnel, *Hydrilla* or *Elodea* plant twigs.

- Arrange the apparatus as shown in the figure.
- Place some twigs of water plant like *Elodea* or *Hydrilla* in a short stemmed funnel and keep it in a beaker containing water.



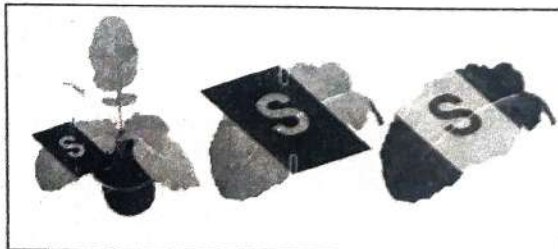
- Invert a test-tube full of water over the stem of the funnel.
- Ensure that the level of water in the beaker is above the level of stem of the inverted funnel.
- Place the apparatus in the sun for at least 2-3 hours.

We can see that in place of water there is air that fills in the set up kept in sun. It is actually the gas that is collected in the test-tube.

Test the gas in the test-tube by inserting a glowing match stick or incense stick which would burst into flames. This shows the presence of oxygen.

iv. Sunlight is necessary to form starch in green plants :

- Take a potted plant with destarched leaves.
- Cover one of its leaves with black paper on which a design is cut. Fix the paper on the leaf in such a manner that light does not enter the dark part.
- Place this potted plant in sunlight for 4 to 5 hours.



- After few hours of exposure to bright sunlight, test the leaf which is covered by black paper for the presence of starch.
- It will be observed that only the parts of the leaf, which could get light through the cut out design, turns blue-black showing the presence of starch.

Chlorophyll and photosynthesis :

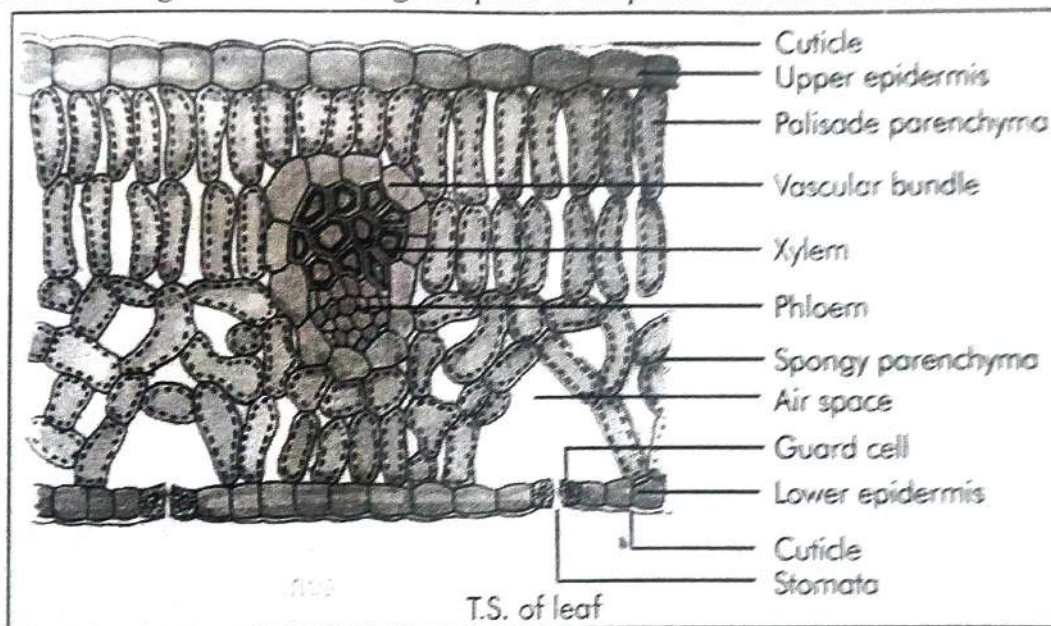
Chlorophyll pigment helps in trapping the solar energy which is required to carry out photosynthesis. So, the more the amount of chlorophyll, there will be more photosynthetic activity and vice-versa.

Where does photosynthesis takes place ?

The exact location of the photosynthetic part or a part containing chlorophyll was not known till another six decades after the discovery of chlorophyll.

In 1883, Julius Von Sachs, observed that chlorophyll is found in organelles within the cell. Such organelles were named as 'chloroplasts'.

These are present in large numbers in the cells (around 40 – 100) of parts like the stomatal guard cells and ground tissues of green parts of the plant. You have studied about Chloroplast.

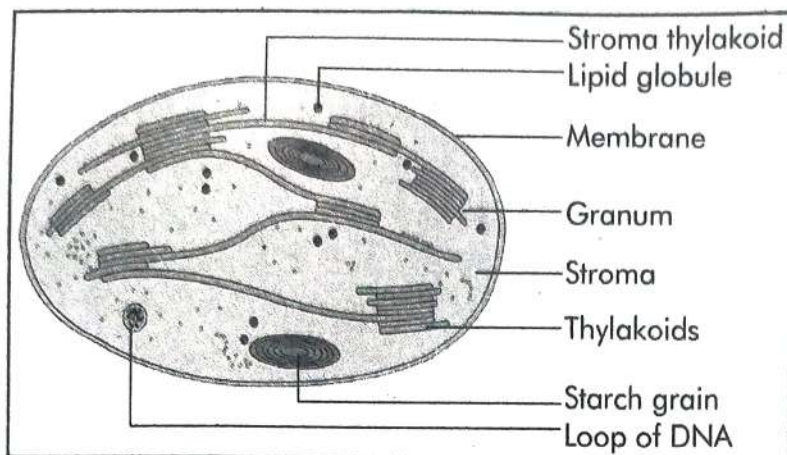


If a cell is broken up, the chloroplasts also break into pieces, so it becomes a very difficult task to isolate them to study the different steps of photosynthesis. It was not until 1954, that Daniel I. Arnon was able to break up plant cells so gently that whole chloroplasts could be obtained for study in the lab.

It has been found that the chloroplast is a membranous structure, consisting of 3 membranes.

The inner most layer forms stacked sack like structures called as grana. It is believed to be a site for trapping of solar energy.

The intermediary fluid filled portion is called as stroma. It is believed to be responsible for enzymatic reactions leading to the synthesis of glucose, which in turn join together to form starch.



Substances found in chloroplast which capture sunlight are called photosynthetic pigments.

Chlorophyll is such a pigment which contains one atom of magnesium.

Two major kinds of chlorophylls are associated with thylakoid membranes. Chlorophyll 'a' is bluish-green in colour and chlorophyll 'b' is yellowish-green colour. Around 250 to 400 pigment molecules are grouped as light harvesting complex or photosynthetic unit in each thylakoid.

Mechanism of photosynthesis :

During photosynthesis several events occur in the chloroplast. The major events are :

- i) Conversion of light energy to chemical energy
- ii) Splitting of water molecule (Photolysis)
- iii) Reduction of carbon dioxide to carbohydrates

Light dependent reaction (Photochemical phase) :

In this reaction light plays a key role. A series of chemical reactions occur in a very quick succession initiated by light and therefore the phase is technically called the photochemical phase or light dependent reaction. The light reaction takes place in chlorophyll containing thylakoids called grana of chloroplasts. Several steps occur in the light dependent reaction.

Step-I : The chlorophyll on exposure to light energy becomes activated by absorbing photons. (Photon is a unit of light energy)

Step-II : The energy is used in splitting water molecule to release O_2 .

The reaction is known as photolysis (photo means light, lysis means breaking). This was discovered by Robert Hill. Hence it is also called Hill's reaction.



Hydrogen produced in photolysis is immediately picked up by special compound NADP (Nicotinamide Adenine Dinucleotide Phosphate) to form NADPH (Reduced NADP). Another energy rich compound ATP (Adenosine triphosphate) is also formed at the end of the light reaction.

Light independent reaction (Biosynthetic phase) :

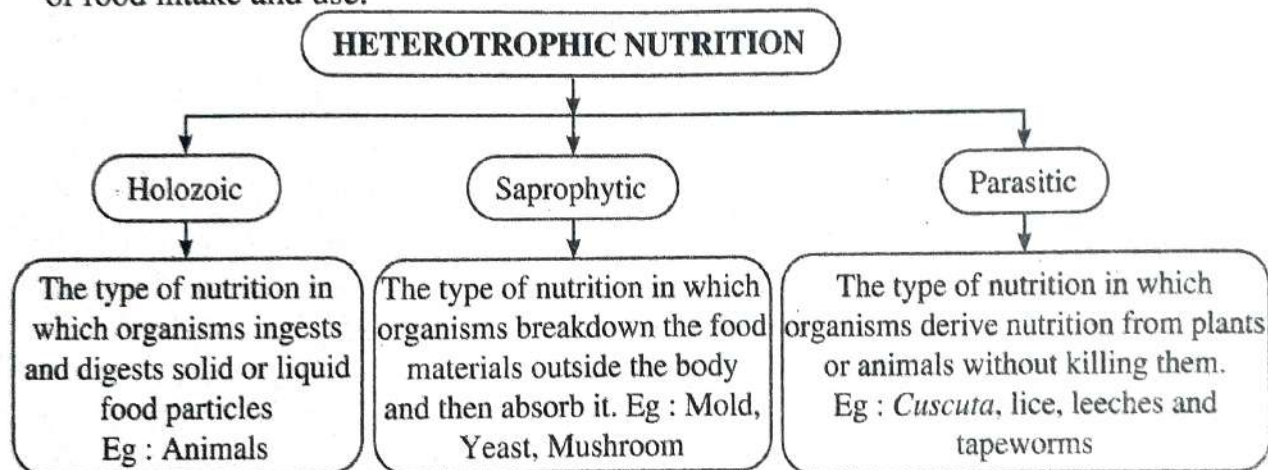
This reaction does not require the presence of light and extension of the phases after day time may occur in some plants. This is also called dark reaction. But the term dark reaction does not mean that it occurs when it is dark at night. It only means that these reactions do not depend on light. In the dark phase the hydrogen combines with CO_2 by utilizing ATP energy and produces glucose ($\text{C}_6\text{H}_{12}\text{O}_6$). This synthesis occurs in a number of steps using certain special intermediate compounds and enzymes. Finally glucose may be converted into storage product starch.

Heterotrophic nutrition :

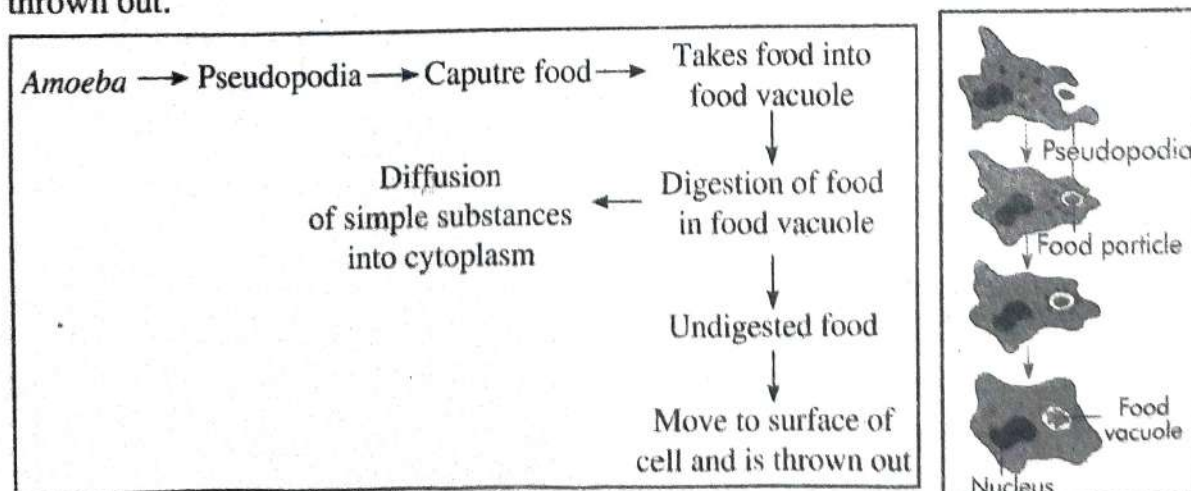
The mode of nutrition in which organisms do not prepare their own food and depend on other organisms for their food is called heterotrophic nutrition.

How do organisms obtain their nutrition :

Depending on the type and availability of food organisms can assort to a range of strategies of food intake and use.



For example, Amoeba takes in food using temporary finger like extensions (pseudopodia) of the cell surface which fuse over the food particle forming food vacuole. Inside the food vacuole, complex substances are broken down into simpler ones. Then diffuse into the cytoplasm. The remaining undigested material is moved to the surface of the cell and thrown out.



In *Paramecium* (unicellular organism) the cell has a definite shape. Food is taken in at a specific spot. Food is moved to the spot by the movement of cilia which covers the entire surface of the cell, where the food is ingested (cytostome).

Parasitic nutrition in *Cuscuta* :

Dodder (genus : *Cuscuta*) is a leafless, twining, parasitic plant belongs to morning glory family (Convolvulaceae).

The dodder plant contains no chlorophyll. *Cuscuta reflexa* has been found to have very little amount of chlorophyll. Hence, it absorbs food through haustoria. Haustoria are root like structures that penetrate the tissue of a host plant and may kill it. The slender, string like stems of the dodder may be yellow, orange, pink, or brown in colour. Its leaves are reduced to minute scales. Its flowers, in nodule like clusters, are made up of tiny yellow or white bell-like petals.

Nutrition in human beings :

Human digestive system is very complex in nature. Different parts are involved and perform different functions by using various digestive juices and enzymes. The alimentary canal is basically a long tube extending from the mouth to the anus.

Passage of food through alimentary canal or gut :

Food is cut and crushed by our teeth in the mouth and mixed with saliva to make it wet and slippery lump called bolus. This process is called as mastication.

This bolus is suitable to pass through oesophagus. Saliva is secreted by three pairs of salivary glands. Saliva mainly contains an enzyme amylase (ptyalin) which help in the breakdown of complex carbohydrates into simple ones.

Digestion : The process of breaking down larger complex substances into simple ones with the help of enzymes is called digestion.

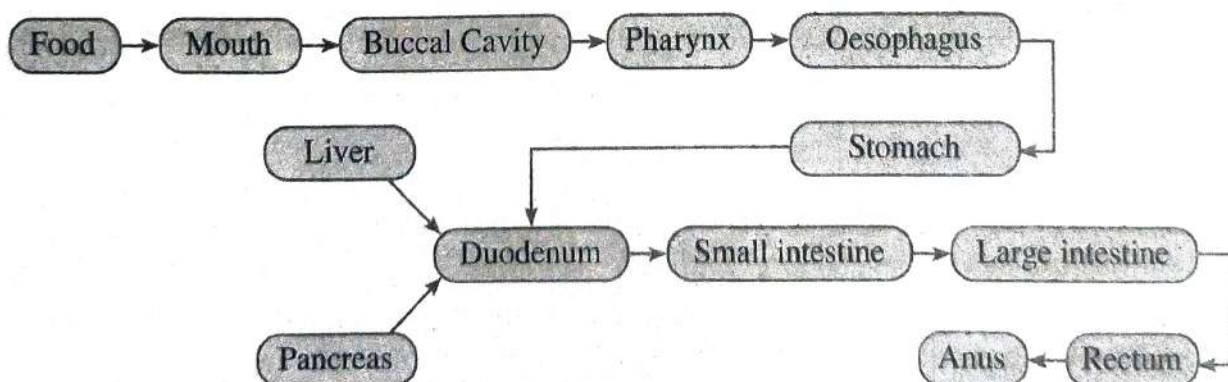
- The soft food mixed with saliva passes through oesophagus or food pipe by wave like movements called peristaltic movement to the stomach.
- At the stomach, food gets churned with gastric juice and HCl rendering the medium acidic. Now the food is in semisolid condition. The digestion of food goes on as most of the proteins are broken down into smaller molecules with the help of enzyme pepsin.
- Food in the form of a soft slimy substance where some proteins and carbohydrates have already been broken down is called chyme and it is passed into the small intestine with the help of sphincters.
- The small intestine is the longest part of the alimentary canal. Its proximal part is called duodenum. It is the site of further digestion of carbohydrates, proteins and fats.
- Fats are digested by converting them into small globule like forms by the help of the bile juice secreted from liver. This process is called emulsification.
- The villi of small intestine absorb the digested food and supply it to every cell of the body through the blood.

- The undigested food passes through the large intestine, where excess water and useful substances are reabsorbed and unwanted materials are egested from the body through anus is called defecation.

Digestive enzymes :

S. No.	Enzyme/ Substance	Secreted by	Secreted into	Digestive juice	Acts on	Products on
1	Ptyalin (Salivary Amylase)	Salivary glands	Buccal cavity	Saliva	Carbohydrates	Dextrins and maltose
2	Pepsin	Gastric glands	Stomach	Gastric juice	Proteins	Peptones
3	Bile (No enzymes)	Liver	Duodenum	Bile juice	Fats	Emulsification (breaking down of large fats into small globules)
4	Amylase	Pancreas	Duodenum	Pancreatic juice	Carbohydrates	Maltose
5	Trypsin	Pancreas	Duodenum	Pancreatic juice	Proteins	Peptones
6	Lipase	Pancreas	Duodenum	Pancreatic juice, Intestinal juice	Fats	Fatty acids and glycerol
7	Peptidases	Intestinal glands	Small intestine	Intestinal juice	Peptides	Amino acids
8	Sucrase	Intestinal glands	Small intestine	Intestinal juice	Sucrose (Cane sugar)	Glucose

Flowchart of human digestive system. (Or) Path of food flow in alimentary canal.



Health aspects of the alimentary canal :

Vomiting is the body's method of ridding itself of unwanted or harmful substances from the stomach. The peristaltic movements of the stomach and oesophagus reverse their

normal direction and the food is expelled. There are many causes of vomiting, but one of the most common is over eating, especially when the food contains a high proportion of fat. Vomiting also occurs when we eat something very indigestible or poisonous.

When we have a greenish vomit usually called as 'bilious' or 'liverish', we get a bitter taste and it is often the result of having over eaten. The liver is unable to cope with the excessive fat and we get a feeling of nausea.

Indigestion is a general term used when there is difficulty in digesting food. Healthy people can usually avoid problems related to digestion by:

- a) having simple, well balanced meal.
- b) eating them in a leisurely manner.
- c) thoroughly masticating the food.
- d) avoiding taking violent exercise soon after eating food.
- e) drinking plenty of water and having regular bowel movements.

Proper functioning of all life processes require adequate amount of food in all living organisms. It is not just the intake of food but its assimilation and expulsion of wastes that play an important role. Intake of fibre rich food avoids constipation.

Deficiency due to malnutrition :

Eating of food that does not have one or more than one nutrients in required amount is known as malnutrition. Poor health, willful starvation, lack of awareness of nutritional habits, socio-economic factors are all the reasons for malnutrition in our country.

Malnutrition may be as follows :

1. Protein malnutrition 2. Calorie malnutrition 3. Protein calorie malnutrition.

Kwashiorkor :

This is due to protein deficiency in diet. Body parts become swollen due to accumulation of water in the intercellular spaces. Very poor muscle development, swollen legs, fluffy face, difficulty in eating, diarrhoea, dry skin are the symptoms of this disease.

Marasmus :

This is due to deficiency of both proteins and calories. Generally this disease occurs when there is an immediate second pregnancy or repeated child births. Lean and weak body, swollen limbs, less developed muscles, dry skin, diarrhoea, etc., are the symptoms of this disease.

Obesity : Apart from the deficiency diseases mentioned above, the constant intake of high calorific food may lead to a condition called obesity. In this condition a person gains extra fat and becomes overweight. Obese children may suffer from problems related to cardiovascular, renal, gallbladder etc.

Vitamin deficiency diseases :

Vitamins are organic substances. They are micro nutrients required in small quantities. Vitamins are classified into two groups. i. Water soluble vitamins (B-complex, vitamin C) and ii. Fat soluble vitamins (vitamin A, D, E and K).

Vitamin	Sources	Deficiency diseases	Symptoms
Thiamine (B ₁)	Cereals, oil seeds, vegetables, milk, meat, fish, eggs	Beri beri	Vomitings, fits, loss of appetite, difficulty in breathing, paralysis.
Riboflavin (B ₂)	Milk, eggs, liver, kidney, green leafy vegetables	Glossitis	Mouth cracks at corners, red and sore tongue, photophobia, scaly skin.
Niacin (B ₃)	Kidney, liver, meat, egg fish, oil seeds	Pellagra	Dermatitis, diarrhoea, loss of memory, scaly skin.
Pyridoxine (B ₆)	Cereals, oil seeds, vegetables, milk, meat, fish, eggs, liver.	Anaemia	Hyper irritability, nausea, vomiting, fits.
Cyanocobalamine (B ₁₂)	Synthesised by bacteria present in the intestine	Pernicious anaemia	Lean and weak, less appetite.
Folic acid	Liver, meat, eggs, milk, fruits, cereals, leafy vegetables.	Anaemia	Diarrhoea, loss of leucocytes, intestinal mucus problems.
Pantothenic acid	Sweet potatoes, groundnuts, vegetables, liver, kidney, egg.	Burning feet	Walking problems, sprain.
Biotin	Pulses, nuts, vegetables, liver, milk, kidney.	Nervous disorders	Fatigue, mental depression, muscle pains.
Ascorbic acid (C)	Green leafy vegetables, citrus fruits, sprouts.	Scurvy	Delay in healing of wounds, fractures of bones.
Retinol (A)	Leafy vegetables, carrot, tomato, pumpkin, papaya, mango, meat, fish, egg, liver, milk, cod liver oil, shark liver oil.	Eye, skin diseases	Night blindness, Xerophthalmia, cornea failure, scaly skin.
Calciferol (D)	Liver, egg, butter, cod liver oil, shark liver oil, Sun-rays stimulate the formation of vitamin D from the sub-cutaneous fat.	Rickets	Improper formation of bones, Knock knees, swollen wrists, delayed dentition, weak bones.
Tocopherol (E)	Fruits, vegetables, sprouts, sunflower oil	Fertility related disorders	Sterility in males, abortions in males.
Phylloquinone(K)	Green leafy vegetables, milk, meat, egg.	Problems related to blood clotting.	Delay in blood clotting, over bleeding.

PRIORITY - 1

2 Marks Questions :

1. What materials are required to prove that oxygen is produced during photosynthesis in the presence of light? (AS₁) (Q.No.2, P.No. 22)
2. Write the chemicals used in experiment conducted to know the presence of starch in leaves. methylated spirit, iodine solution (AS₁) (Q.No. 22, P.No. 11)
3. Write the experimental procedure to prove "sunlight is necessary for photosynthesis". (AS₁) (Q.No.2, P.No.22)
4. What would happen if there are no roughages in the diet? (AS₂) (Q.No.16, P.No.10)
5. What would be the consequences if HCl fail to function? (AS₂) (Q.No.10, P.No.9)
6. Why the potted plant is kept in dark room for three days prior to the 'Black paper experiment'? (AS₃) (Q.No.2, P.No.22)
7. What food items will you include in your meal to take a balanced diet? (AS₇) (Q.No.30, P.No.15)
8. After reading about the diseases due to malnutrition, what type of food habits you cultivate? (Or) What food items will you include in you meal to prevent nutritional deficiency diseases? (AS₇) (Q.No.30, P.No.15)
9. What will happen to protein digestion as the medium of intestine is gradually rendered alkaline? (AS₂) (Q.No. 15, P.No.10)
10. What process do you follow in your laboratory to study the presence of starch in leaves? (Or) Explain the procedure that is to be followed in your laboratory to know (prove) the presence of starch in leaves. (AS₃) (Q.No. 22, P.No. 11)

4 Marks Questions :

1. Explain the necessary conditions for autotrophic nutrition and what are its by-products? (AS₁) (Q.No. 27, P.No. 21)
2. How is the small intestine designed to assimilate / absorb digested food? Explain. (Or) How is the structure of small intestine suitable for the absorption of digested food? (AS₁) (Q.No. 12, P.No. 9)
3. How are fats digested in our bodies? Where does this process takes place? (Or) Write the mechanism of fat digestion in our body. (AS₁) (Q.No. 13, P.No.9)
4. What is the role of saliva in the digestion of food? (Or) (AS₁) What role does saliva play on the food in the oral cavity? (Or) (AS₁) (Q.No. 14, P.No.10)
5. How can you say that carbohydrates are not digested in the stomach? (Or) (AS₁) "Carbohydrates are not digested in the stomach". Justify the statement. (AS₁) (Q.No. 21, P.No.10)

6. How do non-green plants such as fungi and bacteria obtain their nourishment? (Or) Explain the mode of nutrition in non-green plants like bacteria and fungi. (AS₁)(Q.No.18, P.No.11)
7. Explain the process of photosynthesis in detail with the help of a flowchart?
(AS₁)(Q.No5, P.No. 8)
8. If there were no green plants, all life on the earth would come to an end. Justify the statement. (Or) "In the absence of green plants, life on the earth would come to an end". Justify the statement. (Or) (AS₁) Life without green plants on the earth would come to an end. Comment. (AS₁)(Q.No. 29, P. No. 15)
9. Do you think the new reddish leaves of plants also carry out photosynthesis? How?
(AS₇)(TB, P.No.9)
10. Give examples : (AS₁)(Q.No. 3, P.No. 8)
- Digestive enzymes
 - Organisms exhibiting heterotrophic nutrition
 - Vitamins
 - Nutritional deficiency diseases

6 Marks Questions :

- What is malnutrition? Explain some nutritional deficiency diseases. (AS₁)(Q.No.17,P.No. 10)
- Write the differences between : (AS₁)(Q.No. 1, P.No. 6)
 - Autotrophic nutrition - heterotrophic nutrition
 - Ingestion - digestion
 - Chlorophyll - chloroplast
- Give reasons : (AS₁)(Q.No. 2, P.No. 7)
 - Why photosynthesis is considered as the basic energy source for most of living world?
 - Why is it better to call the dark phase of photosynthesis as a light independent phase?
 - Why is it not possible to demonstrate respiration in green plant kept in sunlight?
- How would you demonstrate that green plants release oxygen when exposed to light? (Or) How do you prove that oxygen is produced during photosynthesis in the presence of light? (Or) Explain *Hydrilla* experiment with the help of a neat labelled diagram.
(AS₃)(Q.No. 23, P.No. 12)
- Write an experiment to prove that carbon dioxide is essential for photosynthesis.
(AS₃)(Q.No. 1, P.No. 22)

PRIORITY - 2

2 Mark Questions :

- What happen if chloroplast are absent in the leaf? (AS₂)(Q.No.2, P.No. 21)
- Why do we boil the leaf in methylated spirit before starch test? (AS₃)(Q.No.6, P.No. 16)
- What happens to the plant if the rate of respiration becomes more than the rate of photosynthesis? (AS₂)(Q.No.20, P.No. 11)
- How do you appreciate the role of green plants as food suppliers?
(AS₆)(Q.No. 29, P.No. 15)

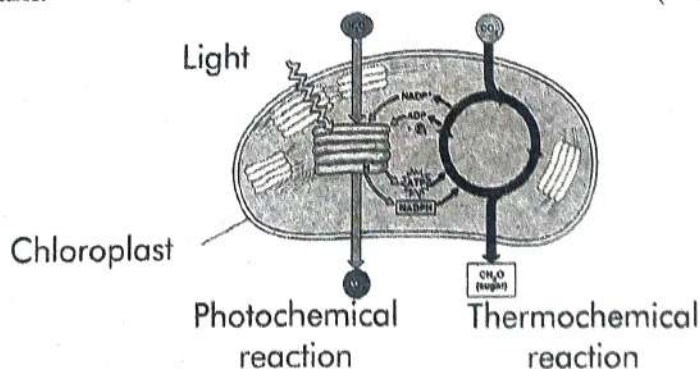
5. What will happen if saliva is not secreted into the buccal cavity? (AS₂) (Q.No. 14, P.No. 14)

4 Marks Questions :

1. Explain the changes that take place during the light independent reactions of photosynthesis. (AS₁) (Q.No. 28, P.No. 14)
2. Describe various diseases caused due to the deficiency of vitamin 'A'. (AS₁) (Q.No. 13, P.No. 17)
3. What food habits are you going to follow after reading the chapter "Nutrition"? Why? (AS₁) (Or) Mention the food habits that are to be followed after reading the chapter, 'Nutrition'. (AS₁) (Q.No.30, P.No. 15)
4. Bhavana wants to collect information about diseases due to malnutrition and vitamins. For this :
 - i) Whom should she meet ?
 - ii) Prepare chart about vitamins, what tabular form she should draw to?
 - iii) What columns does she need to prepare to make a chart on malnutrition?
 (AS₄) (Q.No.2, P.No. 24)
5. Tabulate the information showing the resources, deficiency diseases and symptoms of vitamin B complex. (AS₄) (TB. P.No. 21, Table)

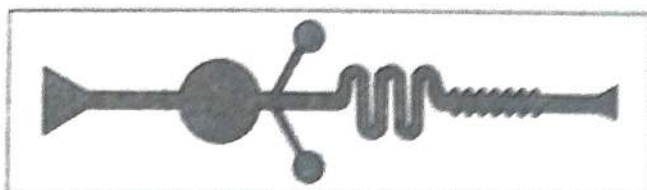
6 Marks Questions :

1. Explain the structure of chloroplast with a neat labelled sketch. (AS₁) (Or) Draw a neat labelled diagram of kitchen of cell. (AS₅) (Q.No. 9, P.No. 9)
2. Observe the following diagram and write a note on light dependent and light independent reactions. (Or) Write a comparative account on light and dark reactions by observing the following diagram. (AS₅) (Q.No.28, P.No. 14)



3. Write about fat soluble vitamins. (AS₁) (Q.No. 17, P.No. 18)
4. Write the experiment of bell jar and mint plant, performed by Priestly to prove that air plays key role in photosynthesis. (AS₃) (Q.No. 3, P.No. 23)

5. Raheem prepared a model showing the passage of the food through different parts of the alimentary canal. Observe this and label its parts. (Or) What does the following picture represent? Observe it and label the parts. (AS, Q.No.27, P.No.13)



PRIORITY - 3

Complete all the remaining question and answers of the lesson.

IMPORTANT DIAGRAMS

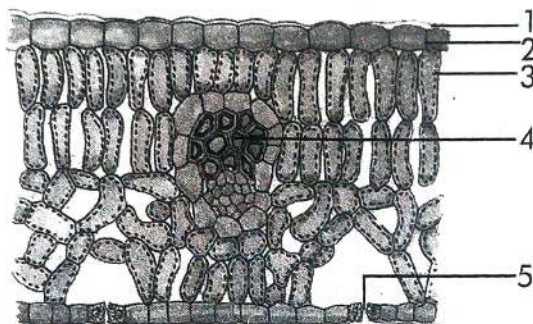
- i) T.S. of leaf ii) Nutrition in *Paramecium* iii) Nutrition in *Amoeba*
iv) Human digestive system v) T.S. of Chloroplast

PRACTICE SHEET

Choose the correct answer :

- The chyme is acidic in stomach but after entering into small intestine, it changes into alkaline due to the action of : [A]
A) Pepsin B) HCl C) Bile juice D) Pancreatic juice
- The H^+ ions produced in photolysis are immediately taken up by a special compound called: [B]
A) ATP B) NADP C) NADPH D) ADP
- Which is the body's method of ridding unwanted or harmful substances from stomach? [B]
A) Chocking B) Vomiting C) Defecation D) Mastication
- In our country, the main source of Niacin is : [D]
i) Kidney ii) Vegetables iii) Oil seeds iv) Sweet potatoes
A) i, ii, iii B) ii and iii C) i and ii D) i and iii
- In the absence of which organ, emulsification process does not take place? [B]
A) Kidney B) Liver C) Small intestine D) Pancreas
- The bile contains : [C]
A) enzymes B) hormones C) pigments D) gases
- $H_2O \xrightarrow[\text{Chlorophyll}]{\text{Light}} H^+ + OH^-$. This change occurs during : [C]
A) Mohl's reaction B) Priestly's reaction C) Hill's reaction D) Harvey's reaction

8. What are the reasons for malnutrition in our country? [D]
 i) Poor health; willful starvation ii) Lack of awareness of nutritional habits
 iii) Socio-economic factors iv) Willful starvation
 A) i and ii B) ii and iii C) Only ii D) i, ii, iii and iv
9. Bacteria present in intestine synthesise _____ vitamin. [B]
 A) A B) B₁₂ C) B₆ D) D
10. The element present in chlorophyll is : [C]
 A) Silver B) Copper C) Magnesium D) Potassium
11. The proximal part of the small intestine is called : [C]
 A) Oesophagus B) Stomach C) Duodenum D) Rectum
12. The below given figure represents the vertical section of a leaf. The functions of parts 4 and 5 respectively are : [D]



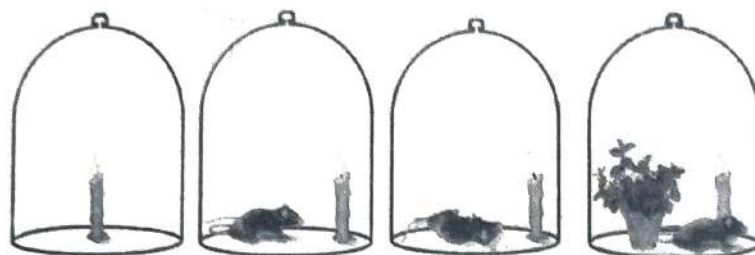
Conduction of water

transpiration

- A) Conduction of food materials and food synthesis.
 B) Conduction of food materials and transpiration.
 C) Conduction of water and food synthesis.
 D) Conduction of water and transpiration.
13. A person is suffering from fatigue, mental depression and muscle pains. This is due to the deficiency of : [B]
 A) folic acid B) biotin C) pyridoxine D) cyanocobalamin
14. Absorption of digested food : Small intestine :: Absorption of water : ? [B]
 A) Stomach B) Large intestine C) Caecum D) Rectum
15. Which of the following is/are 'not' the precaution(s) for the black paper experiment? [D]
 A) Do not boil the leaf in alcohol directly on flame.
 B) Always decolourise the leaf before starch test.
 C) Attach the black paper strip on the surface of leaf tightly so that sunlight does not reach the covered area.
 D) The bottle is kept air tight by applying grease (or) vaseline.
16. Vomiting, fits, loss of appetite, difficulty in breathing, paralysis are the symptoms related to the disease : [B]
 A) Anaemia B) Beri-beri C) Scurvy D) Pellagra

17. The given experimental setup represents :

[A]



- A) Priestly's experiment
C) Glass and rat experiment

- B) Van Neil's experiment
D) Black paper experiment

18. Where do you find more photosynthetic activity in a leaf ?

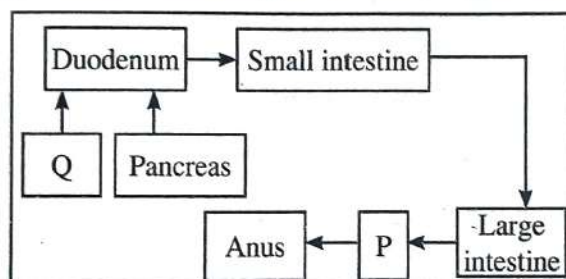
[B]

- A) Upper epidermis
C) Lower epidermis

- B) Palisade parenchyma
D) Spongy parenchyma

19. What does 'P' and 'Q' in the following flowchart represent ?

[D]

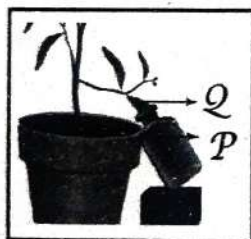


- A) P - Pyloric stomach, Q - Liver
C) P - Duodenum, Q - Liver

- B) P - Large intestine, Q - Oesophagus
D) P - Rectum, Q - Liver

20. Which part of the leaf performs photosynthesis in the given experimental setup ?

[A]



- A) Q
B) P
C) Both P and Q
D) Neither P nor Q

2. RESPIRATION

CHAPTER AT A GLANCE

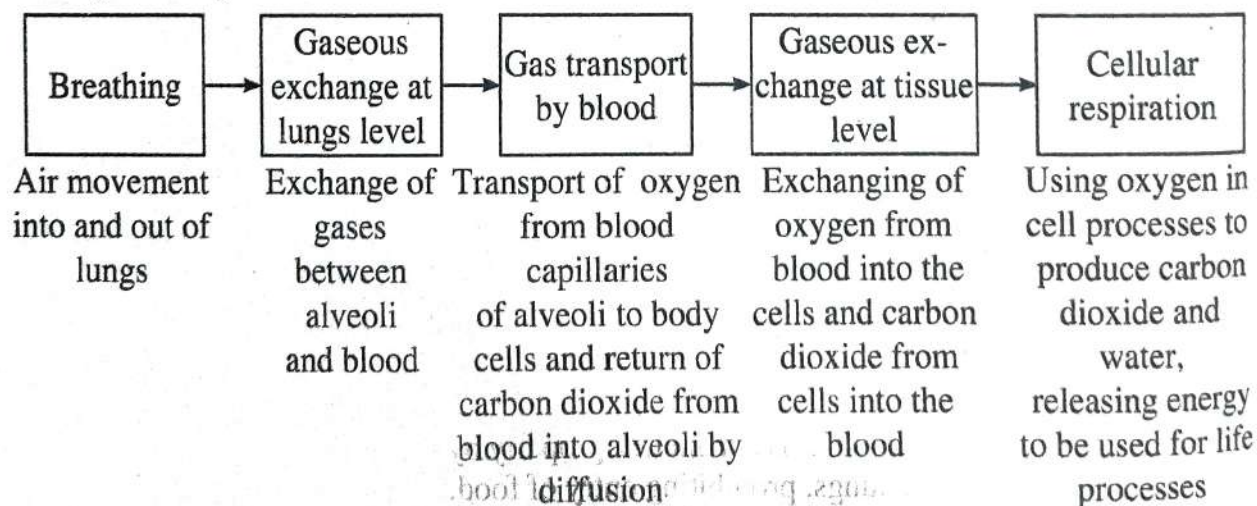
The food provides energy for all the biological activities only after break down through the process known as respiration.

The oxidation of food materials to produce energy is known as respiration. Carbon dioxide and water are released as by products during this process.

The term 'respiration' is derived from Latin word 'respire' meaning 'to breathe'.

Discovery of gases and respiration :

- i) Lavoisier and Priestley did a comprehensive work on properties of gases, their exchange and respiration that we came to know something about how the process of gaseous exchange goes on, in our body.
- ii) Lavoisier also carried out several experiments to understand the properties of gases. In his early experiments, it is clear that Lavoisier thought that the gas liberated on heating powdered charcoal in a bell jar kept over water in a trough was like fixed air.
- iii) In those days carbon dioxide was known as fixed air. The next series of experiments dealt with the combustion of phosphorus in a bell jar. From these studies Lavoisier showed that whatever in the atmospheric air which combined with the phosphorus, was not water vapour.
- iv) His final words are that the substance which combines with the phosphorus is "either air itself, or another elastic fluid present in a certain proportion, in the air which we breathe". This was the respirable air, a component of air that also helped in burning.
- v) The air that we breathe out made lime water milky while that after heating metal did not. He noted that fixed air also made lime water milky.
- vi) He drew immediately the logical conclusion regarding the process of respiration. Either eminently respirable air (O_2) is changed in the lungs to fixed (CO_2) or it may be exchanging.

Steps in respiration :

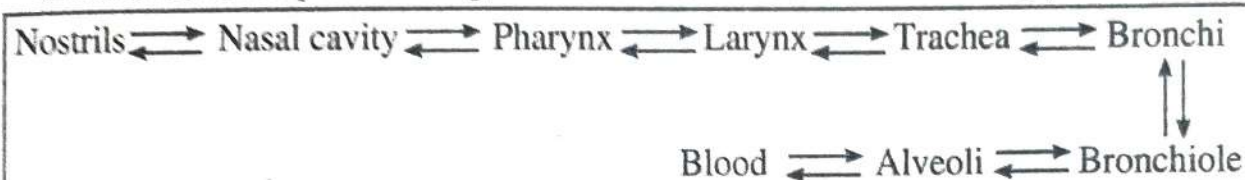
Breathing :

The physical process which involves the exchange of gases is called breathing. It does not require any enzymes.

The exhaled air consists of larger amount of carbon dioxide compared to inhaled air which has more amount of oxygen.

Pathway of air :

In humans, the air passes through the nostrils to alveoli in the following path.



Part of the respiratory system	Function
Nostrils	Facilitates the entry of air into the body and exit of air from the body.
Nasal cavity	Filters the air. Inhaled air passes through the nasal cavity, its temperature is brought close to that of the body, and it takes up water vapour so that it becomes more moist than before.
Pharynx	Warming and moistening goes on in this common passage of digestive and respiratory system.
Larynx (Voice box)	This stiff box contains vocal cords. When air passes out of the lungs and over the vocal cords, it causes them to vibrate. This produces sounds and forms the basis of our speech, song, etc.
Trachea (Wind pipe)	Channelises air to the lungs.
Bronchus	At its lower end, the trachea or the wind pipe divides into two bronchi-one leading to each lung. It helps in the passage of air into bronchioles.
Bronchioles	Each bronchus is further divided into smaller and smaller branches called bronchioles. They pass the air into the structural and functional units of respiratory system i.e., alveoli.
Alveolus	Gaseous exchange takes place here as blood capillaries take up oxygen and expel carbon dioxide.

The alveoli are the structural and functional units of lungs. They are richly supplied with blood vessels and are the sites of gaseous exchange (O_2 and CO_2) that occur between blood and atmosphere.

Epiglottis and passage of air :

The movement of food and air is channelled properly by a flap like valve, the epiglottis that protects the tube to the lungs, prohibiting entry of food.

This valve is partly closed when we swallow food; it deflects food down to the stomach and keeps it out of the trachea or wind pipe which is the route to the lungs.

The epiglottis opens more widely when we take a breath, and air enters the lungs. Nervous regulation is important in guiding the function of epiglottis and passage of food and air.

Mechanism of respiration in human beings :

Respiration in man occurs in two stages :

i) Inspiration

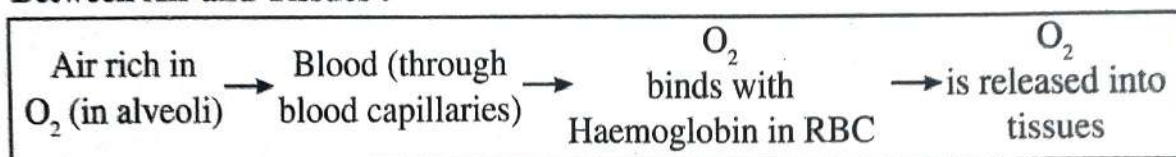
ii) Expiration

INSPIRATION	EXPIRATION
i) It is an active process.	i) It is a passive process.
ii) It is the process of inhalation or breathing air into the lungs.	ii) It is the process of exhalation or breathing air out of the lungs.
iii) It takes place by the contraction of diaphragm and the chest wall muscles.	iii) It takes place by the relaxation of diaphragm and the chest wall muscles.
iv) It increases the volume of thoracic cavity.	iv) It decreases the volume of thoracic cavity.

GASEOUS EXCHANGE (alveoli to capillaries) :

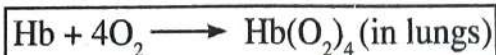
Exchange of gases between alveolus, blood and tissues :

Between Air and Tissues :



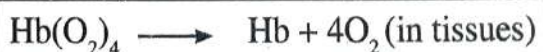
Transportation of oxygen :

The oxygen in the air of lungs combines with the haemoglobin present in blood to form oxyhaemoglobin and carried to the cells.



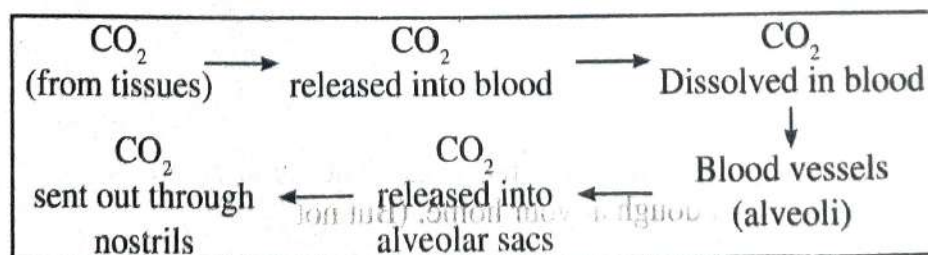
Transport of carbon dioxide :

CO_2 is usually transported in the form of bicarbonates through blood plasma and Carboxy haemoglobin through Hb.



GASEOUS EXCHANGE (capillaries to cells and back) :

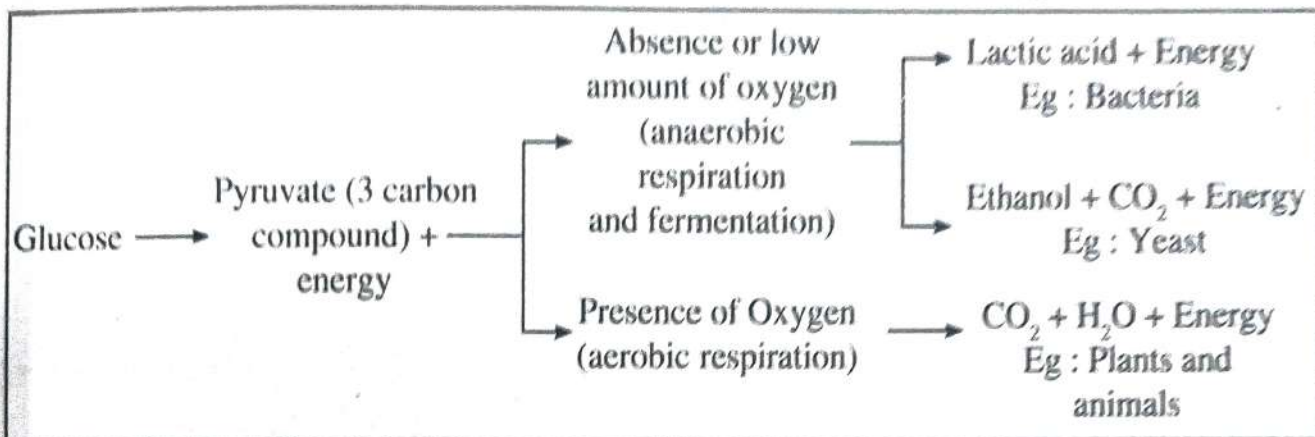
Between Tissues and Air :



Cellular respiration :

Cellular respiration in prokaryotic cells like that of bacteria occurs within the cytoplasm. In eukaryotic cells cytoplasm and mitochondria are the sites of the reactions. The produced energy is stored in mitochondria in the form of ATP (Adenosine tri phosphate). That is why mitochondria are called "power houses of the cell".

ATP is an energy rich compound that is capable of supplying energy wherever needed within the cell. Each ATP molecule gives 7200 calories of energy. This energy is stored in the form of phosphate bonds. If the bond is broken the stored energy is released.

**Oxygen debt :**

The temporary oxygen shortage in the body tissues resulting from strenuous physical activity.

Experiments with yeast :

Yeast grows rapidly if it is supplied with glucose in solution. Indeed, wild Yeasts are normally found growing on the skins of fruits like grapes and apples, from which they derive their food supplies.

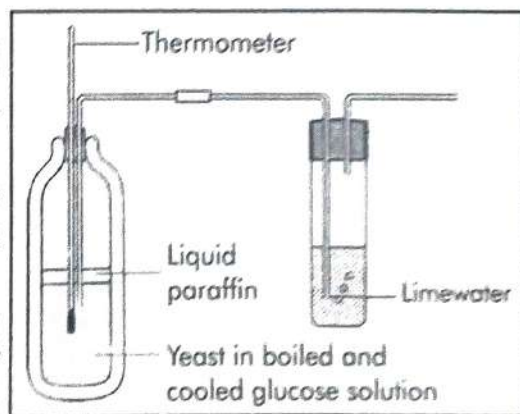
We can remove dissolved oxygen from glucose solution by heating it for a minute, and then cooling it without shaking. Now put in some Yeast.

If you wish to check that the oxygen has been removed from the mixture, add a few drops of diazine green (Janus Green B) solution to the yeast suspension before you pour the liquid paraffin (wax) over it. This blue dye turns pink when oxygen is in short supply around it.

The supply of oxygen from the air can be cut off by pouring one centimetre layer of liquid paraffin on to the mixture.

Fermentation :

If Yeast and sugar solution are left to stand without oxygen for some days, they develop a characteristic smell, caused by production of new compound called ethanol, which has been manufactured by the Yeast from the sugar. The same type of smell you can notice from preserved idly, dosa dough at your home. (But not in refrigerator).



The ethanol can be separated from the Yeast-glucose mixture by the process of fractional distillation since ethanol boils at a lower temperature (70°C) than the sugar solution.

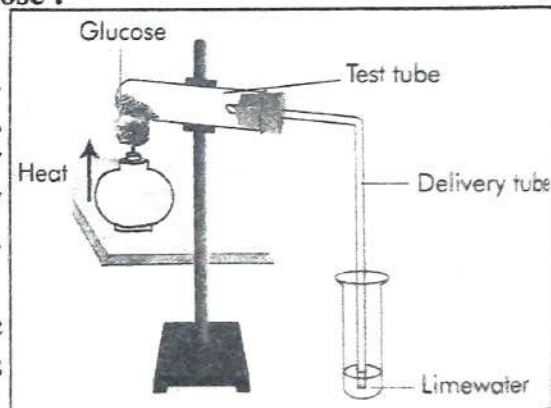
Respiration Vs Combustion :

RESPIRATION (Oxidation of sugars)	COMBUSTION (Burning of sugars in the presence of oxygen)
i) The entire process occurs at the body temperature of an organism. No external energy required.	i) External energy is to be supplied to burn the substance.
ii) No charring and burning. Sugar is oxidised.	ii) Sugar first melts, chars and then burns.
iii) Several intermediate compounds are formed.	iii) No intermediate compounds are formed.
iv) Energy is released at several stages in the form of heat energy and ATP.	iv) Entire energy is released at a time in the form of light and heat energy.
v) It occurs in the presence of water.	v) It occurs in the absence of water.
vi) Enzymes are required.	vi) Enzymes are not required.
vii) It is a controlled process.	vii) It is an uncontrolled process.
viii) Energy is stored in the form of ATP in the body.	viii) Energy is not stored and is released into the atmosphere.

Observing changes during combustion of glucose :

When glucose burns, carbon dioxide and water are produced and energy is released as heat. Combustion of glucose gives us carbon dioxide, water and energy while from the respiratory equation we get the same products. But essentially the processes must differ due to following reasons.

- Glucose must be burnt at high temperature in the laboratory to liberate energy, if it happened in our cells, cells would be burnt.
- Once glucose starts burning we can't stop the process easily, but living cells are able to exercise control over the sort of burning of glucose in the presence of oxygen.
- Water normally stops combustion from taking place while cells contain a lot of water and respiration still goes on.



Heat production by living organisms :

Heat production was a feature of burning glucose or sugar. Living animals and plants usually produce energy in the form of heat. Heat is constantly lost from the body surface thus it must be continuously generated within our bodies to replace what has been lost to keep the body temperature constant. In the course of vigorous activity, a greater amount of heat is generated. We know that we feel hot after some form of strenuous exercise such as running.

Evolution in gases exchanging system :

Exchange of gases is a common life process in all living organisms, but it is not same in all. Single celled organisms the *Amoeba* or multicellular organisms like *Hydra* and *Planaria* obtain oxygen and expel carbon dioxide directly from the body by the process of diffusion.

Respiratory organ	Type of respiration	Organisms
Trachea	Tracheal respiration	Insects
Gills	Branchial respiration	Fishes, Tadpole larva of frog, etc.
Skin	Cutaneous respiration	Earthworm, leech, frog, etc.
Lungs	Pulmonary / Bronchial respiration	Reptiles, birds, mammals, etc.

Respiration in plants :

In plants, exchange of gases takes place through stomata of leaf and lenticels of stem, etc.

Conduction within the plants.

- The stomatal openings leave a series of spaces between the cells inside the plant called 'Air spaces' (Intercellular spaces). These form a continuous network all over the plant.
- The air spaces are lined with water where the oxygen is dissolved and passes through the porous cell walls into the cytoplasm.
- Carbon dioxide produced in cytoplasm passes out into the air spaces by similar process.

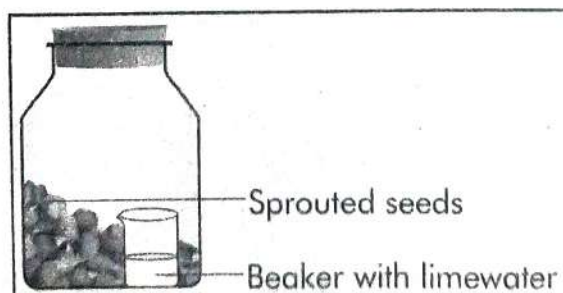
Respiration through roots :

Most plants can aerate their roots by taking in oxygen through the lenticels or through the surface of their root hairs (as their walls are very thin). They obtain oxygen from the air spaces existing between the soil particles. But, plants which have their roots in very wet watered places, such as ponds or marshes, are unable to obtain oxygen. They are adapted to these water-logged conditions by having much larger air spaces which connect the stems with the roots, making diffusion from the upper parts much more efficient.

The mangrove trees of the tropics form aerial roots above the soil and takes in oxygen through these roots.

Evolution of carbon dioxide in respiration :

- Take a handful of moong or bengal gram seeds.



- Soak the seeds in water a day before you perform experiment.
- Keep these soaked seeds in a cloth pouch and tie them with a string tightly.

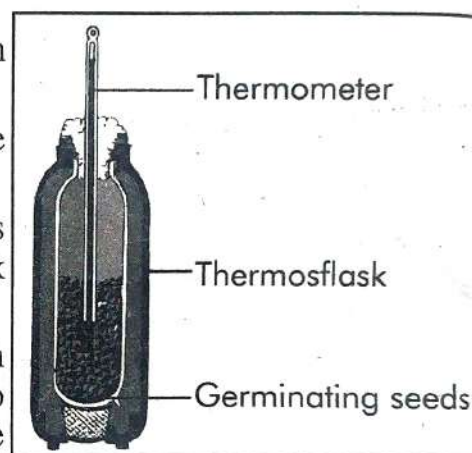
- iv) Keep the cloth pouch in a corner of your classroom.
- v) Next day collect the sprouts / germinated seeds from the pouch. Keep it in a glass bottle/ plastic bottle around 200 ml. capacity.
- vi) Take a small beaker.
- vii) Fill three-fourth of the beaker with lime water and insert it in the beaker carefully.
- viii) Close the plastic bottle tightly.
- ix) Make a similar setup with dry (unsprouted) seeds.
- x) Keep both the sets undisturbed for one or two days.
- xi) During this time observe the colour of lime water in both the sets.

After two days, it can be observed that lime water of the injection bottle or beaker placed in the bottle containing the germinating seeds turns into milky white precipitate.

This indicates that CO_2 is evolved in respiration.

Heat evolved during respiration :

- i) Take a handful of moong or bajra seeds.
- ii) Soak the seeds in water a day before your perform experiment.
- iii) Keep these soaked seeds in a cloth pouch and tie with a string tightly.
- iv) Next day collect the sprouts / germinated seeds from the pouch and put them in a thermos flask and take dry seeds in another thermos flask.
- v) Remove the lid and prepare a cork through which you can bore a hole to insert thermometers into two flasks in such a way that the bulb of the thermometer should dip into the germinating and dry seeds.
- vi) Close the thermos flasks with tight fitting rubber corks.
- vii) Record the initial temperature in both the flasks and record it for every two hours for at least 24 hours.



Constant increase in the temperature is observed in thermometer placed in the germinating seeds.

Photosynthesis Vs Respiration :

PHOTOSYNTHESIS	RESPIRATION
i) It occurs only in green plants and some photosynthetic bacteria.	i) It occurs in all organisms.
ii) Takes place in the presence of sunlight.	ii) Takes place throughout the day (in the presence or absence of light).
iii) A plant can survive without performing photosynthesis for few days.	iii) No organism can survive without respiration even for few minutes.

① Micro organisms are useful in the preservation of food.

PHOTOSYNTHESIS	RESPIRATION
iv) In plants only few cells perform this process.	iv) All living cells of an organism perform this process.
v) It occurs in the chloroplast and is dependent on light.	v) It takes place in the cytoplasm and mitochondria and is independent of light.
vi) In this process light energy is fixed.	vi) In this process energy is released.
vii) Raw materials are carbon dioxide and water.	vii) Use carbohydrates or organic substances and oxygen as raw materials.
viii) Oxygen is liberated and carbon dioxide is utilised.	viii) Oxygen is utilised and carbon dioxide is released.
ix) $6\text{CO}_2 + 12\text{H}_2\text{O} \xrightarrow[\text{Chlorophyll}]{\text{Light}} \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{H}_2\text{O} + 6\text{O}_2 \uparrow$	ix) $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{Energy}$

- During day time, the rate of photosynthesis is usually higher than that of respiration while at night it is just reverse in most plants.
- Temperature, humidity, light intensity etc. seem to affect the ratio of photosynthesis and respiration in plants.

② commercially, micro organisms are used to produce many products like bread. In the preparation of large scale of alcohol.

PRIORITY - 1

2 Marks Questions :

- Write the uses of anaerobic respiration of micro-organisms in our daily life. (AS₇)(Q.No. 5, P.No.40)
- What happens if there is no diaphragm in the body ? (AS₂)(Q.No.15, P.No.42)
- What is the fate of food materials if there is no fermentation ? (AS₂)(Q.No. 3, P.No. 49)
- What happens if the respiratory substrates are unavailable in the organisms ? (AS₁)(Q.No.8, P.No. 46)
- Plants photosynthesise during daytime and respire during the night. Do you agree with this statement? Why ? Why not ? (AS₇)(Q.No. 6, P.No.41)
- What happens if food enters wind pipe instead food pipe? (AS₂) Food sometimes enters the wind pipe and causes choking. How does it happen ? (Or) (AS₁) Sometimes we get severe cough while eating. Why ? (Or) While eating food, if someone remembers us, we get cough. Is it right ? Give reasons. (AS₁)(Q.No. 3, P.No.40)
- Write the observations of combustion of glucose activity. (AS₃) (Q.No. 18, P.No. 43)
- After undergoing strenuous exercise, we feel pain in muscles. What suggestions will you give to overcome this problem. (AS₇) (TB. P. No. 50)
- Write the apparatus and observations of the experiment to prove that CO₂ is a by -product of energy release. (AS₃) (TB, P.No. 50)
- How heat liberated in the body, affects the life processes ? (AS₂)(Q.No. 2, P.No. 49)

deep breathes, breathing exercises, around 15 minutes

2) to increase the lung capacity I can suggest to practice pranayama yoga. 3) Take balanced diet 4) Avoid smoking

X CLASS REVISION PROGRAMME

BIOLOGY

4 Marks Questions :

1. State two similarities and differences between aerobic and anaerobic respiration. (AS₁)(Q.No. 2, P.No.40)
2. After learning the chapter, Respiration what habits you would like to change for proper functioning of lungs. (AS₇)(TB, P. No. 50)
3. How does gaseous exchange takes place at blood level ? (AS₁) (Or) Explain the mechanism of exchange of gases at blood level. (AS₁)(Q.No. 11, P.No.41)
4. Write the adaptations seen in plants living in water logged conditions. (AS₁)(Q.No.11, P.No. 47)
5. How are alveoli designed to maximize the exchange of gases ? (AS₁)(Q.No. 8, P.No.41)
6. Where will the release of energy from glucose in respiration takes place? Mala writes lung while Jiya writes muscles. Who is correct and why ? (AS₁)(Q.No. 9, P.No.41)
7. After a vigorous exercise or work we feel pain in muscles. What is the relationship between pain and respiration ? (Or) Why do we get muscle pains when we play or run for a long time After play or run we feel muscle pain. Justify the statement. (AS₁)(Q.No. 13, P. No.42)
8. Why does the rate of breathing increases while walking uphill at a normal pace in the mountains ? Give two reasons. (AS₁)(Q.No. 4, P.No.40)
9. Raju said stem also respire along with leaves in plants. Can you support his statement Give your reasons. (AS₆) (Or) "It is not only leaves but stem also respire". Justify the statement. (AS₁)(Q.No. 14, P.No.42)
10. Why does a deep sea diver and mountaineer carry oxygen cylinder on his/her back ? (AS₁)(Q.No. 7, P.No.41)

6 Marks Questions :

1. What procedure do you follow to understand anaerobic respiration in your school laboratory? (AS₃)(Q.No. 17, P.No. 43)
2. Explain an experiment to test the production of heat and CO₂ under anaerobic respiration (Or) How do you prove that heat and CO₂ are produced in anaerobic respiration ? (Or) What procedure will you follow to understand anaerobic respiration in your school laboratory ? (AS₃)(Q.No. 17, P.No.43)
3. How can you prove that heat is liberated during respiration? (AS₃)(Q.No. 5, P.No. 51)
4. Explain the mechanism of gaseous exchange at bronchiole level. (AS₁)(Q.No. 12, P.No.42)
5. Draw a block diagram showing events in respiration. Write what you understand about cellular respiration. (AS₃)(Q.No. 22, P.No.44)

PRIORITY - 2

2 Marks Questions :

1. What happens if aerial roots are absent in Mangrove plants ? (AS₂)(Q.No. 11, P.No. 47)

2. Write the uses of anaerobic respiration of microorganisms in our daily life. (AS₆)(Q.No.1, P.No. 54)
3. If you have a chance to meet a pulmonologist what questions are you going to clarify about pulmonary respiration ? (AS₂)(Q.No. 16, P.No.42)
4. What will happen if the respiratory tract is not kept moist ? (AS₂)(Q.No. 5, P.No. 49)
5. "Respiration helps for the survival of organisms". How ? (AS₇)(Q.No. 1, P.No. 55)

4 Marks Questions :

1. Explain the process of conduction of gases in plants. (AS₁) (Q.No.2, P.No. 46)
2. List out the events or steps in respiration. (AS₁)(Q.No. 5, P.No. 46)
3. Write the differences between the respiration in plants and animals. (AS₁)(Q.No. 18, P.No. 48)
4. Distinguish between : (AS₁)(Q.No. 1, P.No.39)
 - a) inspiration and expiration
 - b) respiration and combustion
 - c) photosynthesis and respiration.
5. How do you appreciate the mechanism of respiration in our body ?(AS₆)(Q.No. 23, P.No.45)

6 Marks Questions :

1. How can you prove that CO₂ is evolved during respiration ? (AS₃)(Q.No. 2, P.No. 49)
2. What are the experiments carried out by Lavoisier to understand the property of gases ? (AS₁) (Q.No. 4, P.No. 49)
3. What is the pathway taken by air in the respiratory system? Illustrate with a labelled diagram. (AS₅)(Q.No. 21, P.No.44)
4. Describe the structure of mitochondria with the help of a diagram. (Or) (AS₁) Draw a neat labelled diagram of power house of a cell and explain its structure. (AS₅)(Q.No.3, P.No.54)
5. a) Draw a block diagram to show the breakdown of the glucose in the presence and absence of O₂. (AS₅)(Q.No. 1, P.No.53)
- b) Write the differences between photosynthesis and respiration.

PRIORITY - 3

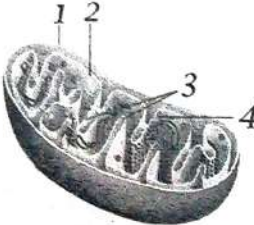
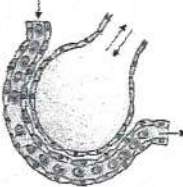
Complete all the remaining question and answers of the lesson.

IMPORTANT DIAGRAMS

- i) Structure of mitochondria
- ii) Respiratory system of man
- iii) Alveolus with blood capillary

PRACTICE SHEET

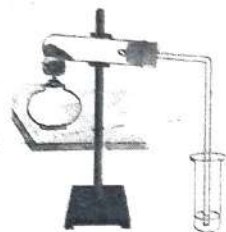
Choose the correct answer :

1. Which of the following organisms can live without oxygen ? [C]
 A) Leech B) Cockroach C) Yeast D) Paramecium
2. 1, 2, 3 and 4 in the following picture indicates : [A]

 A) 1 - Outer membrane, 2 - Inner membrane, 3 - Cristae, 4 - Matrix
 B) 1 - Inner membrane, 2 - Outer membrane, 3 - Cristae, 4 - Matrix
 C) 1 - Outer membrane, 2 - Inner membrane, 3 - Matrix, 4 - Cristae
 D) 1 - Outer membrane, 2 - Matrix, 3 - Cristae, 4 - Inner membrane
3. The air that remains in lungs after a complete exhalation is : [C]
 A) 500 ml. B) 5,800 ml. C) 1200 ml. D) 120 ml.
4. Which of the following gas is called fixed air ? [B]
 A) Nitrogen B) Carbon dioxide C) Oxygen D) Helium
5. In which of the given process lactic acid is released as an end product ? [C]
 A) Fermentation B) Pasteurisation C) Anaerobic respiration D) Aerobic respiration
6. The below given diagram is depicting the : [A]

 A) gaseous exchange between alveoli and blood capillaries
 B) gaseous exchange in tissues
 C) exchange of gases between blood and tissues
 D) exchange of gases between tissues and bronchioles
7. The flattened muscular structure present at the floor of the chest cavity that helps the lungs in moving air into and out of them is : [C]
 A) Skin B) Ribs C) Diaphragm D) Heart
8. Which of the following is an incorrect statement about lungs ? [D]
 A) They are spongy in nature. B) Lungs have tiny chambers called Alveoli
 C) The left lung is smaller than right lung. D) Lungs are protected by single membrane
9. Ethanol can be separated from the mixture of Yeast and glucose by the process of : [B]
 A) decantation B) fractional distillation
 C) fermentation D) destructive distillation
10. Name the 3-carbon compound formed during the process of respiration. [C]
 A) oxalate B) citrate C) pyruvate D) RUBP
11. The Indian ayurvedic physician Patanjali developed a scientific breathing process called : [A]
 A) Yogabhyasa B) Yogasana C) Pranayama D) Dharana

12. Which of the following is the most commonly used sugar for deriving energy in plants, animals and microorganisms ? [B]

A) Starch B) Glucose C) Sucrose D) Fructose

13. The aim of the following experiment is to know that : [B]



- A) Heat is a by-product along with energy
B) CO_2 is a by-product of energy release
C) H_2O is a by-product of energy release
D) Oxygen is a by-product of energy release

14. The respiratory organ in insects is : [B]

A) Skin B) Trachea C) Lungs D) Gills

15. Choose the correct statement from the following : [B]

- i) Respiration is an uncontrolled process.
ii) Several intermediate compounds are formed during respiration.
iii) Respiration occurs in the absence of water.

A) i B) ii C) iii D) i, ii and iii

16. **Statement I :** Deep sea divers carry oxygen cylinder on their back to get oxygen.

Statement II : Humans have lungs, therefore they can't take O_2 from water. [A]

- A) Statement I is True, Statement II is True B) Statement I is False, Statement II is False
C) Statement I is True, Statement II is False D) Statement I is False, Statement II is True

17. Due to the absence of _____ in trachea, it may get collapsed or damaged. [C]

A) Muscular tissue B) Ligaments
C) Cartilaginous rings D) Vocal cords

18. The temporary shortage of oxygen in the body tissues resulting from strenuous physical activity is called : [C]

A) less oxygen B) low oxygen C) oxygen debt D) more oxygen

19. The process of conversion of glucose to pyruvate is called : [C]

A) Phosphorylation B) Gluconeogenesis C) Glycolysis D) Respiration

20. A biochemical compound that readily combines with oxygen and distributes it throughout the human body is : [C]

A) Blood B) RBC C) Haemoglobin D) WBC

3. CIRCULATION

CHAPTER AT A GLANCE

Circulatory system :

The circulatory system, also known as the cardiovascular system, consists of the organs and fluids that transport materials like oxygen and nutrients throughout the body.

Blood vessels circulation :

- It was not until 16th century that we really came to know how our blood vessels functioned. In 1574, an Italian doctor, **Girolamo Fabrici**, was studying the veins in the leg.
- He noticed that they had small valves in them that permitted blood to flow towards heart.
- These were one-way valves that checked backflow of blood. Movement of leg muscle helps in movement of blood through veins.
- **William Harvey** an English man who became a doctor and went to Italy for further education and studied under Fabrici.
- Harvey dissected the hearts of dead people and studied the valves between each atrium and its ventricle.
- Harvey noticed that they were one way valves which allow blood to flow from atrium to the ventricle, but not from ventricle into the atrium.
- Harvey also identified that the blood flows through the heart twice in human beings. Hence, it is called double circulation.
- **Marcello Malpighi** (1628-1694), used the microscope to see tiny blood vessels that were invisible with naked eye. Later, Malpighi studied the wings of bats. He could see blood vessels in their thin membranes under the microscope.
- He could see that the smallest arteries and veins were connected by very fine blood vessels. He called these blood vessels "capillaries" from the Latin word for "hair", because they were as thin as the finest of hairs.

Internal structure of heart :

The heart is a hollow pear shaped organ. It is internally divided into four chambers. Two upper chambers are called Auricles and two lower chambers are called Ventricles.

Internally these four chambers are separated by Walls or Septae.

Name of the septum	Location	Function
Inter-auricular septum	In between right and left auricles / atria	Prevents the mixing of oxygenated blood (left auricle) from deoxygenated blood (right auricle).
Inter-ventricular septum	In between right and left ventricles.	Prevents the mixing of oxygenated blood (left ventricle) from deoxygenated blood (right ventricle).
Auriculo-ventricular septum	In between auricles and ventricles.	Separates the auricles from ventricles.

Auriculo Ventricular Apertures : The right auricle opens into the right ventricle through the right auriculo ventricular aperture while the left auricle opens into the left ventricle through the left auriculo - ventricular aperture.

Valves : These allow the blood to flow only in one direction and prevents the backward flow of blood. These are held in position by tough connective tissue strands called 'chordae tendinae'.

Name of the valve	Location	Allows blood to flow from
Tricuspid valve (has three flaps or cusps)	Right auriculo - ventricular aperture	right auricle to right ventricle
Bicuspid (Mitral) Valve (has two flaps or cusps)	Left auriculo - ventricular aperture	left auricle to left ventricle
Pulmonary valves (3 half moon shaped or semilunar valves)	At the origin of pulmonary aorta in the right ventricle	right ventricle into pulmonary aorta
Aortic valves (3 Semilunar valves)	At the origin of aorta in the left ventricle	left ventricle into the aorta

Blood vessels : There are two types of blood vessels called arteries and veins.

Arteries	Veins
i) Arteries carry oxygenated blood. (except pulmonary artery)	i) Veins carry deoxygenated blood (except pulmonary vein).
ii) The walls of the arteries are thick..	ii) The walls of the veins are thin.
iii) They carry blood from the heart to various body parts (away from heart).	iii) They carry blood from various body parts to the heart (towards heart).
iv) Valves are absent.	iv) Valves are present.

Blood capillaries : Blood capillaries are the microscopic vessels made of single layer of cells. They allow diffusion of various substances which form tissue fluid. They establish continuity between arteries and veins.

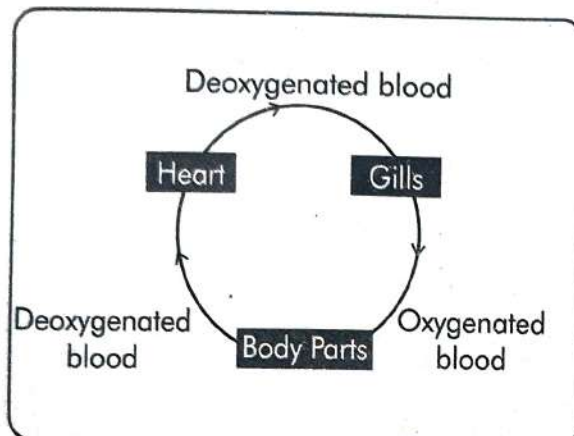
Cardiac cycle : One contraction and one relaxation of atria and ventricles is called one cardiac cycle. It is completed in 0.8 sec.

Steps :

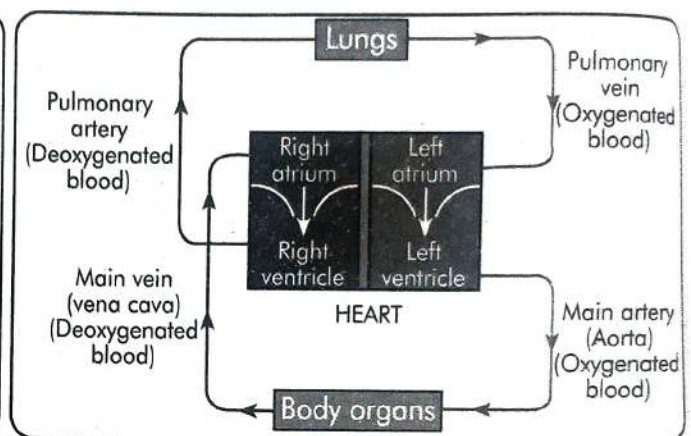
Stage	Event	Heart sounds	Duration
Auricular systole	Auricles contract		0.11 to 0.14 sec
Ventricular systole	Ventricles contract	LUB (closure of AV valves)	0.27 to 0.35 sec
Auriculo-ventricular diastole	Auricles and ventricles relax	DUB (closure of semilunar valves)	0.4 sec

Single and double circulation :

Single Circulation	Double circulation
i) In this, blood passes through heart once in a circulation.	i) In this, blood passes through heart twice in a circulation.
ii) In this, heart pumps blood to the gills for oxygenation.	ii) In this, heart pumps blood to the lungs for oxygenation.
iii) This type of circulation is found in fishes,	iii) This type of circulation is found in amphibians reptiles, birds and mammals.



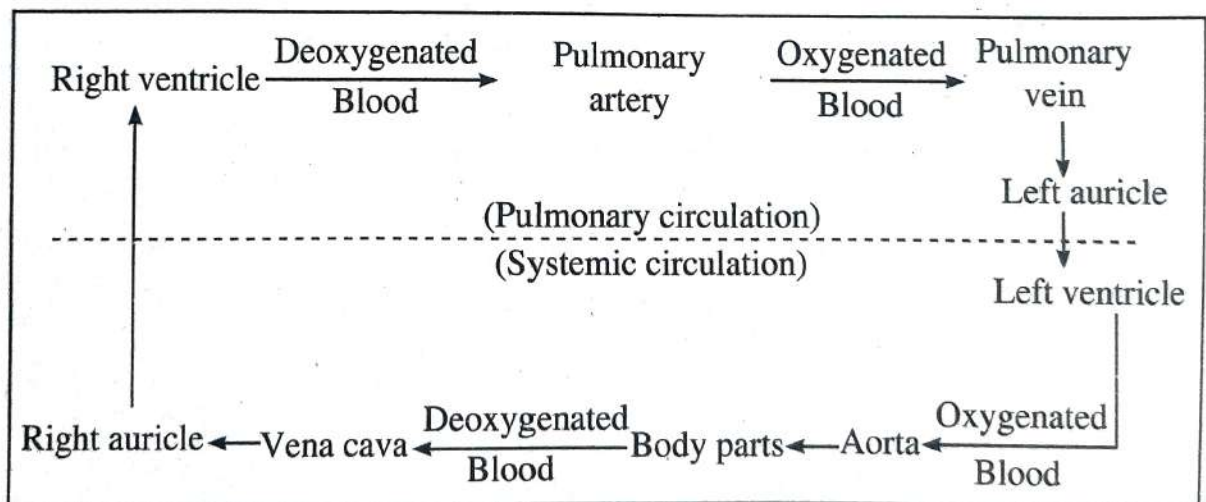
Single Circulation



Double Circulation

Single circulation flowchart : Body parts → Heart → Gills → Body parts

Double circulation flowchart :



Lymphatic system :

- Lymph is a circulatory fluid which also helps in transportation. This is also called tissue fluid. It is formed with some amount of plasma, proteins, inorganic salts. While blood cells pass through the pores present in the walls of capillaries into intercellular spaces, the tissues to from lymph.

- ii) The lymph is similar to plasma except that it is colourless and contains very less proteins.
- iii) From intercellular spaces, lymph goes into lymphatic capillaries. Lymphatic capillaries join to form large lymph vessels which finally open into large veins. Lymph flows only in one direction, that is from tissues to heart through veins.

Path of lymph :

Intercellular spaces → Lymph capillaries → Lymph vessels → Lymph ducts → Large vein → Heart

Functions of Lymph :

- i) Lymph carries digested and absorbed fats from small intestine to different tissues.
- ii) Lymph drains excess fluid from extracellular space back into the blood.
- iii) Lymph protects tissues / cells from infections.
- iv) It also removes waste products from the body cells and drain into blood.

Evolution of the transport (Circulatory) system :

Name of the phylum / organisms	Type of transport / circulatory system
Unicellular organisms (Eg: <i>Amoeba</i>)	Brownian movements
Parazoans (Eg : Sponges)	Use sea water for transportation.
Cnidarians (Eg : <i>Hydra</i> jellyfish)	Gastrovascular cavity
Platyhelminthes (Eg : <i>Fasciola hepatica</i>)	Highly branched digestive system
Nemathelminthes (Eg : roundworms)	Pseudocoelom
Annelids (Eg : Earthworm)	Pulsatile vessel
Arthropods (Eg : Insects)	Pulsatile organs (the heart)

Blood pressure :

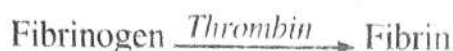
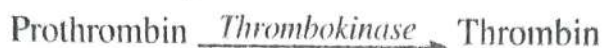
- i) The pressure exerted by blood on the walls of artery is called blood pressure.
- ii) B.P. is always measured in the upper arm artery.
- iii) B.P. varies throughout the body, so a standard place must be used so that a person's blood pressure can be compared over a period of time.
- iv) Doctors measure the blood pressure (B.P.) with a device called sphygmomanometer.

Coagulation of blood :

When you cut yourself, the blood flows out of the wound for only a short time. Then the cut is filled with a reddish solid material. This solid is called a blood clot. If blood do not clot, any one with even a slight wound bleeds profusely.

- i) When the blood flows out, the platelets release an enzyme called thrombokinase.
- ii) Thrombokinase acts on another substance present in the blood called pro-thrombin converting it into thrombin.
- iii) Thrombin acts on another substance called fibrin, that is present in dissolved state converting it into insoluble fibrin.
- iv) The blood cells entangle in the fibrin fibers forming the clot.

- v) The fibrin fibers are attached to the edges of the wound and pull them together. The straw yellowish coloured fluid portion after formation of the clot is serum.



How materials transport within the plant ?

Van Helmont's experiments on plants proved that water containing minerals from the soil are absorbed through roots. The water absorbed by roots and food prepared by leaves are supplied to remaining parts of the plant by vascular bundles having xylem and phloem.

In the root, the xylem tissue is situated towards the exterior while in the stem it is arranged in bundles towards the center.

Experiment to demonstrate how water is absorbed :

Aim : To prove that the water is transported through the xylem.

Apparatus : Glass beaker, red ink, a small balsam plant or Amaranthus plant.

Procedure :

- Take a glass with little water in it. Add a few drops of red ink into the water of glass.
- Uproot a small balsam plant or Amaranthus plant carefully.
- Wash and place it in the beaker of coloured water.

Observation : After a few hours, coloured lines with ascent of the water can be observed in the stem and veins of leaves.

Inference / Conclusion : The observations made above shows that water travels up the stem through narrow tubes in it (xylem vessels).

Absorption of water through root hairs :

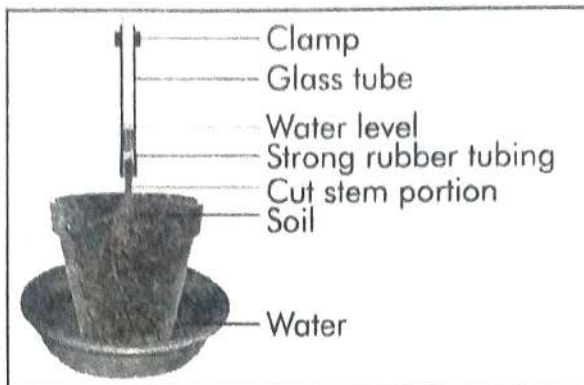
- Roots have small hair like structures called root hairs, originating from cells of single layer of epidermis.
- The cytoplasm of these cells is called cell sap and has dissolved salts.
- Cell sap is also a salt solution and it is much concentrated than the salt solution present in the soil.
- The cell membrane of root hair is a semipermeable membrane and separates the cell sap (concentrated salt solution) from the salt solution in the soil.
- As a result, water flows (along with minerals) from the soil into the cell sap of root hairs.
- Thus, roots absorb water and salts from the soil. The absorbed water and salts will move to the next cell once again by osmosis.

Root pressure :

- The pressure that is developed in the xylem vessels of roots that forces the water upwards is called root pressure.
- Root pressure helps in upward movement of water in the xylem vessels of small plants.

Experiment :

- i) Take a regularly watered potted plant and cut the stem portion 1cm above the ground level.
- ii) Then connect a glass tube by means of strong rubber tubing as shown in the figure.
- iii) The size of glass tube should be equal to the size of the stem. Take care while joining tube and stem being bound tightly, so that water cannot escape from the tube.
- iv) Now, pour some water in the glass tube until water level can be seen above the rubber tube.
- v) Mark the level of water (M_1) in the tube. Keep your arrangement aside for 2 to 3 hours.
- vi) Then observe and mark the water level (M_2) in the tube. The difference between M_2 and M_1 indicates the level of water raised in the stem.



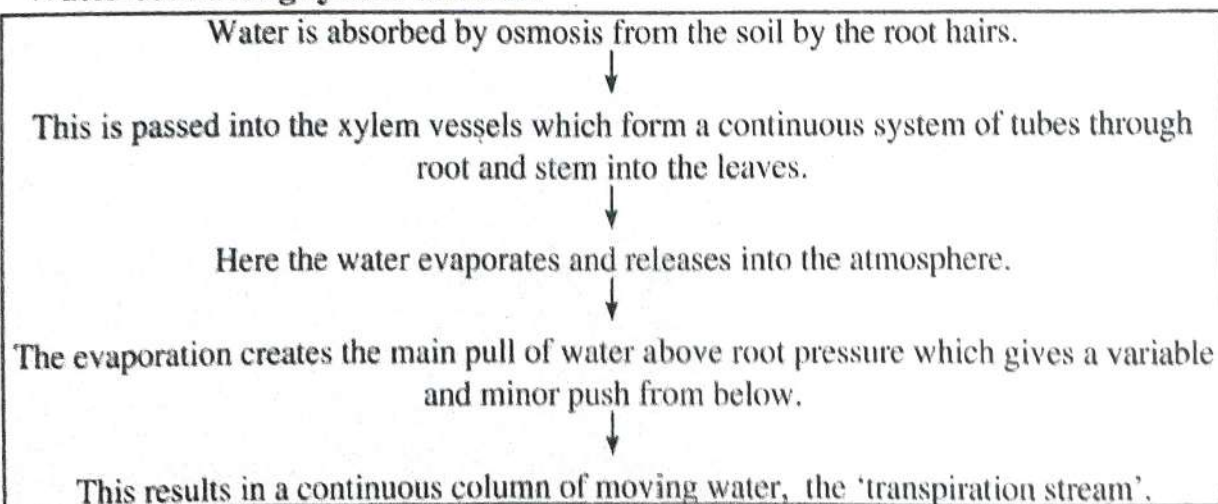
Because of the root pressure, the water level increases in the tube.

The mechanism by which the water travels through the plant :

We have seen that there is a push from below due to root pressure on the columns of water in the xylem vessels, but this is seldom high and in some seasons it is nil.

The evaporation of excess water through leaves is called transpiration. Water evaporates through stomata of leaves and lenticels of stem.

- i) When the leaves transpire, there is a pulling effect on the continuous columns of water in the xylem vessels.
- ii) The top ends of these vessels are surrounded by the leaf's mesophyll cells which contain cell sap, so the water is continuous from the xylem vessels to the walls of the mesophyll cells from which it evaporates into the air spaces causing the pull.
- iii) The water column does not break because of its continuous molecular attraction.
- iv) This is a property of water you demonstrate every time you drink through a straw.

Water-conducting system of a tree.

The amount of water passing through a plant is often considerable.

For example, an oak tree can transpire as much as 900 liters of water per day.

It follows therefore that areas of forest significantly affect the degree of saturation of the air above them, so that when air currents bring air which is already nearly saturated to a forest area, it becomes fully saturated and comes down as rain. This is why forest areas often have a higher rainfall than areas nearby.

Transport of mineral salts :

- i) Mineral salts are necessary for plant nutrition (micro and macro nutrients) and they are obtained from the soil through the root hairs.
- ii) The salts are in the form of electrically charged ions. Sodium chloride (NaCl) is in the form of Na^+ and Cl^- , and Magnesium Sulphate (MgSO_4) occurs as Mg^{2+} and SO_4^{2-} .
- iii) But, they are not absorbed into the root hairs by the simple process of diffusion, but it involves the use of energy by the cytoplasm.
- iv) Once ions are absorbed, the ions travel along with water in the xylem vessels and pass to the growing points of the plants where they are used for growth purpose.
- v) They may also pass laterally from xylem to phloem. Thus, mineral salts are one of the natural factors in plant growth phenomena.

Transport of manufactured food :

- i) Food such as sugar is synthesised in the green parts of plants, mainly the leaves, but this food has to be transported to all the living cells, especially to actively growing cells and the cells which store food.
- ii) Further experiments to illustrate the conduction of sugars by the phloem have been done by removing a ring of bark from a shoot to expose the wood.
- iii) Remove all tissues from the center outwards, including the phloem.
- iv) After a few days, when the tissues above and below the ring were analyzed it was shown that food had accumulated above the ring, but was not present below it.
- v) If it is left for some more time, the stem increases in thickness immediately above the ring, but no growth occurred below it.
- vi) So, any damage to the phloem all around the stem will prevent the food from passing down to the roots and the tree will eventually die.

PRIORITY - 1

2 Marks Questions :

1. What would happen if xylem is removed from the plant ? (AS₂)(Q.No. 1, P.No. 78)
2. What measures do you suggest the elders to maintain the blood pressure ? (AS₇)(Q.No.25, P.No. 71)
3. What will happen if tricuspid and bicuspid valves are absent in the heart? (AS₂)(TB.P.No. 57)
4. What materials do you require to prepare a model stethoscope ? (AS₃)(Q. No. 16, P.No.69)
5. If the valves in veins of the legs fail to stop the flow of blood, what will be the consequences? (Or) What happens if valves are absent in the veins ? (AS₂)(Q.No. 14, P.No.69)

6. What happens if there is no special tissue or organ for transport of substances in plants and animals ? (AS₂)(Q.No. 2, P.No. 79)
7. What is your inference about experiments with aphids ? (Or) How did scientists prove that the food is transported through phloem ? (AS₃)(Q.No. 18, P.No. 70)
8. What would happen if transpiration doesn't take place in plants? (AS₂)(Q.No. 15, P.No.69)
9. What would be the consequences of a deficiency of haemoglobin in our bodies ? (AS₂)(Q.No. 16, P.No.75)
10. What precautions will you take to avoid heart related diseases? (AS₇) (Q.No. 25, P.No. 71)

4 Marks Questions :

1. What is the relationship between blood and plasma ? (Or) How are plasma and blood inter-related ? (AS₁)(Q.No. 2, P.No.66)
2. Explain the three main types of blood vessels in the body ? (AS₁)(Q.No. 4, P.No.66)
3. Write the pathway of blood flow in human heart. (AS₁)(Q.No. 20, P.No. 76)
4. What is root pressure? How it is useful to the plant ? (AS₁) (Q.No. 11, P.No.68)
5. Phloem is a food source for some animals. How do you justify this statement ? (AS₁)(Q.No. 12, P.No.68)
6. What do you want to compare with transportation in blood vessels in man ? (Or) What can circulatory system in man be compared with ? (AS₇) (Q.No.22,P.No. 71)
7. After reading the lesson transportation, what precautions would you suggest to your elders about oedema. (AS₇)(Or) List out the precautions that are needed to prevent oedema. (AS₇)(Q.No. 25, P.No.71)
8. What is transport system ? How is this helpful to an organism ? (AS₁)(Q.No. 1, P.No.66)
9. What is hypertension ? How is it caused ? (AS₁)(Q.No.29, P.No. 77)
10. Write the differences between xylem and phloem. (AS₁) (Q.No. 9(c), P.No. 67)

6 Marks Questions :

1. Write differences between :
a) systole - diastole b) veins - arteries (AS₁)(Q.No. 9(a & b), P.No.67)
2. Explain the way how plants get water by osmosis through root hairs. (Or) Explain the way how the plants absorb water from soil. (AS₃)(Q.No. 10, P.No.68)
3. Write an experiment to show the movement of water molecules from roots to shoot. (AS₃) (Q.No.2, P.No.79)
4. What is coagulation of blood ? Write its mechanism. (Or) Explain the process of blood coagulation. (AS₁)(Q.No. 12, P.No.74)
5. What is cardiac cycle ? Explain the stages of it. (AS₁)(Q.No. 14, P.No.74)

PRIORITY - 2

2 Marks Questions :

1. What happens if transport system fails to function ? (AS₂)(Q.No. 1, P.No. 66)
2. What is the relation between transportation and transpiration ? (Or) (AS₁)
"Transportation and transpiration are related in plants". Justify the statement.
(AS₂)(Q.No.24, P.No. 76)
3. John prepared stethoscope with paper cup and plastic tube. Write down the procedure of preparation. (AS₃)(Q.No. 16, P.No.69)
4. What can circulatory system in man be compared with ? (AS₇)(Q.No.22, P.No. 71)
5. What happens if the walls of the arteries are same as veins ? (AS₂)(Q.No. 11, P.No.73)

4 Marks Questions :

1. How do you feel about transportation of water in huge trees ? (AS₆)(Q.No. 2, P.No.83)
2. What is blood pressure (B.P.)? /How is it measured ? Mention the normal B.P. of a healthy human. (AS₁)(Q.No.30, P.No. 77)
3. Write the differences between the transportation in humans and plants. (AS₁)(Q.No.27, P.No. 77)
4. "Lumen size is bigger in the veins than arteries". Give reason. (AS₁) (Q.No.21, P.No. 76)
5. a) When do we hear the sounds 'lub' and 'dub' of heart ? (AS₁) (Q.No. 22, P.No. 76)
b) Which type of blood vessels carry blood away from the heart and towards the heart? (Q.No. 3, P.No. 66)

6 Marks Questions :

1. How do you observe internal and external features of a mammalian heart ? (AS₃)(Q.No.3,P.No. 80)
2. Enlist the differences between : (AS₁)
a) right and left auricles (Q.No. 2, P.No.72)
b) right and left ventricles (Q.No. 6, P.No.72)
3. Write a note on human lymphatic system. (AS₁)(Q.No. 10, P.No.73)
4. Draw a block diagram to explain single and double circulation. Write differences between them. (AS₅)(Q.No. 20, P.No.70)
5. Prepare a block diagram showing from water absorption by roots till transpiration by leaf. (AS₅)(Q.No. 21, P.No.71)

PRIORITY - 3

Complete all the remaining question and answers of the lesson.

IMPORTANT DIAGRAMS

- i) Internal structure of heart
- ii) T.S. of artery and T.S. of vein
- iii) Single and double circulation

PRACTICE SHEET

Choose the correct answer :

1. If the pulse rate of a healthy individual is 72 times per minute, then his heart beat rate per minute will be : []
 A) 70 B) 75 C) 71 D) 72
2. Aphids excrete sugar out of the body as a sticky syrup called : []
 A) Wax B) Honey dew C) Latex D) Resins
3. Xylem and Phloem are present in : []
 A) Cambium B) Pericycle C) Endodermis D) Vascular bundle
4. Match the following : []
 a) Atrial contraction i) 0.11 - 0.14 seconds
 b) Ventricular contraction ii) 0.27 - 0.35 seconds
 c) Auriculo-ventricular relaxation iii) 0.8 seconds
 A) a - iii, b - i, c - ii B) a - iii, b - ii, c - i C) a - i, b - ii, c - iii D) a - i, b - iii, c - ii
5. Foresters encourage predators as foxes, badgers, hawks as predatory to keep down the population of : []
 i) Voles ii) Rabbits iii) Grey squirrels
 A) i and ii B) ii and iii C) i and iii D) i, ii and iii
6. The movements shown by unicellular organisms like *Amoeba* are called : []
 A) Peristaltic movements B) Brownian movements
 C) Wave like movements D) None of the above
7. Match the following : []
 a) Cnidarians i) Sea water
 b) Platyhelminthes ii) Pseudocoelom
 c) Nematelminthes iii) Pulsatile vessel
 d) Sponges iv) Highly branched digestive system
 e) Annelids v) Gastrovascular cavity
 A) a - v, b - iv, c - ii, d - i, e - iii B) a - v, b - ii, c - iv, d - i, e - iii
 C) a - iv, b - v, c - ii, d - i, e - iii D) a - iv, b - ii, c - iii, d - i, e - v
8. The human circulatory system can move one ml. of blood from heart to foot in about: []
 A) 20 Sec B) 30 Sec C) 60 Sec D) 70 Sec

9. The type of circulation which supplies nutrients to the tissues directly is called :
A) pulmonary circulation B) open type of circulation
C) closed type of circulation D) branchial circulation
10. The yellowish straw coloured fluid portion after formation of the blood clot is :
A) Serum B) Pus C) Plasma D) Lymph
11. Transport of mineral salts in plants is carried out by means of :
A) diffusion B) passive transport C) active transport D) transpiration
12. How much percentage of the world population is suffering with thalassemia minor?
A) 2.5 % B) 3.5% C) 4.5% D) 6.5%
13. In which type of genetic disorder the blood may not coagulate ?
A) Septicemia B) Thalassemia C) Haemophilia D) Pernicious anaemia
14. In humans blood passes from one part to the other through tube like structures called:
A) cells B) tissues C) organs D) vessels
15. Choose the incorrect statement from the following :
A) Lymph is a circulatory fluid which also helps in transportation
B) It drains excess fluid from extracellular spaces back into the blood
C) Lymph protects tissues / cells from infections
D) Erythrocytes are present in the lymph
16. An oak tree can respire as much as _____ litres of water per day.
A) 700 B) 900 C) 800 D) 600
17. Single circulation is seen in animals such as :
A) Amphibians B) Reptiles C) Birds D) Fishes
18. The rhythmic throbbing of arteries which we feel at the wrist region just below the thumb is called :
A) Heart beat B) Blood pressure C) Pulse D) Circulation
19. Which system transports the tissue fluid into the main blood stream ?
A) Venal system B) Arterial system
C) Lymphatic system D) Circulatory system
20. In small plants like herbs upward movement of water to the leaves is helped by :
A) Osmosis B) Diffusion C) Root pressure D) Transpiration

4. EXCRETION

CHAPTER AT A GLANCE

Excretion is the term coined for all the biological process involved in separation and removal of wastes or non useful products from the body. (In Latin 'ex' means out, 'crenere' means shift.)

- i) Living beings need energy for their survival and to perform activities either building up of body material (anabolism) or its breakdown (catabolism), collectively called metabolic activities.
- ii) Organisms use different substances for metabolic activities. Different products are generated as a result of these metabolic activities.
- iii) Different kinds of materials are produced out of various metabolic activities. Some of these may be harmful for the organism and are removed from their body or packed and stored in some other forms. These are all the wastes produced in the body of an organism. Other metabolic activities generate nitrogenous wastes which are to be removed along with salts, excess water and several other materials.

Excretion in Human Beings :

A number of reactions take place during various metabolic activities, many useful substances and energy is produced. At the same time many other things happen such as toxic wastes may be produced, water content may increase, ionic balance in the body may be disturbed. The waste products include carbon dioxide, water, nitrogenous compounds like ammonia, urea, uric acid, bile pigments, excess salts etc. The most poisonous of all waste products of metabolism is Ammonia.

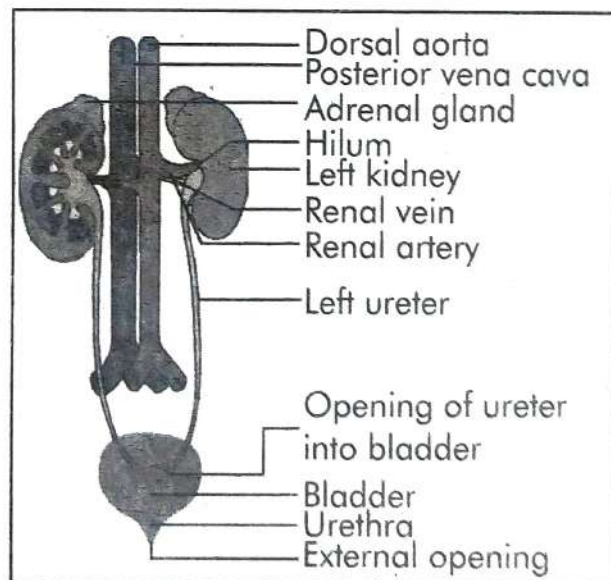
Excretory System in Human being :

In human beings excretion mainly occurs through a urinary or excretory system consisting of a pair of kidneys, a pair of ureters, urinary bladder and urethra.

Kidneys :

In Human beings there is a pair of bean shaped, reddish brown structures in the abdominal cavity attached to dorsal body wall one on either side of the back bone. The right kidney is placed slightly lower than the left kidney. The size of the kidney is 10 cm in length, 5-6 cm in breadth, and 4 cm in thickness. Each kidney is convex on the outer side and concave on the inner side.

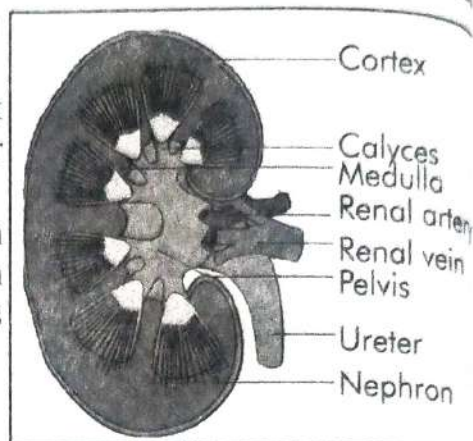
The position of the right kidney is lower than the left kidney due to the presence of liver above. The inner side of each kidney has a fissure or hilus for the entry of a renal artery, exit of a renal vein and an ureter. Renal artery brings oxygenated blood loaded with waste products and renal vein carries deoxygenated blood.



The waste products generated in various organs of the body are filtered and removed by the kidneys.

L.S. of kidney shows two distinct regions. Dark coloured outer zone called the cortex and pale inner zone called medulla.

Each kidney is made up of approximately more than one million (1.3 to 1.8 million) microscopic and thin tubular functional units called nephrons or uriniferous tubules.

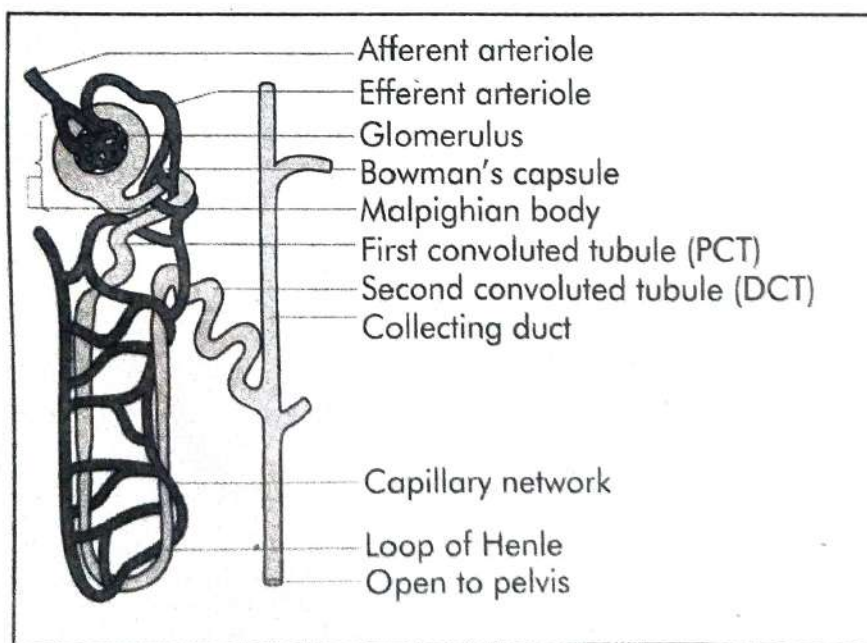


Structure of nephron :

Each nephron has basically two parts. One is malpighian body and other is renal tubule.

Malpighian body: It consists of a blind cup shaped broader end of nephron called Bowman's capsule and bunch of fine blood capillaries called glomerulus. The Bowman's capsule and glomerulus are together called malpighian capsule or renal capsule. Glomerulus develop from afferent arteriole. It gives rise to an efferent arteriole. Because of the narrower outlet (efferent arteriole) pressure exerts in the glomerulus. It functions as a filtration unit. Bowman's capsule which accommodates one glomerulus is lined by a single layer of squamous epithelial cells called podocyte cells. There are fine pores between podocyte cells to allow passage of materials filtered out of glomerulus.

Renal tubule : It has three parts. First or Proximal Convoluted Tubule (PCT), Loop of Henle, which is U shaped, Second or Distal Convoluted Tubule (DCT). Distal convoluted tubules open into a collecting tube. Collecting tube form pyramids and calyces which open into the pelvis. Pelvis leads into the ureter. All the parts of the renal tubule are covered by a network of peritubular (around tube) capillaries formed from efferent arteriole. The peritubular capillaries join to form renal venule, which joins the other venules to form finally the renal vein.



Mechanism of urine formation :

Formation of urine involves 4 stages. They are :

- i) **Glomerular filtration:** The blood from renal artery flows into afferent arteriole of glomerulus. This blood is filtered under high pressure. As a result waste materials along with some water and useful substances are filtered out. They enter into the Bowman's capsule. This is called glomerular filtration. This filtrate is called as 'primary urine'.
- ii) **Tubular reabsorption:** Primary urine which is in glomerulus is almost equal to blood in chemical composition except the presence of blood cells. Peritubular capillaries present around the Henle's loop reabsorb essential substances and excess water present in the primary urine.
- iii) **Tubular secretion:** After reabsorption of essential substances and water urine travels through the loop of Henle. From peritubular capillaries, present around the loop of Henle, waste materials left unfiltered in the blood during glomerulus filtration are secreted into the loop of Henle.
- iv) **Formation of concentrated urine :**

Urine that reaches into the collecting tubule from the loop of Henle is further concentrated in the presence of hormone vasopressin. Deficiency of vasopressin causes excessive dilute urination. This condition is called as 'diabetes insipidus'.

Ureters :

There are a pair of whitish, narrow distensible and muscular tubes of 30 cm length. Each ureter arises from hilus of the kidney. It moves downward and obliquely opens into the urinary bladder. Ureter carries urine from the kidney to the urinary bladder. The movement of urine in the ureter is through peristalsis.

Urinary bladder :

It is a median, pear shaped and distensible sac. It is situated in the pelvic region on the ventral side of the rectum in the abdomen. It stores 300 - 800 ml of urine temporarily which is brought by two ureters.

Urethra :

It is a tube that takes urine from urinary bladder to outside. The opening of urinary bladder into urethra is guarded by a ring of muscles or sphincter. Urethra is 4 cm long in females and open in to vestibule where as in males it is about 20 cm long. Its opening is separate in females but is in common with the reproductive tract in males (urino-genital duct).

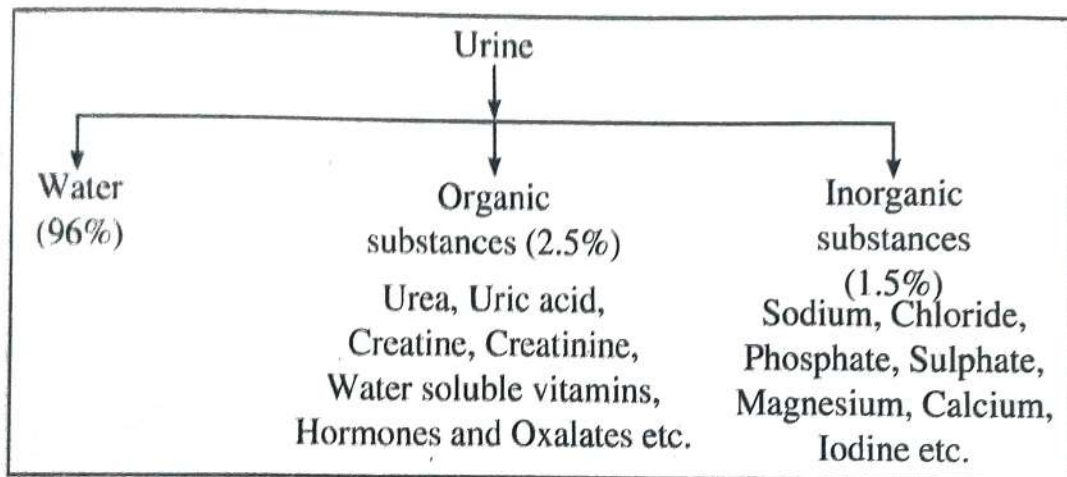
Micturition :

The urine is temporarily stored in the bladder. There are two sets of circular sphincter muscles in the bladder. When the bladder is filling up both these muscles are constricted, so the exit is closed. However as the pressure of the urine increases the walls of the bladder are stretched and this triggers off an automatic reflex action which causes the upper sphincter to relax. But the lower sphincter is under the control of brain. So urine can still be retained until this muscle is relaxed too. Control of urination is not possessed by very young children but is gradually learned. Urge for micturition occurs when urinary bladder is filled with 300 - 400 ml of urine. The stretched bladder stimulates nerve endings to develop the reflex. However, urine can be retained in the urinary bladder till it gets filled up

to the maximum capacity of 700 - 800 ml. At this time the urge becomes painful and leads to voluntary micturition. Total amount of urine excreted per day is about 1.6-1.8 litres. This quantity increases with larger intake of fluids like water, fruit juices and decreases with lesser intake.

Composition of urine :

Urine is a light yellowish transparent fluid produced by urinary system. Urine has ambr color due to presence of urochrome. Composition of normal urine varies considerably depending on several factors.

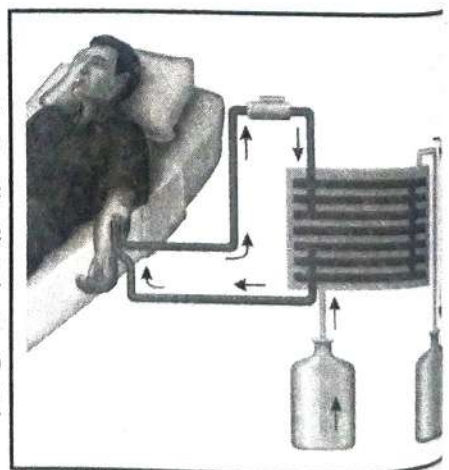


It is acidic (pH=6.0) in the beginning but becomes alkaline on standing due to breakdown of urea to form ammonia.

Complete and irreversible kidney failure is called 'End Stage Renal Disease' (ESRD). When kidneys stop working completely, our body is filled with extra water and waste products. This condition is called uremia. Our hands or feet may swell. You feel tired and weak because your body needs clean blood to function properly.

Dialysis Machine (Artificial kidney) :

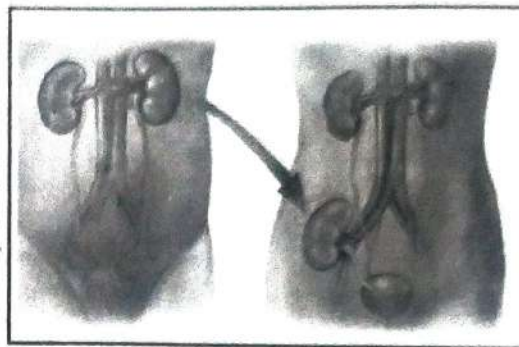
Kidneys are vital organs for survival. Several factors like infections, injury, very high blood pressure, very high blood sugar or restricted blood flow to kidneys leads to accumulation of poisonous wastes in the body and leads to death. Dialysis machine is used to filter the blood of a person when both kidneys are damaged. The process is called 'haemodialysis'. In this process blood is taken out from the main artery, mixed with an anticoagulant, such as heparin, and then pumped into the apparatus called dialyzer. In this apparatus blood flows through channels or tubes.



These tubes are embedded in the dialyzing fluid. The membrane separates the blood flowing inside the tube and dialyzing fluid (dialysis), which has the same composition as that of plasma, except the nitrogenous wastes. Each dialysis session lasts for 3 to 6 hours. This method is being used for thousands of uremic / kidney failure patients all over the world.

Kidney transplantation :

The best long term solution for kidney failure (acute renal failure) is Kidney transplantation. A functioning kidney is used in transplantation from a donor preferably a close relative. The kidney that is received by a recipient must be a good match to his body, to minimize the chances of rejection of transplanted kidney by the immune system of the recipient. Modern clinical procedures have increased the success rate of such complicated technique. Now a days the process of organ donation helps a lot of kidney failure patients. Organs are collected from brain dead patients. Then transplanted to the recipient. This is called organ donation.

**Other pathways of excretion (accessory excretory organs) :**

Lungs, skin, liver and large intestine have their own specific functions but carry out excretion as a secondary function.

Lungs: In respiratory process lungs remove carbon dioxide and water vapour.

Skin: It consists of large number of sweat glands richly supplied with blood capillaries, from which they extract sweat and some metabolic wastes. Since the skin sends out plenty of water and small amount of salts, it serves as an excretory organ. Sebaceous glands in skin eliminate sebum which contains waxes, sterols, hydrocarbons and fatty acids.

Liver: The life span of RBCs is 120 days and they are destroyed in liver. It produces bile pigments (bilirubin, biliverdin and urochrome) which are metabolic wastes of haemoglobin of dead R.B.Cs. Urochrome is eliminated through urine. Biliverdin and bilirubin are excreted through bile along with cholesterol and steroid hormones, extra drug, vitamins alkaline salts etc. Liver is also plays a key role in urea formation.

Large Intestine : Excess salts of calcium, magnesium and iron are excreted by epithelial cells of colon (large intestine) for elimination along with the faeces.

Small amount of nitrogenous wastes are also eliminated through saliva and tears.

Excretion in other organisms :

Different organisms use varied strategies in excretion. Specific excretory organs are absent in unicellular organisms. These organisms remove waste products by simple diffusion from the body surface into the surrounding water. Fresh water organisms like Amoeba, Paramecium possess osmoregulatory organelle called contractile vacuole. It collects water and waste from the cell and slowly moves in the cytoplasm, reaches the surface and bursts to release its content to outside. The main excretion takes place through body surface (Diffusion).

Name of the phylum /Class	Excretory system
Protozoa	Simple diffusion from the body surface into the surrounding water
Porifera and coelenterates	Water bathes almost all their cells
Platyhelminthes	Flame cells

Name of the phylum /Class	Excretory system
Nematoda	Renette cell
Annelids	Nephridia
Arthropoda	Green glands, Malpighian tubules
Mollusca	Metanephridia
Echinodermata	Water vascular system
Reptiles, Aves and Mammals	Kidneys

Excretion in plants :

Plants do not have specific organs to excrete these wastes. Plants breakdown waste substances at much slower rate than in animals. Hence accumulation of waste is also much slower. Green plants in darkness and plants that do not possess chlorophyll produce carbon dioxide and water as respiratory waste products. Oxygen itself can be considered as a waste product generated during photosynthesis, that exits out side through stomata of leaves and lenticels of stem.

Plants can get rid of excess water by a process like transpiration and guttation. Waste products may be stored in leaves, bark, and fruits. When these dead leaves, bark, and rip fruits fall off from the tree then waste products in them are get rid off. Waste gets store in the fruits in the form of solid bodies called Raphides. However several compounds are synthesized by the plants for their own use especially for defense. Many plants synthesize chemicals and store them in roots, leaves, seeds, etc., for protection against herbivores. Most of the chemicals are unpleasant to taste. Hence, herbivores usually do not prefer to eat such plants. Some of the chemicals are toxic and may even kill the animal that eat them.

Some of the plants secrete chemicals when injured. These chemicals seal the wound and help the plant to recover from an injury. Some of the plants release attractants for other organisms which will help the plants for pollination, seed dispersal or even in their nutrition. For example, plants having root nodules secrete chemicals to attract rhizobia in the surroundings of the roots and form a symbiotic relationship with the rhizobium. These compounds are called secondary metabolites.

The biochemical substances produced in plants are of two types, primary metabolites and secondary metabolites. The materials like carbohydrates, fats and proteins are primary metabolites. The materials which do not require for normal growth and development are secondary metabolites. Eg : Alkaloids, Tannins, Resins, Gums, and Latex, etc. Though plants produce these chemical for their own use man found the usage of these chemicals for own benefits. They are generally coloured and fragrant.

Alkaloids:

These are nitrogenous by-products and poisonous. These are stored in different parts of plants. Common alkaloids in plants and their uses are given below.

Alkaloid	Plant	Part	Uses
Quinine	<i>Cinchona officinalis</i> (Cinchona)	Bark	Antimalarial drug
Nicotine	<i>Nicotiana glauca</i> (Tobacco)	Leaves	Insecticide, Stimulant
Morphine, Cocaine	<i>Papaver somniferum</i> (Opium)	Fruit	Pain killer
Reserpine	<i>Rauwolfia serpentina</i> (Snake root)	Root	High B.P
Caffeine	<i>Coffea arabica</i> (Coffee plant)	Seed	Central nervous system stimulant
Nimbin	<i>Azadirachta indica</i> (Neem)	Seeds, Barks, Leaves,	Antiseptic
Scopolamine	<i>Datura stramonium</i>	Fruit, flower	Sedative
Pyrethroids	<i>Chrysanthemum sps</i>	Flower	Insecticides

Gums : Plants like Neem, *Acacia* oozes out a sticky substance called gum.

Name of the secondary metabolite	Plant	Part	Uses
Tannins (carbon compounds)	<i>Cassia</i> , <i>Acacia</i>	Stored in different parts of the plant	Tanning of leather & used in medicines.
Resins (Eg : Pinus)	Gymnosperms	Resin passages of inner bark of the plant.	Used in varnishes
Gums	Neem, <i>Acacia</i>	Cut portions of stem or branches	Adhesives, binding agents Preparation of medicines, food, etc.
Latex (Sticky milky white substance)	<i>Hevea braziliensis</i> (Rubber plant) <i>Jatropha</i>	Stem	Preparation of Rubber, Source of biodiesel

Excretion Vs Secretion :

Excretion and secretion are the same in nature. Since both are involved in transportation and elimination of unwanted components from the body. Excretion is the removal of materials from a living being, while secretion is movement of material from one point to other point. So secretion is active while excretion is passive in nature. For example, Humans excrete materials such as tears, urine, Carbon dioxide, and sweat while secretion on other hand, includes enzymes, hormones, and saliva. Plants too excrete wastages through roots into its surroundings and falling off leaves and bark. Secretions occur in the plant body in different forms eg: latex, resins, gums etc.

PRIORITY - 1

2 Marks Questions :

1. Imagine what happens if waste materials are not sent out of the body from time to time.
(AS₂)(Q.No. 13, P.No.103)
2. What happens if plants does not shed their leaves and bark periodically ?
(AS₂) (Q.No. 8, P.No. 109)
3. Write the application of alkaloids.
(AS₇) (Q.No. 16, P.No. 103)
4. You read about "brain dead" in the chapter excretion. What discussion would you like have, why you think so?
(AS₆)(Q.No. 22, P.No.106)
5. What procedure you should follow to collect gum from trees ? (AS₃) (Q.No.15, P.No. 10)
6. How are gums useful in our daily life ?
(AS₇) (Q.No.1, P.No. 115)
7. Write the measures that you take in your daily life to keep your kidneys healthy.
(AS₇)(Q.No. 24, P.No. 107)
8. When you visit a cold country you will not sweat easily. Why ?
(AS₇) (Q.No. 5, P.No. 109)
9. Kavya is an ESRD patient. Manya says it is Oedema and Daksha says uremia. Who correct and why?
(AS₂)(Q.No. 25, P.No. 112)
10. What happens if Loop of Henle fails to function ?
(AS₂) (Q.No. 17, P.No. 112)

4 Marks Questions :

1. How are waste products excreted in *Amoeba* ?
(AS₁)(Q.No. 2, P.No.99)
2. Name different excretory organs in human body and excretory materials generated by them. (AS₁) (Or) Tabulate the accessory excretory organs of human body and the excretory products excreted by them.
(AS₁) (Q.No. 3, P.No.99)
3. Do you find any relationship between circulatory system and excretory system? What is it?
(AS₁)(Q.No. 8, P.No.100)
4. Write a brief note on composition of urine.
(AS₁)(Q.No. 1, P.No.108)
5. Write about the importance of secondary metabolites like, Tannins and Resins in everyday life.
(AS₇) (Q.No. 3, P.No. 109)
6. Write a brief note on accessory excretory organs of human beings. (AS₁) (TB, P.No. 8)
7. a) Where is the transplanted kidney fixed in the body of a kidney failure patient ?
(AS₁)(Q.No. 19, P.No. 112)
b) Can a donor of kidney survive the life with single kidney without any complication? Give your reason.
(AS₁)(Q.No. 6, P.No.108)

8. Study the following table and answer the questions that follow. (AS₁) (TB)

Alkaloid	Plant	Part	Uses
Quinine	<i>Cinchona officinalis</i>	Bark	Anti-malarial drug
Nicotine	<i>Nicotiana glauca</i>	Leaves	Insecticide
Caffeine	<i>Coffea arabica</i>	Seeds	Central nervous system stimulation
Nimbin	<i>Azadirachta indica</i>	Bark, leaves, seeds	Antiseptic
Scopolamine	<i>Datura stramonium</i>	Fruit, flower	Sedative
Reserpine	<i>Rauwolfia serpentina</i>	Root	Medicine for snake bite

- Why does drinking coffee make us energized?
 - Name the alkaloid which acts as cancer causing agent?
 - Why do we add neem leaves to bathing water when a person is suffering from skin diseases?
 - Name the plant which gives sedative.
9. How plants manage the waste materials? (AS₁) (Q.No. 5, P.No. 99)
10. Deepak said that 'Nephrons are functional and structural units of kidneys'. How will you support him? (AS₁) (Q.No. 4, P.No. 99)

6 Marks Questions :

- What is meant by excretion? Explain the process of formation of urine. (AS₁) (Q.No. 1, P.No. 98)
- How do you study the external and internal features of a kidney? (AS₁) (Q.No. 1, P.No. 113)
- Why do some people need to use a dialysis machine? Explain the principle involved in it. (AS₁) (Q.No. 6, P.No. 99)
- Give reasons : (AS₁) (Q.No. 9, P.No. 100)
 - Always vasopressin is not secreted.
 - When urine is discharged, in beginning it is acidic in nature later it become alkaline.
 - Diameter of afferent arteriole is bigger than efferent arteriole.
 - Urine is slightly thicker in summer than in winter.
- Write the differences between : (AS₁) (Q.No. 10, P.No. 101)
 - Functions of PCT and DCT
 - Kidney and artificial kidney

PRIORITY - 2

2 Marks Questions :

- What happens if dialysis is not invented? (AS₁) (Q.No. 1, P.No. 113)

2. List out the things that make you amazing in excretory system of human being. (AS₆)(Q.No: 21, P.No.106)
3. What happens if excretory system fail to function ? (AS₂)(Q.No. 22, P.No. 112)
4. What values can be inculcated after reading this lesson excretion ? (AS₆)(Q.No. 12, P.No.115)
5. To keep our kidneys healthy for long period, what questions will you ask a nephrologist/urologist? (AS₂)(Q.No. 14, P.No.103)

4 Marks Questions :

1. a) Blood is filtered in Bowman's capsule of nephron. For the filtration of blood some pressure is needed. How is this pressure created ? Explain. (AS₁)(Q.No.23, P.No. 111) 3.
- b) What are Raphides ? (AS₁)(Q.No. 24, P.No. 112)
2. Write a short note on Micturition. (AS₁)(Q.No. 2, P.No.108) 4.
3. We people have very less awareness about organ donation. To motivate people, write few slogans about organ donation. (AS₆)(Q.No. 23, P.No.107) 5.
4. Tabulate the information related to excretory system or organ found in different phyla animal kingdom. (AS₄)(TB P.No.91, Table- 4) 6.
5. What are alkaloids ? Which alkaloids are harmful for us ? (Or)(AS₁)(Q.No. 26, P.No.111)

6 Marks Questions :

1. What are the gum yielding trees in your surroundings ? What procedure should you follow to collect gum from trees ? (AS₃)(Q.No. 15, P.No.103) 7.
2. What is meant by osmoregulation? How is it maintained in human body ? (AS₁)(Q.No. 7, P.No.100) 8.
3. Describe the structure of nephron with the help of diagram. (AS₁)(Q.No. 18 P.No.104)
4. Draw a neat labelled diagram of human excretory system show its path using a block diagram. (AS₃)(Q.No. 19, P.No.105) 9.
5. Write the differences between : (AS₁)(Q.No. 10, P.No.101)
- a) Excretion and Secretion.
- b) Primary metabolites and secondary metabolites.

PRIORITY - 3

Complete all the remaining question and answers of the lesson.

IMPORTANT DIAGRAMS

- i) Human excretory system
- ii) L.S. of kidney (Or) Internal structure of kidney
- iii) Structure of nephron (Or) Uriniferous tubule

PRACTICE SHEET

Choose the correct answer :

1. Which of the following part of a kidney contains the lowest concentration of urea ?
A) Loop of Henle B) Renal vein []
C) Bowman's capsule D) Glomerulus
2. In which of the following organisms water baths almost all their cells ? []
i) Coelenterata ii) Porifera iii) Protozoa
A) only i B) only ii C) i, ii and iii D) i and ii
3. Pick the odd one. []
A) PCT B) DCT C) Renal cortex D) Loop of Henle
4. Main function of kidney is : []
A) digestion B) respiration C) secretion D) excretion
5. Malpighian tubules are the excretory organs in : []
A) earthworm B) housefly C) flatworm D) hen
6. Statement I : Urinary bladder can store a maximum of 700 - 800 ml. of urine.
Statement II : Urge for micturition occurs when urinary bladder is filled with 400 - 700 ml. of urine. []
A) Statement I is True, Statement II is True B) Statement I is False, Statement II is False
C) Statement I is True, Statement II is False D) Statement I is False, Statement II is True
7. Which one among the following is an accessory excretory organ ? []
A) Liver B) Stomach C) Small intestine D) Heart
8. What is the source of waste products that are excreted by the urinary system ?
A) Excess food B) Undigested fiber []
C) Pollutants in the water and food D) Metabolic processes
9. Sequence of steps in urine formation in a nephron is : []
A) Glomerular filtration → Tubular reabsorption → Tubular secretion
B) Tubular reabsorption → Tubular secretion → Glomerular filtration
C) Tubular secretion → Glomerular filtration → Tubular reabsorption
D) Tubular reabsorption → Concentration of urine → Tubular secretion
10. The process where blood pressure forces water and other dissolved substances from the blood into the glomerular capsule is : []
A) tubular reabsorption B) concentration of urine
C) glomerular filtration D) tubular secretion
11. The initial pH of urine is : []
A) 5 B) 6 C) 7.5 D) 6.5

12. Find the incorrect function of the urinary system.
 A) Removal of waste products from the blood stream.
 B) Storage and excretion of urine.
 C) Regulation of leucocyte and platelet production.
 D) Regulation of blood volume and indirectly, blood pressure.
13. Which sequence correctly represents the path of urine after it leaves the kidneys?
 A) Ureters → urinary bladder → urethra
 B) Urinary bladder → ureters → urethra
 C) Urethra → urinary bladder → ureters
 D) Urinary bladder → urethra → ureters
14. In a typical adult kidney, the renal _____ is subdivided into 8 to 15 distinct conical or triangular structures known as renal _____.
 A) cortex; papillae
 B) medulla; pyramids
 C) medulla; columns
 D) cortex; pyramids
15. Arrange the following in the correct sequence to trace the path of filtrate through renal tubule.
 1) proximal convoluted tubule
 2) distal convoluted tubule
 3) ascending limb of nephron loop
 4) descending limb of nephron loop
 A) 1, 2, 3, 4
 B) 2, 3, 4, 1
 C) 1, 4, 3, 2
 D) 3, 2, 1, 4
16. From which part of the renal tubule 75% of the water and virtually all nutrient electrolytes, and plasma proteins are re-absorbed into the blood?
 A) Proximal convoluted tubule
 B) Distal convoluted tubule
 C) Ascending limb of nephron loop
 D) Descending limb of nephron loop
17. The hollow space at the centre of kidney where urine is collected after its formation is called:
 A) Renal pelvis
 B) Glomerulus
 C) Distal convoluted tubule
 D) Urethra
18. Identify the function of Sedative.
 A) Induces sleep
 B) Insecticide
 C) Antiseptic
 D) Pain killer
19. Urge for micturition occurs when urinary bladder is filled with _____ ml. of urine.
 A) 300 - 800
 B) 300 - 400
 C) 700 - 800
 D) 1600 - 1800
20. The yellow pigment derived from haemoglobin that is broken down and excreted by kidney is:
 A) uric acid
 B) urochrome
 C) creatine
 D) creatinine

5. COORDINATION

CHAPTER AT A GLANCE

A system of control and coordination is essential in living organisms so that the different body parts can function as a single unit to maintain homeostasis as well as respond to various stimuli.

In animals, the nervous system and hormonal systems are responsible for control and coordination.

Responding to stimuli :

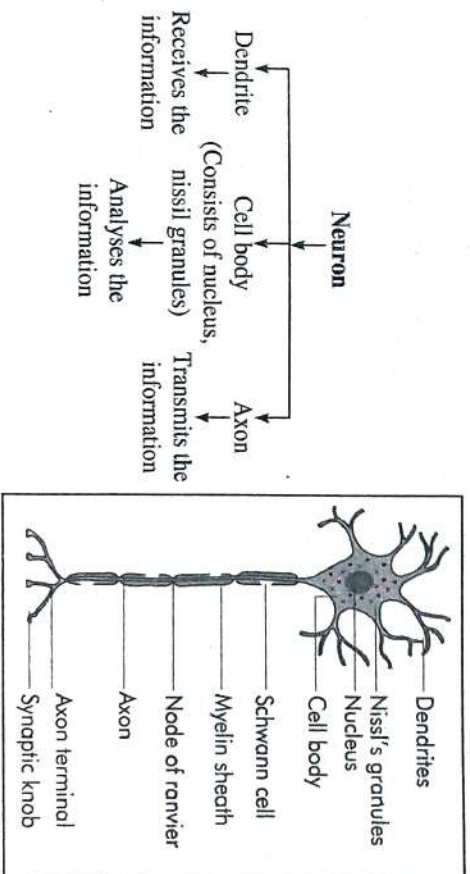
Changes in the surroundings or signals of changes or stimulus are reasons for responses by living organisms. All living organisms respond to stimuli. The cat may be running because it saw a mouse. Plants grow towards the sunlight. We start sweating when the surroundings are hot and humid. The ability to react to particular stimulus in a particular situation must be of great importance in ensuring the survival of the organism.

Integrating pathways - Nervous coordination :

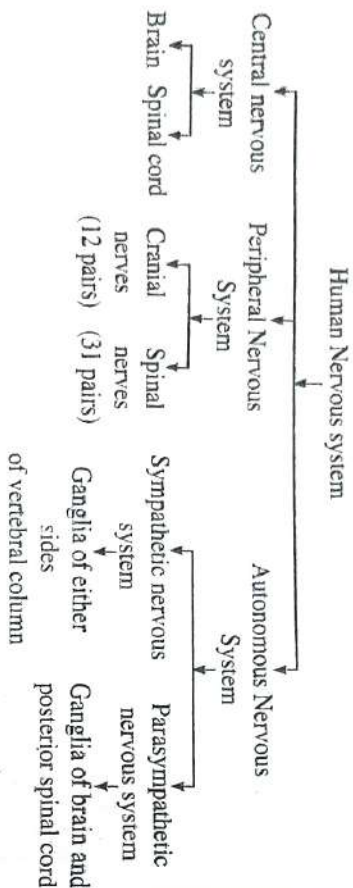
Galen, a Greek physiologist (A.D. 129-200) made one notable observation. One of his patients, having suffered a blow on the neck when falling from his chariot, complained loss of feeling in the arm while still retaining normal muscular control of its movement. Galen concluded that nerves were of two kinds – those of sensation and those of action. According to him the blow in the neck had damaged the nerves of sensation but had not affected its action.

Structure of Nerve cell :

Neurons are specialized cells of the nervous system, which are considered as the basic units of nervous system. They use electrical and chemical signals for the transmission of information.



- Receptors are specialised tips of the nerve fibres that collect the information to be conducted by the nerves.
- Nerve impulses travel in the following manner from one neuron to the next :
 - Dendrites → Cell body → Axon → Nerve terminals → Synapse → Dendrites of next neuron
- The axon is surrounded by a specialized insulatory sheath called myelin sheath. This sheath is interrupted at regular intervals called nodes of Ranvier. The myelin sheath is made up of Schwann cells and chiefly consists of fatty material.
- Chemicals released from axon tip of one neuron (nerve terminals), travel across the synapse or neuromuscular junction to reach the next cell (neuron or muscle fibre).
- Nerve impulses from many neurons interact to carry out the complex process of thinking.
- Three divisions of the nervous system are Central nervous system, Peripheral nervous system and Autonomous nervous system.



Types of neurons :

Neurons are classified mainly into three different types :

S.No.	Type of Neuron	Function
1	Sensory (Afferent)	Carry messages towards the central nervous system.
2	Motor (Efferent)	Carry messages from the central nervous system (Brain or Spinal cord) to other parts of the body.
3	Association neuron	Carry messages from the central nervous system (Brain or Spinal cord) to other parts of the body.

- Spinal cord controls the reflex actions and conducts messages between different parts of the body and brain.
- Reflex action is an automatic, rapid and immediate reaction to a stimulus and is below the level of consciousness. No thinking is involved in reflex action.
- Reflex arc is the neural pathway that mediates a reflex action. (Or) The structural and functional unit of a reflex action.

Pathway of reflex arc :

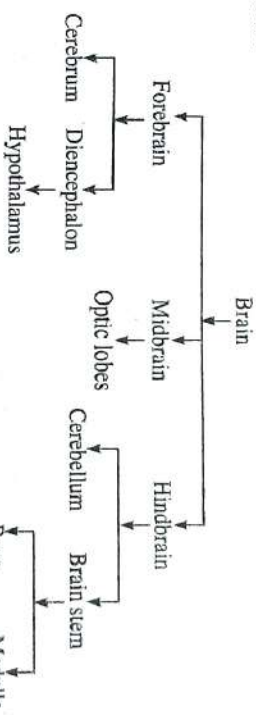
Receptor → Sensory neuron → Relay neuron (Interneuron) → Motor neuron → Effector

- The sensory neurons of reflex arc, synapse in the spinal cord activates the spinal, motor neurons without delay to execute a quick action, especially in case of emergencies. The brain also receives the information while the reflex action occurs.

S.No.	Component of reflex arc	Function
1	Receptor	Receives stimuli and generates impulses
2	Sensory (Afferent) Neuron	Carries information from the receptor to the interneurons in the spinal cord
3	Interneurons	Processes the information and generates responses
4	Motor (Efferent) neuron	Carries the information from the spinal cord to the effector organ
5	Effector	Receives the information from the effector neuron and shows the appropriate response

The 3 main parts of the brain are forebrain, midbrain and hindbrain.

The largest part of the brain, the forebrain, is the main thinking region. It is made up of cerebrum, hypothalamus and thalamus. Cerebellum, pons and medulla constitute the hindbrain.



Cerebrum is the largest part of the brain whereas the cerebellum is the second largest part of the brain.

Part of the brain	Functions
Olfactory lobes	Functions are concerned with sense of smell.
Cerebrum	i) Site of mental abilities, controls thinking, memory, reasoning, perception, emotions and speech. ii) Interprets sensations and responds to cold, heat, pain and pressure.
Forebrain	i) Relay centre for sensory impulses, such as pain, angry, happiness. ii) Reflex centre for muscular activities. iii) Centre for certain emotion iv) Centre for water balance, blood pressure, sleep and hunger. v) The hypothalamus controls the pituitary gland.

Midbrain	Optic lobes	i) It relays motor impulses from the cerebral cortex to the spinal cord and relays sensory impulses from the spinal cord to the thalamus. ii) Show reflexes for sight and hearing.
Hindbrain	Cerebellum	i) Maintains posture, equilibrium and muscle tone. ii) Co-ordinates voluntary movements initiated by cerebrum.
	Medulla oblongata	i) Contains centre for cardiac, respiratory and vasomotor activities (vasomotor = actions a blood vessel which alter its diameter). ii) Coordinates reflexes like swallowing, coughing, sneezing and vomiting.

Brain is protected by a bony box called cranium, within which three layers of fluid-filled membranes are present for absorbing shock.

Peripheral nervous system consists of cranial nerves and spinal nerves and assists in transmitting information between central nervous system and rest of the body.

Reflex actions, voluntary actions and involuntary actions are the various types of responses shown by the nervous system.

The sense organs detect changes in the surroundings and pass this information to the central nervous system, which after processing the information, acts through the muscles.

The movements of muscle tissues are brought about by the contraction and relaxation of the contractile proteins in response to nerve impulses.

Spinal cord :

Spinal cord is a long and cylindrical structure.

It extends from the back of the hindbrain to the back of the stomach or lumbar region, through the neural canal of the vertebral column.

Unlike the brain the white matter is towards periphery while grey matter is towards the centre of the spinal cord.

The myelinated axons leave the spinal cord from both sides of the vertebral column.

Animals were shown to have the ability to respond to stimuli even when the brain was removed. 'Leonardo da Vinci' (1452-1519) and 'Stephen Hales' (1677-1771) both recorded the survival of frogs whose brain had been removed.

The animal still produced muscular movements if its skin was pinched or pricked. Both observers further recorded that the animal died as soon as spinal cord was damaged by pushing a needle down it.

The vertebrae of the vertebral column protects the spinal cord from injuries.

Peripheral nervous system :

The nerves attached to the spinal cord have two types of connections or roots – One to the back or dorsal side and other to front or the ventral side of cord. The experimental work of two men, Charles Bell in Scotland and Francois Magendie in France, in the early nineteenth century, showed that these roots have different functions. If the dorsal roots of

an experimental animal were cut the animal made no obvious reaction. If, however, the ventral roots were even lightly touched, the muscles to which the nerve was connected twitched violently. The ventral root evidently controlled muscular activity, the dorsal root did not.

Autonomous Nervous System (ANS):

The PNS can involuntarily control several functions of regions like our internal organs, blood vessels, smooth and cardiac muscles. So it is called autonomous nervous system. It has voluntary control of muscles of some areas of skin and the skeletal muscle as well.

Certain involuntary functions are controlled by autonomous nervous system in our body. For example reduction and expansion of the pupil of our eye. When we enter a dark room we cannot see anything immediately. Slowly we are able to see the things around us in the room. This is because of increase in diameter of pupil, which allows more light into the eyes. When we come out of the dark room into broad day light the diameter of the pupil decreases allowing less light to fall on retina of the eyes. Both these functions occur under the influence of the autonomous nervous system.

The autonomous nervous system has two parts - sympathetic and parasympathetic, which cause physical reactions opposite to each other.

Sympathetic Nervous System (SNS) :

The sympathetic system is formed by the chain of ganglia on either sides of the vertebral column and the associated nerves.

Parasympathetic Nervous System (SNS) :

The para sympathetic system is formed by the nerves arising from the ganglia of the brain and the posterior part of the spinal cord.

Coordination without nervous system :

We shall now study about coordination that may not require nervous control.

The story of insulin :

In 1868 Paul Langerhans, Professor of Pathology at the University of Freiburg in Germany, working on the structure of the pancreas, noted certain patches of cells quite different in appearance from the normal tissue cells of the organ and richly supplied with blood vessels. They are named as Islets of Langerhans (islets stands for islands) and it produce a secretion which directly liberated into the blood.

In Latin 'insula' means an island. The name insulin was coined for the secretion, even though the method of isolation was not discovered.

Insulin helps to maintain blood sugar levels and treat the disease called sugar or diabetes mellitus.

The various endocrine glands in humans are hypothalamus, pineal gland, pituitary gland, thyroid gland, parathyroid glands, thymus, pancreas, adrenal glands, ovary (in female) and testis (in males).

Some important hormones and their functions in human body:

Name of the gland	Location	Hormone secreted	Response of body to hormone
Pituitary	Floor of brain	1. Somatotrophic	Growth of bones
		2. Thyrotrophic	Activity of thyroid gland
		3. Gonadotrophic	Activity of ovary and testis
		4. Adrenocorticotrophic hormone	Stimulates secretion from adrenal cortex
		5. Luteinising hormone	In males - secretion of testosterone In female - Ovulation, development of corpus luteum and secretion of progesterone
Thyroid	Neck	6. Follicle stimulating hormone	In male - spermatogenesis In female - growth of granular follicles, estrogen secretion, milk production and secretion.
		7. Vasopressin	Regulate absorption of water from the kidney tubules
		Thyroxine	General growth rate and metabolic activity
Ovary	Lower abdomen	Oestrogen	Growth of the uterus and stimulation of the pelvis Control of the 28 days menstrual cycle in females
		Progesterone	Development of uterus, embryo implantation, development of mammary glands
Testis	Scrotum	Testosterone	Growth of hair on face, muscular development depending of voice, growth of penis, testis, four and 1/2 inch length of male sex organ.
Adrenal	Superior kidney	Adrenaline	Increases heart-beat rate. Rise in blood pressure. Dilatation of the primary arteries. Dilatation of the pupil of the eye.
Pancreas	Near duodenum	Insulin	Decrease glucose percentage in blood Maintain glucose level in blood convert glucose - glycogen.
		Glucagon	Increase glucose percentage in blood Converts the glycogen into glucose

Feedback mechanisms are present to regulate the hormone action

Character	Nervous system	Endocrine system
Mode of communication	Electrical impulses	Chemical substances
Speed of communication	Very quick	Slow
Can reach	Only cells connected by nervous system	All cells of the body
Continuity	Cannot continuously transmit impulses	Can act steadily and persistently

Plants lack nervous and muscular system.

Plants respond to stimuli by showing two types of movements- growth independent and growth dependent.

Growth independent movements are usually quicker than growth dependent ones and involve the use of electrochemical signals by the plant. To achieve this movement, the plant cells change shape by altering their water content.

Growth dependent movements or tropic movements are slow, occurring either towards or away from the stimulus.

Tropic movements are shown in response to environmental factors such as light, gravity, touch, water and chemicals.

Nastic movements are non-directional responses to stimuli e.g. temperature, humidity, touch, light and are usually associated with plants.

Plant roots are positively geotropic and negatively phototropic whereas plant leaves are usually negatively geotropic and positively phototropic.

Pollen tubes show chemotropism by growing towards the ovules.

In addition to electrochemical signals, plants and animals use hormones for growth and coordination.

MOVEMENTS IN PLANTS

Phototropism	Geotropism	Thigmotropism	Chemotropism	Heterotropism
Growth movement of shoot towards light	Growth movement of root towards gravity	Growth movement of tendrils towards touch make	Growth movement of pollen tube towards chemicals	Growth movement of roots towards available water
			growth of pollen tube towards ovule	

- Charles Darwin and his son Francis Darwin performed some experiments on phototropism.
- They stated that when seedlings are freely exposed to a lateral light some 'influence' is transmitted from upper to the lower part causing the material to bend.

3 In 1926, the Dutch plant physiologist F.W. Went succeeded in separating this 'influence from the plant that produced it. He named that influence as auxin (Greek word auxin means to increase).

4 Auxins cause the bending of plant stem towards light as well as the curling of plant tendrils around a support.

5 Important plant hormones are auxins, gibberellins, cytokinins, abscisic acid and ethylene.

Plant Hormones	Uses
Auxins	Cell elongation and differentiation of shoots and roots.
Cytokinins	Promote cell division, promotion of sprouting of lateral buds, delaying the ageing in leaves, opening of stomata.
Gibberellins	Germination of seeds and sprouting of buds; elongation of stems; stimulation of flowering; development of fruit, breaking the dormancy in seeds and buds.
Abscissic acid	Closing of stomata; seed dormancy, promoting ageing of leaves.
Ethylene	Ripening of fruit

PRIORITY - 1

2 Marks Questions :

1. What will happen if efferent nerves are degenerated? (AS₂)(Q.No. 9, P.No. 136)
2. What do you learn by doing the activity - "holding a falling stick"? (AS₃)(TB, P.No. 101)
3. How does autonomous nervous system functions in every day life? (AS₂)(Q.No. 1, P.No. 140)
4. Write the applications of phytohormones. (AS₂)(Q.No. 1, P.No. 138)
5. What will happen if peripheral nervous system fails to function? (AS₂)(Q.No. 15, P.No. 136)
6. What happens if brain and spinal cord are non functional? (AS₂)(Q.No. 1, P.No. 137)
7. Rahul has taken a potted plant and covered its base tightly and hanged it upside down. He observed it for a week. What observations would have been made by Rahul? (AS₂)(Q.No. 20, P.No. 129)
8. What will happen to a potted plant kept near window in the room? (AS₂)(Q.No. 17, P.No. 128)
9. What happens if all functions of the human body is controlled only by brain? (AS₂)(Q.No. 18, P.No. 128)
10. If you accidentally touch a very sharp surface with your feet, what is your response? Explain the phenomena. (AS₂)(Q.No. 2, P.No. 140)

4 Marks Questions :

1. Describe the structure of neuron with the help of a diagram. (AS₂)(Q.No. 1, P.No. 136)

2. Explain the mechanism of reflex arc. (AS₂)(Q.No. 2, P.No. 140)

3. Do you think body's team work maintains functioning of our body? Justify your answer with an example. (AS₂)(Q.No. 2, P.No. 124)

4. Read the following sentences and compare with endocrine glands. (AS₂)(Q.No. 24, P.No. 130)

Pheromones are chemical substances secreted by organisms. These act as chemical signals secreted by exocrine glands. Pheromones are used as signals by the members of same species. Honey bee secretes pheromones that attract other bees to the location of food.

5. What is synapse? How it is useful in transferring information? (AS₂)(Q.No. 5, P.No. 124)

6. Explain trophic movements with suitable examples. (AS₂)(Q.No. 3(a), P.No. 134)

7. State whether the following actions are voluntary actions, reflex actions or conditioned reflex. (AS₂)(Q.No. 13, P.No. 127)

i) Blinking of eyes

iii) Playing on the key board

iv) Salivating when food is put in the mouth

v) We close our ears when we hear unbearable sound. (AS₂)(Q.No. 16, P.No. 128)

8. How does a neuron differ from an ordinary cell in structure? (AS₂)(Q.No. 11, P.No. 127)

9. Give an example to show how hormones can influence visible changes in your body. (AS₂)(Q.No. 10, P.No. 127)

10. What suggestions will you give to diabetic patients to maintain proper glucose levels in blood? (AS₂)(Q.No. 14, P.No. 136)

6 Marks Questions :

1. Give an example of coordination in your body where both hormonal and nervous controls function together. (AS₂)(Q.No. 3, P.No. 124)
2. Man is the most intelligent animal. What could be the fact that helped us to reach such a conclusion? (AS₂)(Q.No. 13, P.No. 127)
3. What procedure do you follow to understand the effect of plant growth hormones (in agar medium) in the terminal portion of the tip of stem (coleoptile)? (AS₂)(Q.No. 22, P.No. 129)
4. How does phototropism occur in plants? (Or) Define phototropism and explain with an example. (AS₂)(Q.No. 7, P.No. 125)
5. You are walking in the traffic. Suddenly you heard a loud sound. How coordination takes place in this situation among respective organs? Draw a block diagram to explain this situation. (AS₂)(Q.No. 28, P.No. 132)

PRIORITY - 2

2 Marks Questions :

1. What would be the consequences if testosterone fails to function? (AS₂)(Q.No. 13, P.No. 136)

2. It is very interesting to watch a creeper entwining its tendrils to the support. How do you appreciate it?
(AS₁)(Q.No.31, P.No. 132)
3. What may happen if anger persists for a longer period? (Or)(AS₂)
(AS₂)(Q.No.2, P.No. 137)
4. How does anger come down?
(AS₂)(Q.No.16, P.No. 137)
5. What happens if neurons are not inter connected with each other?
(AS₂)(Q.No. 5, P.No.135)
6. What happens in case of spinal cord injury?

4 Marks Questions :

1. Insulin plays a major role in diabetes. Give reason.(Or) Diabetic patients take insulin injections. Give reason.
(AS₁)(Q.No.14, P.No. 136)
2. Distinguish between :
a) Central nervous system & peripheral nervous system.
b) Receptor and Effector
(AS₁)(Q.No.18, P.No. 137)
3. Write a note on "The story of insulin".
(AS₁)(Q.No. 8, P.No. 126)
4. Give an example and explain how plants may immediately respond to stimulus.
(AS₁)(Q.No. 14, P.No. 127)
5. The axon of nerve cell in hand is shorter than the axon of nerve cell in leg. Do you support this statement? Why?
(AS₁)(Q.No.12, P.No. 136)

6 Marks Questions :

1. What is peripheral nervous system? Mention its functions. (Or) How does the peripheral nervous system work? Explain. (Or) Explain the functioning of peripheral nervous system in human body.
(AS₁)(Q.No.15, P.No. 136)
2. Write the differences between :
i) Auxins - ABA
ii) Auxins - Gibberellins
(AS₁)(Q.No.9, P.No. 126)
3. Suggest an experiment to show how roots grow away from light in most plants.
(AS₁)(Q.No.26, P.No. 131)
4. Draw a picture representing connection between dendrite - dendrite, axon - dendrite. Why do they connect like that?
(AS₁)(Q.No.6, P.No. 125 a & b)
5. Distinguish between :
a) Stimulus and Response
b) Afferent and efferent nerves / sensory nerves and motor nerves.

PRIORITY - 3

Complete all the remaining question and answers of the lesson.

IMPORTANT DIAGRAMS

PRACTICE SHEET

- i) Nerve cell ii) Reflex arc iii) Brain

Choose the correct answer :

1. Which of the following is known as 'gland of emergency'?
A) Pituitary B) Parathyroid C) Thyroid D) Adrenal
2. Iodine is necessary for the production of :
A) Parathormone B) Vasopressin C) Glucagon D) Thyroxine
3. Calcium and phosphate levels in blood and bones are regulated by :
A) Thyroxine B) Prolactin C) Parathormone D) Thyroid
4. Medulla oblongata coordinates reflexes like :
i) Coughing and sneezing ii) Centre for water balance
iii) Emotion and speech iv) Vomiting
A) i and iii B) ii and iii C) i, ii and iii D) i and iv
5. Who recorded the survival of frogs in which the brain had been destroyed?
A) Francois Magendie and Starling B) Charles Bell & F.W. Went
C) Leonardo Da Vinci and Stephen Hales D) Stephen Hales and Francois Magendie
6. The scientist who coined the term hormone is :
A) Darwin B) Lamarck C) Straling D) Langerhans
7. Auxins in plants are synthesised at :
A) Nodes B) Petiole C) Meristems D) Internodes
8. Response of plants to gravity is known as :
A) Geotropism B) Hydrotropism C) Thigmotropism D) Phototropism
9. What acts upon a blood vessel and alters its diameter?
A) Vasomotor B) Allomotor C) Presomotor D) Nanomotor
10. Parts of the brain below the cerebrum are together known as :
A) Mid brain B) Cerebellum C) Diencephalon D) Medulla oblongata
11. The correct pathway of impulse in a reflex arc is :
A) Detector → Afferent neuron → Spinal cord → Effector muscle → Efferent neuron
B) Detector → Efferent neuron → Spinal cord → Afferent neuron → Effector muscle
C) Effector → Afferent neuron → Spinal cord → Efferent neuron → Detector
D) Detector → Afferent neuron → Spinal cord → Efferent neuron → Effector muscle
12. Identify the function of para sympathetic nervous system.
A) Increase blood pressure B) Inhibits heart beat
C) Inhibit pancreatic activity D) Inhibit salivation

13. The phytohormone which helps in survival of leaves on the plant for a longer time is:
 - A) Auxins
 - B) Gibberellins
 - C) Ethylene
 - D) Cytokinins
14. Ripen stigma secretes sugary substance to attract pollen grains. This response is known as:
 - A) Hydrotropism
 - B) Chemotropism
 - C) Phototropism
 - D) Thigmotropism
15. The path of electric impulse in a neuron is:
 - A) Dendrite → Cell body → Cyton → Axon terminals → Axon
 - B) Axon → Axon terminals → Cyton → Dendrites
 - C) Axon terminals → Axon → Cell body → Dendrites
 - D) Dendrites → Cell body → Axon → Axon terminals
16. Which of the following parts of brain consists of optic lobes?
 - A) Cerebrum
 - B) Midbrain
 - C) Cerebellum
 - D) Medulla
17. The neuron that carries the messages towards the central nervous system from detectors is:
 - A) Motor neuron
 - B) Afferent neuron
 - C) Efferent neuron
 - D) Ventral neuron
18. Daksha bought a dozen of raw mangoes and kept them along with two ripe mangoes. After two days she found that all the mangoes were ripened. This is due to the release of:
 - A) Cytokinins
 - B) Auxins
 - C) Ethylene
 - D) Gibberellins
19. Which of the following pair of plant hormones are growth inhibitors?
 - A) Auxins and Gibberellins
 - B) Gibberellins and Cytokinins
 - C) Abscissic acid and Ethylene
 - D) Auxins and Cytokinins
20. Peripheral nervous system consists of nerves that arise from:
 - A) only brain
 - B) only spinal cord
 - C) both brain and spinal cord
 - D) neither brain nor spinal cord

6. REPRODUCTION

CHAPTER AT A GLANCE

In plants and animals reproduction is a necessary life process for continuation of life by the production of offsprings.

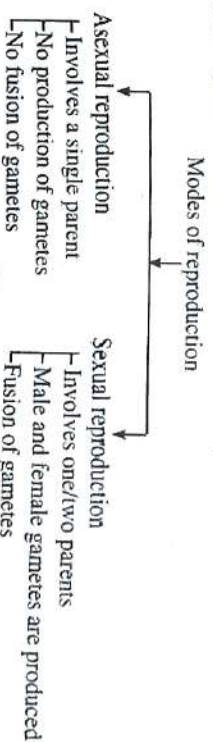
Organisms are capable of giving rise to offsprings by the process of reproduction.

Some organisms may reproduce differently in different situations.

The time required to reproduce also varies from organism to organism. Even within the organism there could be certain environmental conditions that would make the process of reproduction faster.

Reproduction :

Reproduction is defined as a biological process in which an organism gives rise to young ones (offspring) similar to itself. Reproduction ensures the species survival from generation to generation.

Modes of reproduction :

Organisms can reproduce asexually in many ways.

i) **Fission** : Single celled organisms, such as *Paramecium* and bacteria, reproduce by splitting into two or more offsprings.

ii) **Budding** : A growth on the body as a bud that grows to form nearly identical copy of parent. When the bud totally grows then it separates from the parent and survives independently. Eg: Yeast, *Hydra*.

iii) **Fragmentation** : This happens only in the simplest, such as some flatworms, moulds, lichens, *Spirogyra* etc. Fragmentation is a common mode of reproduction in algae, fungi and many land plants.

iv) **Parthenogenesis** : This is a process of reproduction where there is a shift from sexual to asexual mode of reproduction. In this process generally the female gamete develops into zygote without fertilization. This strange kind of reproduction occur in bees, ants and wasps.

v) **Parthenocarp** : Now a days we are able to develop seedless fruits like watermelon, grapes etc. Naturally in some plants like banana ovary directly develops into fruit without the process of fertilization this phenomenon is called as "Parthenocarp".

vi) **Regeneration** : Many organisms have the ability to give rise to new individuals from their body parts. This is similar to fragmentation.

vii) **Vegetative propagation** : In higher plants, vegetative propagation may be natural or artificial.

viii) **Natural propagation** :

Leaves : In *Bryophyllum* small plants grow at the edge of leaves.

Stems : Aerial weak stems like runners and stolons, when they touch the ground give off adventitious roots. When the connection with the parent plant is broken, the stem portion with the adventitious roots develops into an independent plant. Some examples for propagation by stem are by stolons, bulbs, corms, tuber, etc.

Runner : - Oxalis, Hydrocotyl; **Stolons** - Jasmine, Strawberry; **Bulbs** - Onion, Corn - Colocasia, **Rhizome** - Turmeric, Ginger, **Tuber** - Potato.

Root - Roots of Murraya, Guava, Millingtonia etc., produce radical buds that grow

Artificial propagation :

i) **Cutting** : Some plants grow individually when a piece of parent plant having bud is cut from the existing plant. Eg: Rose, Hibiscus

ii) **Layering** : A branch of the plant with at least one node is bent towards the ground and a part of it is covered with moist soil leaving the tip of the branch exposed above the ground. After some time, new roots develop from the part of the branch buried in the soil. Eg: Nerium, Jasmine.

iii) **Grafting** :

Two plants are joined together in such a way that two stems join and grow as a single plant. One which is attached to soil is called stock and the cut stem of another plant without roots is called scion.

Grafting is used to obtain a plant with desirable characters.

This technique is very useful in propagating improved varieties of various flower and fruit. Eg: Mango, citrus, apple, rose.

The cutting, layering and grafting are the traditional methods for the artificial propagation of plants. Examples of plants produced in this manner are Banana, Pineapple, Orange, Grape, Rose, etc.

Spore formation :

Rhizopus produces hundreds of microscopic reproductive units called spores. When the spore case (also called sporangium) bursts, the spores spread into air. These airborne spores land on food or soil, under favourable conditions like damp and warm conditions, they germinate and produce new individuals. Eg : Fungi like *Rhizopus*, *Mucor* etc.

Bacteria and non-flowering plants such as ferns and mosses reproduce by the method of spore formation.

Sporophyll : Ferns also produce spores. On the lower surface of the leaf you find clusters of dot like structures called sporangia. These contain spores.

Asexual Reproduction :

Mode of reproduction	Examples
Binary Fission	Bacteria, <i>Paramecium</i>
Budding	Yeast
Fragmentation	Flatworms, moulds, lichens, spirogyra, algae, fungi
Parthenogenesis	Ants, bees, wasps,
Parthenocarry	Banana, Watermelon (seedless) etc.
Regeneration	Planaria
Spore formation	Sporophyll, fern

Vegetative propagation :

Mode of reproduction	Examples	Mode of reproduction	Examples
Leaves	<i>Bryophyllum</i>	Corms	Ginger, <i>Colocasia</i>
Stolons	Jasmine, Strawberry	Tuber	Potato (eyes)
Bulbs	Onion (<i>Allium cepa</i>)	Roots	Murraya, Guava, Millingtonia etc.

Artificial propagation :

Mode of reproduction	Examples
Cutting	Rose, Hibiscus
Layering	Nerium
Grafting	Mango, Citrus, Apple, Rose

Sexual reproduction : sexual reproduction is a way of reproduction where fusion of gametes takes place, by a process called fertilisation.

Fertilisation may occur either outside the body of the female organism (external fertilisation) or inside the female organism's body (internal fertilisation).

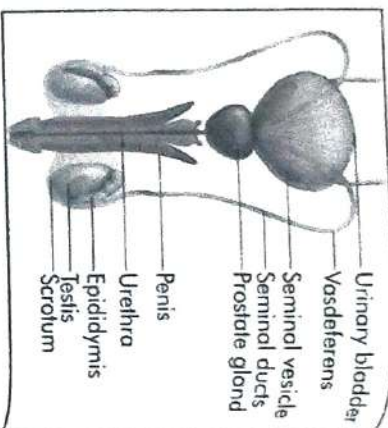
External fertilisation is common in aquatic animals like most of the fishes and amphibians. The female lays a vast number of eggs in water and male releases some millions of sperms on them in water. As the chance of fertilisation is controlled by nature which occurs externally, hence it is inevitable to give rise to vast number of eggs and sperms (gamete).

Reproduction in a placental mammal - Human beings :

While talking about placental mammals especially human beings special reproductive organs have developed in males and females to carry out reproduction.

Male reproductive system :

- In human males, the two testes are located in pocket like structure outside the body wall called the scrotum.
- Each testis has several lobules and each lobule contain several seminiferous tubules.
- One prostate, two cowper's glands are accessory glands in male reproductive system secretes a fluid called semen.
- The sperm cell is a flagellated structure with long tail. This helps them to move towards the ovum.



The development of the male reproductive organs is regulated by the male sex hormone called testosterone.

Secondary sexual characters are also controlled by the male sex hormones.

Men produce sperm, from the age of about 13 or 14 years.

Part of the male reproductive system	Function
Testes	Produce the sperms (Male gametes)
Vasa efferentia	Collect spermatozoa from seminiferous tubules
Epididymis	Stores spermatozoa temporarily
Vasa deferentia	Collect the sperms from epididymis
Seminal vesicles	Produce seminal fluid and supply energy to sperms
Prostate gland & Cowper's glands	Secrete semen which provide nutrients for sperms
Ejaculatory duct	A duct from seminal vesicles joins the vas deferens
Urethra (Urino-genital duct)	Common passage for sperms and urine

The passage of spermatozoa :

Seminiferous tubules → vasa efferentia → epididymis → vasa deferentia → ejaculatory duct → urethra

The sperm :

Structure	Function
Head	i) The sperm has a head which bears acrosome. ii) It helps the sperm in penetrating into ovum. iii) In the middle of the head male nucleus is present. It fuses with the female nucleus.
Middle piece	i) Mitochondria middle piece produce energy for the movement of the sperm.
Tail	i) It propels the sperm.

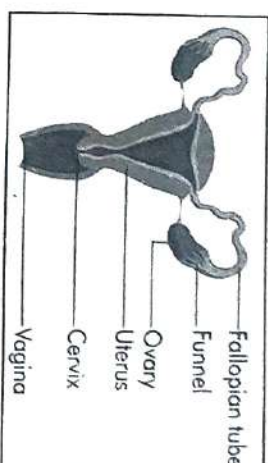
The sending out of semen from male's body is called ejaculation. From normal fertility 60% of sperm should have normal shape and size. And 40% of them must have vigorous motility.

Men produce sperm, from the age of about 13 or 14 years, and can go on doing so during most of their lifespan. Their power to do so decreases as they grow older.

Female reproductive system :

Part of the female reproductive system	Function
Ovaries	Produce the ova (Female gametes)
Fallopian tube / Oviduct	Collects the ova and site for fertilisation.
Uterus	Implantation of embryo & Embryonic development

The ova develop in tiny cellular structures called follicles, which at first look like cellular bubbles in the ovary. They are called graafian follicles.



Each follicle contains a single ovum which is formed after the process of cell division (meiosis).

The release of the egg or ovum is called ovulation.

Fertilization occurs as the ovum passes through the oviduct thus begins a new life, fertilization with sperm would lead to formation of a mass that might grow to form a baby.

During the development of the embryo, tiny finger like projections grow from the surface of the outer membrane called chorion. Tissues of the chorion and the adjacent part of the uterine tissue make up the placenta.

Placenta is a tissue formed by the cells from the embryo and the mother. It is formed at around 12 weeks of pregnancy and becomes an important structure for nourishment of the embryo.

Another embryonic membrane, the amnion, grows around the embryo itself. The cavity within the amnion becomes filled with fluid called amniotic fluid.

Another membrane called allantois, which originates from the digestive canal of the embryo forms the major part of a tube like structure called umbilical cord.

From the third month of pregnancy the embryo is called foetus.

- Pregnancy lasts, on an average, 9 months or 280 days. This period is called gestation period.

Embryonic layer		Function
Chorion		Gas exchange
Amnion		Protection
Allantois		Storage of nitrogenous waste
Yolk sac		Nourishment (Birds & Reptiles)
		No specific function in placental mammals.

Child birth :

- Childbirth begins when the muscle layers of the uterus starts a rhythmic contract and relax, these actions are felt as labour pains.
- Generally, at this stage, the sac (amnion) around the baby breaks, and its fluid contents are released. The umbilical cord leading from the baby to the placenta, is tied off and cut by the doctor. The muscular contractions of the uterus continue until they push out the tissue of the placenta, which are commonly called the "afterbirth."
- During the last part of pregnancy, a watery lymph like fluid called 'colostrum' accumulates in the mammary glands.
- For the first few days after the baby is born, the mammary glands secrete only colostrum.
- **Sexual reproduction in plants :** Flowers are the essential parts of sexual reproduction in flowering plants.

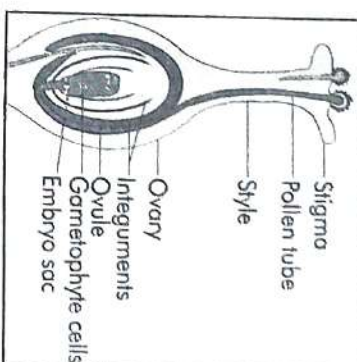
Part	Description	Function
Calyx	Calyx is the outermost whorl of the flower. It is made up of individual units called sepals which are green and leaf like.	Calyx protects the bud stage.
Corolla	It lies inner to calyx. It is made up of individual units called petals which are generally brightly coloured.	Its bright colour attracts insects and birds.
Androecium	Androecium is the male reproductive part. Its individual units are called stamens. They lie inner to corolla. A stamen consists of filament and anther.	Stamens produce pollen grains containing male gametes.
Gynoecium	Gynoecium is the inner most whorl. It is the female reproductive part of the flower. It is also called carpel. A carpel consists of stigma, style and ovary.	Ovary has ovules which contain female gametes.

- Flowers having both the stamen and carpel are bisexual like *Datura*.
- Stamens (male portion called androecium) produce male sex cells called pollen grain.
- Carpels (female portions called Gynoecium) produce female sex cells called ovules inside ovary.

- The plants having flowers where male reproductive cells of stamen of the flower fertilise the female reproductive cell of the carpel of the same flower is called self-pollination.
- The male cells of flower of a plant fertilise the other female flower on the same or different plant of the same group then this type of pollination is called cross pollination.

Structure of the ovule :

- An ovule is an egg-shaped structure attached by a stalk to the inner side of the ovary.
- At the center of each ovule is a microscopic embryo sac filled with food and water. The embryo sac is composed of gametophyte cells.



- The majority of flowering plants have an embryo sac consisting of 7 cells and 8 nuclei.
- One is a large central cell containing 2 nuclei. These are called polar nuclei. The other cell is the egg. It is located at the end of the embryo sac closest to the opening through which the pollen tube enters.
- One of the two sperms fuses with the egg to form a zygote.
- The zygote will develop into an embryonic plant within the ovule.
- Union of one male nucleus with the egg, and the second nucleus with the secondary nucleus is called "double fertilisation".
- Double fertilization occurs only in flowering plants.
- The seed produced after fertilisation contains the future plant or embryo that develops into a seedling under appropriate conditions. The process is called germination.

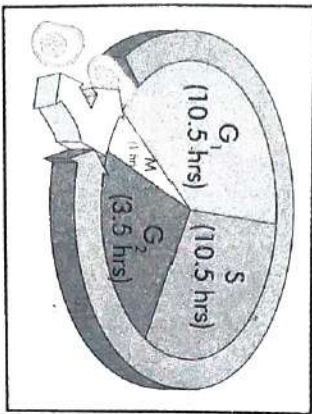
Cell division and continuation of life :

- Continuation of life starts from cells either those of the general body or the sex cells (gametes).
- Virchow (1821-1902) a proponent of cell theory is given the credit for the phrase *Omnis cellula e cellula*, or cells arise from pre-existing cells, indicates the importance of cell division in the creation of new cells.
- In cell division the cell divides into two halves with equal number of chromosomes which are similar to parent cell and are diploid in nature.

- The chromosomes number always remained the same.
- Cell division is always preceded by mitosis.
- In case of man egg cells and sperm cells like other cells, must contain 46 chromosomes.
- August Weismann (1834-1914) a biologist hypothesised that :
 1. In successive generations, individuals of the same species have the same number of chromosomes.
 2. In successive cell division the number of chromosomes always remain constant.
- The scheme of mitotic division was confirmed in 1904 by Theodore Boveri (1862-1915).
- **Cell division in Human beings :**
- Cell is the structural and functional unit of life of any organism.
- The process of cell division is same in unicellular organisms and highly evolved in multicellular organisms like human being.

Cell cycle :

- The process of cell division is called 'Mitosis', which is completed in 40 to 60 minutes (this is the time of active division).



- The period between two cell divisions is called 'Interphase'. Interphase can be divided into three phases.

Phase	Duration	Event
G ₁	10.5 hrs	This is the linking period between the completion of mitosis and the beginning of DNA replication (Gap 1 phase). The cell size increase during this period.
S	10.5 hrs	This is the period of DNA synthesis (Synthesis phase) leading duplication of chromosomes.
G ₂	3.5 hrs	This is the time between the end of DNA replication and the beginning of mitosis. (Gap 2 phase). Cell organelles divide and prepare chromosome for mitosis.

Mitosis :

Stage	Description
Prophase	i) Chromosomes contract, spiral and become visible even in light microscope and nucleoli become smaller (chromatin to chromosomes). ii) Chromosomes split lengthwise to form chromatids, connected by centromeres. iii) Nuclear membrane breaks down. iv) Centrosome, containing rod-like centrioles, divide and form ends of spindle (probably animal cells only). (Note : No pairing of chromosomes as in meiosis)
Metaphase	i) Chromosomes move to spindle equator, centromeres attached to spindle fibres. ii) Centromeres split, separating the chromatids.
Anaphase	i) Spindle fibres attached to centromeres contract, pulling chromatids towards poles ii) Chromatids elongate, become invisible, (replication at this stage to become chromosomes). iii) Nuclear membranes form round daughter nuclei.
Telophase	i) Cell membrane pinches in to form daughter cells (animals) or new cell wall material becomes laid down across spindle equator (plants). ii) Nucleus divides into two and division of cytoplasm starts.

Meiosis :

Unlike mitosis which is a continuous process for division in most cells. Meiosis occurs only during the formation of gametes in sexual reproduction. Meiosis has two phases. The second phase of meiosis is similar to normal mitosis, but chromosomes do not duplicate, more over the chromosomes are distributed equally to each cell. Thus the four daughter cells have just the number of chromosomes of the parent cell. These are haploid (containing only one set of chromosome). Thus this division is also called reduction division.

Reproductive health :

The diseases can be transmitted from person to person in a variety of ways. Since the sexual act is a very intimate connection of bodies, it is not surprising that many diseases can be sexually transmitted. These include bacterial infections such as Gonorrhea and syphilis, and viral infections such as AIDS (Acquired Immuno Deficiency Syndrome). These diseases spread by unsafe sexual contacts, using infected devices, infected blood transfusion, from an infected mother to child.

The government established Anti Retroviral Therapy (ART centres) to supply medicine to HIV patients. Medical and health, family health departments AIDS control projects implementing various programmes like ASHA (Accredited Social Health Activist), Red Ribbon Express, etc., to create awareness in society on the risks and symptoms of AIDS.

Birth control methods :

Many ways have been devised to avoid pregnancy.

- The prevention of pregnancy in women by preventing fertilisation is called contraception.
- Any device or chemical (drug) which prevents pregnancy in woman is called a contraceptive.
- Some devices not only prevents fertilisation but also prevent transmission of some sexually transmitted diseases (STD) like gonorrhoea, syphilis and AIDS.
- Chemicals in the forms of pills are induced either orally or inserting into female reproductive organ - vagina. It contains hormones which stop the ovaries from releasing ovum into the oviducts.
- The use of intra-uterine device called copper-T, loop etc. are also very effective in preventing pregnancy.
- Surgical methods of birth control are available for males as well as females. In males, small portion of vas deferens (sperm ducts) is removed by surgical operation and both ends are tied properly. This method is called vasectomy. In females a small portion of oviduct (fallopian tube) is removed by surgical operation and the cut ends are tied. This prevents the ovum from entering into the oviducts. This method is called tubectomy.

Fighting against social evil**Teenage motherhood :**

Child birth is even more complicated. Understanding it and getting prepared for it needs maturity of the mind and body. Thus a girl only after 18 years of age can be said to be prepared for the same. Most of the times this age is also dangerous to the girl. According to the department of family welfare 21% of teenage mothers die during delivery. So girls below 18 years of age should not marry.

Stop female foeticide :

Today's girl child may become a great scientist, a famous doctor, a top class engineer, a dedicated administrative officer, a world renowned economist, a wonderful teacher of an unmatched world leader of tomorrow. Due to reckless female foeticide the male female child sex ratio is declining at an alarming rate in some sections of our society. Stop female foeticide! Save the girl child.

PRIORITY - 1**2 Marks Questions :**

1. Write the precautions to be taken while preparing the temporary slide of *Rhizopus*.
(AS₂)(TB, P.No. 128)
 2. How will you get the desired useful traits by using grafting method?
(AS₂)(Q.No. 16, P.No. 19)
- Write the procedure to be followed to observe the various parts of an embryo of a dicot seed.
(AS₂)(TB, P.No. 138)

4. What happens to the wall of uterus during menstruation? (AS₂)(Q.No. 15, P.No. 10)
5. What happens if karyokinesis and cytokinesis fail during cell division.
(AS₂)(Q.No. 10, P.No. 17)
6. What precautions will you take to keep away from various sexually transmitted diseases?
(AS₂)(Q.No. 28, P.No. 15)
7. "All unicellular organisms undergo only mitotic cell division during favourable conditions". Do you support this statement? Why?
(AS₂)(Q.No. 16, P.No. 11)
8. Vicky's father wants to grow a single plant having two desirable characters - colourful flowers and big fruits. What method will you suggest him and why?
(AS₂)(Q.No. 17, P.No. 11)
9. What happens if reproduction doesn't takes place in the living organisms?
(AS₂)(TB, P.No. 123)
10. What would be the consequences if there is no meiosis in organisms that reproduce sexually?
(AS₂)(Q.No. 26, P.No. 11)

4 Marks Questions :

1. What are the different modes of asexual reproduction? Cite them with examples.
(AS₂)(Q.No. 5, P.No. 9)
2. How does reproduction help in providing stability to population of a species?
(AS₂)(Q.No. 13, P.No. 10)
3. Draw a neat labelled diagram of human spermatozoa and mention the function of acrosome.
(AS₂)(Q.No. 2, P.No. 27)
4. Write the differences between stamen and carpel.
(AS₂)(Q.No. 3(b), P.No. 8)
5. Why do fish and frog produce a huge number of eggs each year? (Or) (AS₂)
Fishes and frogs release huge number of eggs into water? Give reason.
(AS₂)(Q.No. 1, P.No. 7)
6. What are the advantages of sexual reproduction?
(AS₂)(Q.No. 12, P.No. 10)
7. What is tissue culture? Mention the uses of it.
(AS₂)(Q.No. 37, P.No. 23)
8. What are the disadvantages of child marriages? (AS₂) (Or) Will you encourage child marriages? Explain the disadvantages of it.
(AS₂)(TB, P.No. 145)
9. How does binary fission differ from multiple fission?
(AS₂)(Q.No. 18, P.No. 20)
10. Write the differences between Grafting and Layering.
(AS₂)(Q.No. 3(a), P.No. 8)

6 Marks Questions :

1. In what ways does sexual reproduction differ from asexual one? State at least three reasons.
(AS₂) (Or) Write the differences between sexual reproduction and asexual reproduction.
(AS₂)(Q.No. 6, P.No. 9)
2. During interphase the genetic material (DNA) makes its copy in order to distribute equally to the daughter cells. What is interphase? Explain the stages of interphase.
(AS₂)(Q.No. 20, P.No. 20)

- Explain the different methods of contraception.
- Explain the process of fertilisation in plants.
- Write the differences between mitosis and meiosis.

(AS₁)(Q.No.38, P.No.23)
(AS₁)(Q.No.4, P.No.8)
(AS₁)(Q.No.14, P.No.10)

PRIORITY - 2

2 Marks Questions :

- What happens if acrosome is absent in sperm cell ?
(AS₂)(Q.No.7(i), P.No.9)
- When the foetus is growing inside the uterus it needs nutrients. What provides these nutrients ?
(AS₁)(Q.No.9, P.No.10)
- Uproot an onion plant and take a thin section of its root tip. Stain it and observe under microscope. Draw as you see and identify the stages of the cell division.
(AS₂)(Q.No.18, P.No.11)
(AS₂)(Q.No.19, P.No.20)
- What happens if pollination does not occur in a flower ?
(AS₂)(Q.No.27, P.No.15)
- How will you appreciate cell division that helps in perpetuation of life ?

4 Marks Questions :

- Mention the importance of different embryonic membranes in the development of foetus.
(AS₁)(Q.No.11, P.No.17)
- The first milk from mother is very essential for the child. Give reason.(Or) (AS₁)
It is very important to feed colostrum to the new born baby. What is colostrum ? In what way does colostrum helps the newborn baby.
(AS₁)(Q.No.8, P.No.17)
(AS₁)(Q.No.7, P.No.9)
- Write a brief note a parthenogenesis.
(AS₁)(Q.No.3, P.No.26)
- If a woman is using a copper-T, will it help in protecting her from sexually transmitted diseases ? How?
(AS₁)(Q.No.1, P.No.29)
- Tabulate the post fertilisation changes in plants.

6 Marks Questions :

- Prepare a flowchart to explain the process of sexual reproduction in plants. (AS₁)(Or) With the help of flowchart explain the process of sexual reproduction in plants.
(AS₂)(Q.No.24, P.No.14)
- What is mitosis ? Explain the stages of mitosis with the help of labelled diagrams. (Or) Make a flowchart to show the cell cycle and explain cell division describing different stages of mitosis.
(AS₁)(Q.No.21, P.No.12)
(AS₁)(Q.No.12, P.No.17)
- How is placenta formed ? State any two functions.
(AS₁)(Q.No.5, P.No.28)
- How do you observe pollen grain under microscope ?
(AS₁)(Q.No.1, P.No.24)

Complete all the remaining question and answers of the lesson.

PRIORITY - 3

IMPORTANT DIAGRAMS

- Male reproductive system
- Sperm cell
- Female reproductive system
- Structure of flower
- Process of fertilisation
- Structure of ovule
- Life cycle of flowering plant

PRACTICE SHEET

Choose the correct answer :

- Which of the following is an example of Runner ?
i) Jasmine ii) Ginger iii) Oxalis iv) Hydrocotyl
A) i and ii B) ii and iii C) iii and iv D) ii and iv
- Which of the following is an example for non-endospermic seed ?
A) Bean B) Corn C) Castor D) Maize
- Contraceptive method which prevents the spread of STDs is :
A) Spermicides B) Condoms C) Copper - T D) Tubectomy
- The stage of mitosis in which nuclear membrane breaks down is :
A) Prophase B) Metaphase C) Anaphase D) Telophase
- The result of sexual reproduction cannot show the following.
A) Recombination of genetic material
B) Production of new characters in the offspring
C) Production of identical offspring
D) Occurrence of reduction division before gamete formation
- Lichens reproduce by :
A) Budding B) Fragmentation C) Fission D) Conjugation
- Calcasia* propagates by means of :
A) Root B) Bulbs C) Corms D) Tuber
- The outer membrane of growing embryo is :
A) Amnion B) Chorion C) Allantois D) Placenta
- The period between two cell divisions is called :
A) G₁ phase B) S phase C) Interphase D) G₂ phase
- The fluid secreted by accessory glands of the male reproductive system is :
A) Colostrum B) Lymph C) Seminal plasma D) Semen

11. Bacteria responsible for curdling of milk is called :
 A) *Lactobacillus* B) *Streptococcus* C) *Leishmania* D) *Plasmodium*
12. Which of the following is correct pathway of travel for sperm released from the testes?
 A) Seminal duct, epididymis, urethra B) Urethra, epididymis, seminal duct
 C) Epididymis, urethra, seminal duct D) Epididymis, seminal duct, urethra
13. Double fertilisation is a characteristic feature of :
 A) Fungi B) Angiosperms C) Gymnosperms D) Algae
14. Arrange the stages of cell-cycle in correct order.

 A) 4, 1, 2, 3 B) 2, 3, 4, 1 C) 4, 2, 3, 1 D) 1, 3, 4, 2
15. Regeneration is similar to _____ in few aspects.
 A) Budding B) Parthenogenesis C) Fragmentation D) Grafting
16. Fishes and Amphibians show similarity in :
 i) releasing male and female gametes into water
 ii) external fertilisation
 iii) fertilised eggs growing into embryo
 A) only i B) only ii C) i and iii D) i, ii and iii
17. Find the incorrect one about meiosis.
 A) Helps in formation of Gametes B) Parent cell divides twice
 C) Reduction division D) Formation of two daughter cells
18. Identify the odd one.
 A) Gonorrhoea B) Syphilis C) AIDS D) HIV
19. Which of the following is true about pollen grain ?
 A) A pollen grain is produced by stigma.
 B) A pollen grain fuses with seed during fertilisation.
 C) When a pollen grain lands on an anther of flower of the same species, a pollen tube is formed.
 D) The male cells in a pollen grain pass traits of parents to the offspring.
20. Identify the correct pathway in the life cycle of flowering plant. []
 A) Mature plant → Fertilisation → Zygote → Embryo → Seed
 B) Seedling → Mature plant → Fertilisation → Seed → Embryo
 C) Zygote → Embryo → seed → Seedling → Germination
 D) Mature plant → Fertilisation → Zygote → Seed → Embryo

7. COORDINATION IN LIFE PROCESSES

CHAPTER AT A GLANCE

Human body is a wonderful machine. It is a complicated structure than it appears. Different life processes in living organisms like respiration, digestion, blood circulation, excretion, nervous system etc., are inbuilt in our body at their specific places and carry out their specific functions in a coordinated manner.

Every process is dependent on other to keep the body in good condition. To understand this concept we analyze how digestive system is coordinated with other systems as an example. We shall study the digestive system from feeling hungry to utilization of food, illustrating the inter connected processes going on in our body.

A major cause for feeling hungry lies in the physiology of blood circulation. Levels of different substances are generally maintained in the blood mainly by our digestive system. One of the major substances is glucose. When its levels in the blood fall, we get hunger pangs in stomach.

This again involves production of a series of proteins, some of which are hormones like Ghrelin. Secretion of the hormone "Ghrelin" occurs in the stomach when it goes empty. Ghrelin is secreted from certain cells in the wall of the stomach. Hunger contractions (hunger pangs) start to occur in the stomach due to hunger generating signals that reach the brain from the stomach due to the secretion of this hormone. It is believed that the Diencephalon in forebrain and vagus nerve (10th cranial nerve) plays an important role in carrying these signals to the brain. Hunger pangs continue up to 30-45 minutes. Increase in ghrelin levels results in sensation of hunger and motivation to consume food. When you feel your stomach is full and there is no need of food any more, another hormone leptin is secreted that suppresses hunger.

Relation between taste and smell :

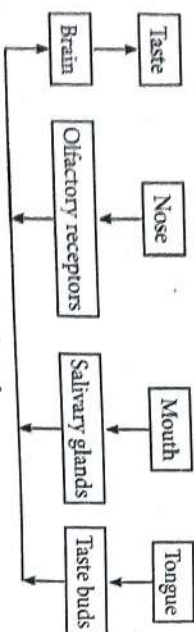
Taste and smell are intimately entwined. This close relationship is most apparent in how we perceive the flavors of food. Anyone with severe cough and cold can not make out the difference in tastes of certain food items. Actually, what is really being affected is the flavor of the food, or the combination of taste and smell. That's because only the taste, not the food odors, are being detected. Taste itself is focused on distinguishing chemicals that have a sweet, salty, sour, bitter, or umami taste (umami is Japanese for 'savory'). However, interactions between the senses of taste and smell enhance our perceptions of the foods we eat.

When we smell, the air borne substances get dissolved in the watery film of nasal mucus. The chemoreceptors in nose are other wise called olfactory receptors which trigger signals in the form of nerve impulses to the brain where smell is detected. Similarly as we take food into our mouth the taste buds send signals to the brain. Picking up the slight differences in smell the food tastes are identified in our brain. Sometimes mouth starts watering just by

hearing the name like tamarind / lime / mango etc., In general, we prefer the food material, which is attractive to our eyes, and flavor to nose, then we taste it. Therefore, when we eat, without our knowledge, we use our sight, nose and tongue for selecting food for ingestion. Russian scientist Pavlov has conducted experiments and found that even the thought of food will water your mouth (conditioned reflexes).

Sugar crystals over the tongue :

Tongue is sensory in function and contains taste buds. These taste buds are tiny papillae with an opening on top. Within them there are several taste sensitive cells. Any food substance when placed on the tongue gets dissolved in the saliva secreted by salivary glands in the mouth. When the tongue is pressed against the palate the food substance is pressed against the opening of the taste bud letting it to reach the taste cells and triggering taste signals. Finally the taste is recognized in the brain.



Dental formula explains the arrangement of teeth.

Type of teeth	Number	Shape	Function
Incisors	8	Sharp edges	Cutting the food
Canines	4	Sharp and pointed edges	Tearing the food
Premolars	8	Blunt and nearly flat surface	Grinding the food
Molars	12	Blunt and nearly flat surface	Grinding the food

The circular muscles of the mouth enable the food to be pushed into the oral cavity and to be moved around. As the food cannot be swallowed directly, the teeth grind, chew and shred it. This process is called mastication. For this purpose the surface muscles of the jaw help in biting and chewing actions, and move the jaw up, down, forward and backward during food mastication.

The teeth help in cutting and grinding while tongue movements evenly spread out the food and help in mixing it with saliva. The muscles of the mouth enable the food to be pushed in the oral cavity and to be moved around. The fifth cranial nerve has been found to control the movement of muscles in the jaw.

Action of saliva on flour (atta) :

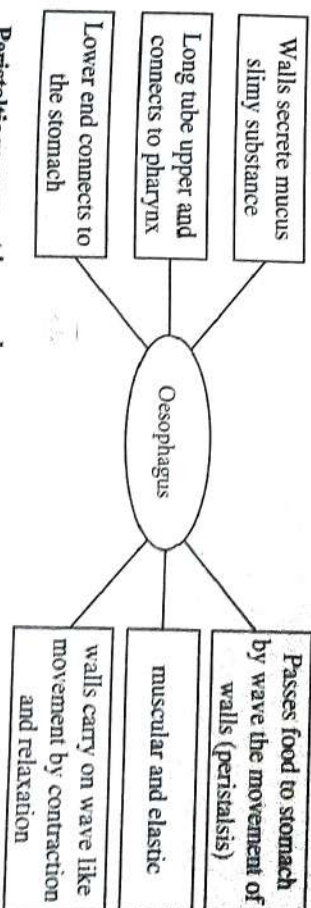
Take a test tube half filled with water and add a pinch of flour to it. Shake the test tube well till the flour gets mixed. Take a few drops of this in a watch glass and test for the presence of starch by putting a drop of diluted iodine solution in it. A blue black color confirms the presence of starch.

Now divide the mixture into two equal halves by transferring it to another test tube. Note that both the test tubes have the same amount of solution. Add a teaspoon of saliva to one of the test tubes and mark it. Do not add anything in the other test tube. After some time (45 minutes), add a drop of dilute Tincture Iodine solution to test tubes containing the solution. Under the action of autonomous nervous system saliva is secreted by three pairs of the salivary glands to moisten the food to make chewing and swallowing easier. As a result of chewing, food forms into a slurry mass called 'bolus' that is transported into the oesophagus by the action of swallowing with the help of the tongue.

The enzyme salivary amylase in the saliva breaks down the large starch molecule into smaller subunits usually into sugars. The mechanism for swallowing is also under nervous coordination and its control center is somewhere in the brain stem (medulla oblongata and others). During mastication food size becomes convenient to swallow.

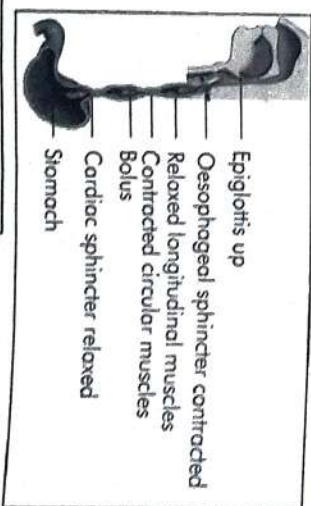
Travel of food through oesophagus :

The oesophagus receives the food pushed by the swallowing action of the mouth. The following schematic representation shows some functional and structural attributes of oesophagus.



Peristaltic movement in oesophagus :

The walls of the food pipe secrete a slippery substance called mucus. Mucus lubricates and protects the oesophageal walls from damage. This helps the food bolus to slide down easily just as the oiled potatoes that move in the tube. Besides this, the saliva in the bolus also aids in easy movement of food, which moves into the stomach.



The wall of the oesophagus is made up of two kinds of smooth muscles. The inner layer consists of circular muscles and the outer layer of longitudinal muscles. Contraction of the circular muscles results in narrowing of the oesophagus just behind the bolus. So the food is squeezed downwards.

Contraction of the longitudinal muscles in front of the bolus widens the tube. This results in shortening of that particular part of the oesophagus. Contraction and relaxation of these muscles bring in a wave like motion that propels the food bolus into the stomach by the action called as "peristalsis". This is involuntary and under the control of autonomous nervous system.

Stomach the mixer and digester :

When the food is in the oral cavity, the nerves in the cheek and tongue are stimulated, these carry messages in the form of nerve impulses to the brain. These messages are transmitted from the brain, to the wall of the stomach, and stimulate the gastric glands to produce gastric juice. These secretions are stimulated by the nervous system. The contraction of the stomach muscles squeeze and mix the food with the acids and juices of the stomach. These digestive juices turn the food into a smooth porridge like consistency called chyme. Some large protein molecules are also broken down here.

As the process of digestion in the stomach nears completion, the contractions of the stomach decrease. This prompts the muscles, called as pyloric sphincter at the opening of the stomach and the first part of the small intestine or duodenum, to relax. This opens the pathway into duodenum releasing the partially digested food (chyme) in small quantities into the duodenum.

		
Pyloric valve closed	Pyloric valve closed	Pyloric valve slightly opened
Propulsion : Peristaltic waves move food from one part to the other.	Grinding : The most vigorous peristalsis and mixing action occur close to the pylorus.	Retropulsion : Small amounts of chyme is pushed into the duodenum, simultaneously forcing most of it back into the stomach.

Peristalsis involves the contraction of the muscle behind the food and the relaxation of the muscle in front of the food giving rise to a thrust that pushes the food forward through the digestive canal. A wave of contraction followed by relaxation in muscles helps in forward movement of food.

We observe that digestion of food starts from the mouth. While the food passes through the gut the food settles for some time for digestion at certain locations.

Our stomach is not like a bag with specific volume. It is like a pouch which is elastic in nature. The size of the stomach increases based on the food that we intake. Digestive juices are produced depending on the quantity of food material. If the stomach would produce same amount of digestive juices irrespective of the food quantity the walls of stomach would be destroyed.

We also know that stomach secretes strong acids during digestion. The HCl secreted by the walls of the stomach is strong enough to digest the hard bones as well.

Travel of food from the stomach to the intestine :

Now food is a soup like mixture when it leaves the stomach and enters the small intestine. When the food enters the intestine the acidic nature of the chyme initiates the production of hormones like secretin and cholecystokinin which stimulate pancreas, liver and walls of small intestine to secrete pancreatic juice, bile juice and succus entericus.

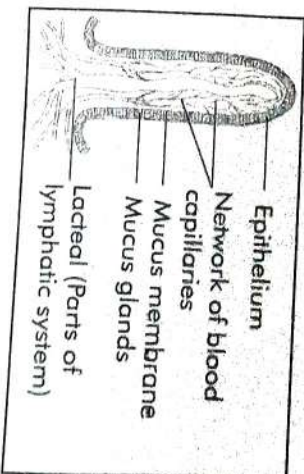
The absorption of nutrients by villi in the small intestine is a very selective process. The walls of the intestine allows only tiny nutrient particles to pass through.

Paper tube and folded papers :

The inner surface of the small intestine contains thousands of finger like projections called villi.

These villi increase the surface area so that the food retained in the folds can remain longer thereby enhancing absorption.

The digestive tract is unique among internal organs because it is exposed to a large variety of physiochemical stimuli from the external world in the form of ingested food. As a consequence, the



intestine has developed a rich store of coordinated movements of its muscular apparatus along with neural apparatus to ensure the appropriate mixing and propulsion of contents during digestion, absorption, and excretion.

The neural apparatus of our digestive tract comprises of such a vast and complicated network of neurons that it has been nicknamed by scientists as the second brain. When the unwanted waste material (stools / faeces) reach the large intestine. The peristaltic waves move the stool into the rectum.

The left side of the colon acts like a storage tank of faeces. Water gets reabsorbed and the remaining wastes usually hard mass gets stored in the last part (Rectum) of the large intestine. This smelly yellowish faecal mass usually called as stool is later expelled out of the body through the anus.

There are two muscular sphincters helping the exit of stool. One control and the other is under voluntary control. These muscular and closing of the aperture of the canal which are called as

voluntary in opening

If energy has to be obtained from food, it has to be oxidised. For this purpose respiration has to go on. During inhalation oxygen moves across the walls of the alveoli and enters the blood. From here it enters the red blood cells and gets distributed throughout the cells of our body. At the same time carbon dioxide from the blood moves into the alveoli of the lungs and breathed out during exhalation. Nutrients in the cells get oxidized and energy is released.

During respiration we breathe continually by inhaling and exhaling air. This is an involuntary process controlled by the medulla oblongata of the autonomic nervous system (ANS). During respiration the movement of inter costal muscles/diaphragm moves the rib cage inflating and deflating the lungs. Air containing more of oxygen enters the blood stream through lungs. If the oxygen has to reach the tissues it has to be circulated through blood. Hence the process of digestion which is a complex process involves many organs and organ systems. Though digestion occurs in the food canal, co-ordination of respiration and blood circulation is necessary. Otherwise oxidation of food and transport of substances which is vital for energy releasing process will not take place. This may lead to the shutdown of systems interdependent on each other.

PRIORITY - 1

2 Marks Questions :

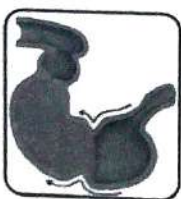
1. If you were suffering from a severe cold, you would feel that the food is tasteless. Why?
(AS₂)(Q.No. 3, P.No. 55)
2. What is your inference in the experiment to prove the role of the palate in recognizing taste?
(AS₃)(Q.No. 21, P.No. 48)
3. Write the procedure followed in an activity to show peristaltic movements in oesophagus/movement of bolus in oesophagus.
(AS₃)(Q.No. 3, P.No. 56)
4. What happens if the stomach is structured like a tube as oesophagus?
(AS₂)(Q.No. 5, P.No. 55)
5. What will happen if we swallow food directly without mastication? (AS₂)(Q.No. 4, P.No. 55)
6. What are the observation of experiments conducted by Ivan Pavlov on conditional reflexes?
(AS₃)(Q.No. 1, P.No. 56)
7. Suggest any two important habitual actions to your friend that he/she need to follow while eating food, keeping in view of the chapter "Coordination in life processes".
(AS₃)(Q.No. 31, P.No. 52)
8. What happens if we swallow food directly without mastication? (Or) Give any two suggestions to your friend to keep the digestive tract healthy. (AS₂)(Q.No. 4, P.No. 55)
9. What conclusion can be made from the potato cycle tube experiment?
(AS₃)(Q.No. 3, P.No. 56)
(AS₃)(Q.No. 5, P.No. 54)
10. How do we detect the smell of an agarbatti (Incense stick)?
(AS₃)(Q.No. 5, P.No. 54)

4 Marks Questions :

1. a) What is anal sphincter?
(AS₃)(Q.No. 3, P.No. 54)
b) Why do we salivate during a nap of day time?
(AS₃)(Q.No. 2, P.No. 59)
2. a) What are papillae? Name different kinds of papillae present on the tongue.
(AS₃)(Q.No. 4, P.No. 54)
b) We can't identify taste when food is very hot. Give reason. (AS₃)(Q.No. 6(b), P.No. 43)
(AS₃)(Q.No. 1, P.No. 42)
3. What do you mean by hunger pangs?
(AS₃)(Q.No. 2, P.No. 42)
4. What are the organ systems involved in digestion of food which we eat?
(AS₃)(Q.No. 2, P.No. 42)
5. Why only small quantities of food should pass from stomach to duodenum?
(AS₃)(Q.No. 8, P.No. 55)
6. Is there any reason for the intestine to be coiled with many folds? In what way it is helpful during the process of digestion?
(AS₃)(Q.No. 11, P.No. 45)
7. How can you say that mouth is a munching machine?
(AS₃)(Q.No. 8, P.No. 44)
8. How do you appreciate stomach as a churning machine? How does this coordination go on?
(AS₃)(Q.No. 29, P.No. 52)
9. How can you justify the enteric nervous system as the second brain of the gut?
(AS₃)(Q.No. 13, P.No. 45)
10. Tabulate the information about different types of teeth in human body with their number and function.
(AS₃)(Q.No. 1, P.No. 57)

6 Marks Questions :

1. Draw a neat labelled diagram showing a peristaltic movement in oesophagus. Explain the importance of mucus on the walls of food pipe.
(AS₃)(Q.No. 24, P.No. 49)
2. Observe the given part of the digestive system. What is it? What is its role during digestion?
(AS₃)(Q.No. 5, P.No. 43)



3. Write the procedure involved in the acid and leaf experiment to understand the concept "how the stomach gets protected from its own acid secretions". Compare the observations with the changes that takes place in human digestive system. (AS₃)(Q.No. 2, P.No. 56)
4. a) The mere smell or sight of food stimulates hunger. Describe the process through a neat diagram.
(AS₃)(Q.No. 26, P.No. 50)
b) Draw a block diagram showing sensation of taste from taste buds (taste buds) to brain.
(Q.No. 23, P.No. 49)

5. Write the differences between the following :
 a) bolus - chyme
 b) mastication - rumination

PRIORITY - 2

2 Marks Questions :

1. What should be the pH of the mouth ? What happens if it decreases ?
 (AS₁)(Q.No. 3, P.No.55)
2. If your taste buds were affected what would happen to your interest in having food ?
 (AS₁)(Q.No. 2, P.No.55)
3. What happens if salivary ducts are blocked/ closed ?
 (AS₁)(Q.No. 18, P.No.46)
4. What experiment should you perform to understand action of saliva on flour ? Explain its procedure and apparatus used.
 (AS₁)(Q.No. 17, P.No.47)
5. What would be the consequences if sphincters fail to function ? (AS₁)(Q.No.4, P.No.42)

4 Marks Questions :

1. What is the function of peristalsis in the following parts ?
 a) oesophagus
 b) stomach
 c) small intestine
 d) large intestine
 (AS₁)(Q.No. 12, P.No.45)
2. Collect the information related to feeling hunger from your school library and prepare a note on it.
 (AS₁)(Q.No. 22, P.No.48)
3. Write the differences between small intestine and large intestine.
 (AS₁)(Q.No. 7(b), P.No. 44)
4. During the journey of food from mouth to stomach through oesophagus, how does muscular system coordinate in this process ?
 (AS₁)(Q.No. 10, P.No.45)
5. Write the procedure of tissue paper experiment to understand the concept of reabsorption of water from waste materials in large intestine.
 (AS₁)(Q.No. 4, P.No.57)

6 Marks Questions :

1. Draw a schematic diagram of villus in small intestine and label the parts.
 (AS₁)(Q.No. 25, P.No.50)
2. Give reasons :
 a) If glucose level falls in blood, we feel hungry.
 b) Small intestine is similar to a coiled pipe.
 (AS₁)(Q.No. 6, P.No.43)
3. List out the sphincter muscles of the food canal you have observed and give a brief description.
 (AS₁)(Q.No. 16, P.No.46)
4. Suggest a simple experiment to prove the role of palate in recognising taste.
 (AS₁)(Q.No. 21, P.No.48)

5. What is mastication ? Explain the role of different sets of teeth in this process.
 (AS₁)(Q.No. 9, P.No.44)

PRIORITY - 3

Complete all the remaining question and answers of the lesson.

IMPORTANT DIAGRAMS

- i) Peristaltic movement of bolus
- ii) Structure of villus

PRACTICE SHEET

Choose the correct answer :

1. The nerve that controls the movement of jaw muscles :
 A) 1st cranial nerve B) 3rd cranial nerve C) 4th cranial nerve D) 5th cranial nerve
2. The pH beyond of is :
 A) alkaline B) slightly acidic C) neutral D) highly acidic
3. Read the following statements.
 P : Our stomach is like a bag with specific volume.
 Q : The size of stomach increases based on food we intake.
 A) P and Q are true B) P and Q are false C) P is false, Q is true D) P is true, Q is false

4. We secrete _____ of saliva per day.
 A) 1.5 - 2.5 litres B) 1.5 - 2 litres C) 1 - 1.5 litres D) 2.5 - 3 litres

5. Which of the following statement is correct about oesophagus ?

- A) Muscular and non - elastic
- B) It's lower end connects to the pharynx.
- C) It is also called wind pipe.
- D) It is Walls secrete mucus.

6. The inner surface area of the small intestine is increased for the more effective absorption because of :

- A) crista B) villi C) lactae D) blood capillaries

7. Which of the following helps in the expulsion of stool ?

- A) Gastric sphincter B) Pyloric sphincter C) Anal sphincter D) Cardiac sphincter

8. Hormone that stimulates stomach to secrete gastric juice is :

- A) Gastrin B) Rennin C) Cholecystokinin D) Secretin

9. Small amounts of chyme is pushed into the duodenum, some of it back into the stomach. This phenomenon is :

- A) Propulsion B) Grinding C) Retention D) Absorption

10. Identify the scientist who measured the temperature of the stomach during digestion is maintained constantly. []
 A) Ivan Pavlov B) St. Martin C) Dr. Beaumont D) Dr. Bateson
11. Inner layer of the digestive canal is protected from the harmful effect of HCl by :
 A) water in the food B) enzymes C) mucus D) hormones []
12. If our teeth arrangement ratio is 2 : 1 : 2 : 3, in this, one (1) represents : []
 A) Incisors B) Canines C) Molars D) Premolars
13. Find out the false statement from the following. []
 A) Digestive juices are produced depending on the quantity of food material.
 B) Mucus lubricates and protects the oesophageal walls from damage.
 C) Sometimes mouth starts watering just by hearing the name like tamarind/lemon/mango, etc.
 D) The walls of oesophagus secrete juice containing hydrochloric acid (HCl) and other digestive juices.
- Whose experiments revolutionised the concepts of digestion ? []
 A) Dr. Martin B) Dr. Beaumont C) Dr. Bateson D) Dr. Mecklead
- The reason(s) for the formation of ulcers in stomach(s) are : []
 i) indigestion ii) stress and strain iii) bacteria
 A) i and ii B) ii and iii C) i and iii D) i, ii and iii
16. Sphincter that is present between the stomach and duodenum is : []
 A) sphincter of oddi B) pyloric sphincter
 C) anal sphincter D) cardiac sphincter
17. The process that is useful for rumination in domestic animals like cow and buffalo is : []
 A) peristalsis B) reverse peristalsis C) emulsification D) retropulsion
18. Emulsification in digestive system is due to presence of : []
 A) saliva B) gastric juice C) peptic juice D) bile juice
19. Swallowing of food is under control of : []
 A) Cerebrum B) Diencephalon C) Midbrain D) Medulla oblongata
20. Read the following statements and find the false one. []
 A) The mucus lining of the stomach protects it from damage by its own acids.
 B) The mechanism for swallowing is coordinated by the swallowing centre in the fore brain.
 C) Food canal is also known as gastro-intestinal tract.
 D) Human digestive system involves both the muscular and nervous systems.

8. HEREDITY - EVOLUTION

CHAPTER AT A GLANCE

Differences in characters within very closely related groups of organisms are referred to as variations. Often a new character in a group may lead to variations which may also get inherited.

Over centuries variations and their role in nature have been studied by naturalists. During early 19th century, a lot of work was done by several scientists. Some of these studies will help us to understand how variations occur and are transferred from one generation to the next. We shall study a detailed account of experimental evidences provided by Mendel in the early 19th century who is known as 'Father of genetics'.

In 1857 Gregor Johann Mendel started working on the problem of how variations were passed from one generation to the other. Mendel did not do his experimental work either in a University or in a Laboratory. As he was a monk in a monastery he simply did his experiments in the monastery garden. He worked for over seven years after which he presented the conclusions from his experimental data in the form of a detailed research paper.

Mendel made many careful observations of plants and found that pea plants would be most suitable to carry on further experimentation. Then he planned and designed the experiments to find out the answers to questions that came to his mind. He had worked on nearly 10,000 pea plants of 34 different varieties.

Observing pea plants carefully, Mendel noted that they differ from one another in many ways. For example plants were tall or dwarf, seed shape round or wrinkled, seed coat (cotyledon) colour yellow or green. Thus, Mendel had chosen 7 pairs of contrasting characters for his study.

Character	Description
Colour of the flower	Purple or white
Position of the flower	Either in axial or terminal in position
Colour of the seed	The seeds are either yellow or green colour
Shape of the seed	The seeds are either round (Smooth surface) or wrinkled
Shape of the pod	The completely developed pod is either smooth, full and constricted.
Colour of the pod	All the ripe pods are yellow in colour. Unripe pods are either yellow or green in colour.
Length of the stem	When the plants are grown in the same environment some stems are all (6 to 7 feet) and some stems are dwarf (3/4 feet to 1 1/2 feet).

The results of Mendel's F_1 crosses for seven characters in pea plants.

Character	Dominant trait	Recessive trait	F_2 generation dominant : recessive	Ratio
Flower colour	Purple	White	705 : 224	3.15 : 1
Flower position	Axial	Terminal	651 : 207	3.14 : 1
Seed colour	Yellow	Green	6022 : 2001	3.01 : 1
Seed shape	Round	Wrinkled	5474 : 1850	2.96 : 1
Pod shape	Inflated	Constricted	882 : 299	2.95 : 1
Pod colour	Green	Yellow	482 : 152	2.82 : 1
Stem length	Tall	Dwarf	787 : 277	2.84 : 1

Mendel had chosen garden pea as material for his experiments because it has following advantages.

- 1) Well defined characters,
- 2) Bisexual flowers,
- 3) Predominantly self fertilization,
- 4) Suitable for cross pollination
- 5) It is an annual plant

Mendel hypothesized that characters were carried as traits and an organism always carried a pair of factors for a character. He also hypothesized that distinguishing traits of the same character were present in the population of an organism. He assumed that the traits shown by the pea plants must be in the seeds that produced them. The seeds must have obtained these traits from the parent plants.

Mendel carried out several experiments to find out answers to questions like How do parent plants pass on their traits to the seeds? Will the seeds from tall plants always produce new tall plants?

Mendel did the experiment with two pure breeds of peas for seed colour contrasting characters yellow and green, and they are represented as yellow with 'Y' and green with 'y'. He started cross fertilizing pure breeds having contrasting characters. Mendel started with two pure breeds of peas with different properties. We here take as an example the characteristic colour of the pea seeds yellow and green. Cross pollinating a pure breed of yellow and green gives first filial (F_1) generation (Mendel called it as first filial or progeny) of first generation parents. filial means progeny.)

All pea seeds were Yellow. These pea plants on self pollination gave Second Filial (F_2) generation Mendel self pollinated these (F_2) pea plants too, and found out that [About 75% yellow (seeds) peas and about 25% green (seeds) peas.]

Third Filial (F_3) generation By self pollination of these (F_2) pea plants he got F_3 generation. He found out that

- 1) A set of peas (about 25%) gave only yellow seed giving pea plants.
- 2) Rest of the yellow seed giving pea plants gave about 50% yellow and about 25% green seed giving pea plants.
- 3) 25% pea plants gave only green seed giving pea plants. (25%)

Mendel made some assumptions by which he could explain his observations.

Assumption 1: Every pea plant has two 'factors' which are responsible for producing a particular character or trait. The determining agent responsible for each trait is called a factor.

Mendel carefully chose the plants which did not produce a mixed result (pure). In our example of yellow and green peas, a pure breed (parental stage) will have both the 'factors' of the same type.

A pure breed (parental) yellow seed giving pea will have both the 'factors' of the same type. Let us denote them by 'Y'.

A pure breed (parental) green seed giving pea will have both the 'factors' of the same type. Let us denote them by 'y'.

Assumption - 2: During reproduction one 'factor' from each parent is taken to form a new pair in the progeny.

Assumption - 3: One of these will always dominate the other if mixed together. The trait expressed in F_1 generation is called dominant. While the other which did not express in F_1 is called recessive.

Assume that 'Y' (the one causing yellow colour) is a dominating 'factor'. That means if 'Y' and 'y' come together 'Y' will dominate. Then the pea seeds will be always yellow in colour. From assumption - 2, the breed after cross pollination will have one factor from pure breed yellow (Y) and one from the pure breed green (y).

That is, all the peas will have the paired factor 'Yy' and by assumption-3 all the peas will be Yellow as 'Y' factor is dominant.

♂	Y	y	Yy (Yellow)
♀	Y	Yy	Yy (Yellow)
	Y	Yy	Yy (Yellow)
	Y	Yy	Yy (Yellow)

All the pea plants are yellow (F_1 - Generation). The trait expressed in F_1 is dominant, unexpressed is recessive. This is the law of dominance.

Self pollination in F_1 - Generation

On self pollinating these peas (ones with Yy factor), the new breed have any combinations of 'Y' and 'y'.

♂	Y	y
♀	Y	YY
	Y	YY
	y	Yy
	y	Yy

It can be YY, Yy, yy or yy. All of them are in equal ratio.

F₂-Generation

So in this heap we will get approximately equal number of YY, Yy, yy and yy peas. But any pea that has a Y factor will be yellow. Any pea that has both yy will be green. Since all combinations are equally likely:

- 1) YY will be approximately 25% and is yellow.
- 2) yY will be approximately 25% and is yellow, Yy will be approximately 25 % and is yellow
- 3) yy will be approximately 25% and is green.

Some seeds appear yellow in colour in F₁ generation. When these seeds were sown some of the plants produced green coloured seeds. So we can't determine internal character based on external visible character.

Phenotype : Externally visible characters are called phenotype and ratio is called phenotypic ratio.

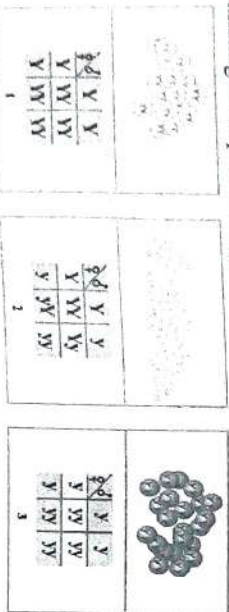
In the above example in F₂ generation we can clearly observe that 75 percent are yellow seed producing pea plants and 25 percent are green ones. The monohybrid phenotypic ratio is 3 : 1.

Genotype : The genetic constitution for a particular character or factor is Genotype and ratio is Genotypic ratio.

Genetically, in 75 percent yellow seed producing pea plants only 25 percent pea plants produce yellow seeds that are pure breeds (YY) and are 'homozygous' that is to have the same factors for representing a character.

Remaining 50 percent yellow seed producing pea plants are (Yy) heterozygous. The remaining 25 percent green seed producing pea plants are pure (yy) homozygous type. The monohybrid genotype ratio is 1:2:1 (25:50:25).

On self-pollinating these peas of F₂ generation we get,



- 1) The YY peas on self pollination will give only yellow (YY) peas. This was explained with the experimental result that this set gave 100% yellow peas.
 - 2) The Yy or yY peas on self pollination give about 75% yellow peas and about 25% green peas. This situation is same as step 2 ratio 3:1.
 - 3) The green peas that contain yy factors will give only green peas.
- In nature there are many factors responsible for different properties.

Dihybrid cross :

This can be applied to Mendel's experiment when two pairs of contrasting characters are taken into consideration.

- 1) Colour of peas-yellow and green symbolically indicated as 'Y', 'y'.
- 2) Shape of peas-Round and wrinkled symbolically indicated as 'R', 'r'.

The plants with yellow and round seeds (pure) were crossed with those having green and wrinkled seeds (pure).

All pea seeds that were yellow and round each pea will have factors 'YYRR'.

Since Yellow colour (Y factor) and round (R factor) are dominant traits. All the pea seeds will be round and yellow (F₁ generation). When these plants are self pollinated.

In F₂ generation he got some seeds which were round and yellow (YYRr or YYRR), some seeds were round and green (yyRR or yyRr), some seeds were wrinkled and yellow (Yyrr or Yyrr), and some seeds were wrinkled and green (yyrr).

Mendel explained the process of inheritance of more than one pair of characters. This law is known as **Law of independent assortment**.

Mendel's Laws :

Mendel propounded that, among a pair of closely related 'alleles' or factors for a character, only one expresses itself in the first generation as one of the allele is dominant over the other. This is so evident that it came to be called as Mendel's **Law of Dominance**.

He also stated that, every individual possesses a pair of alleles (assuming only a pair is present) for any particular trait and that each parent passes a randomly selected copy (allele) of only one of these to an offspring. The offspring then receives its own pair of alleles for that trait one each from both parents. This is what Mendel called 'segregation' and it is the **Law of Segregation**.

Traits that may be passed on from one generation to the next are called as heritable traits. We have studied some of them for the pea plant, in the experiments conducted by Mendel.

Parent to progeny

A person resembles his grandfather, a girl seems to be a photocopy of her aunt, generally we hear such comments. These similarities are the result of inherited traits transmitted from parent to progeny.

According to Mendel traits like the colour of seed, seed coat, length of stem etc. were heritable traits from parental generation.

Transmission or passing of characters or traits from parent to offsprings is called 'Hereditry'. The process in which Traits are passed from one generation to another generation is called 'Inheritance'.

How do traits get expressed?

Mendel hypothesised that each character or trait is expressed due to a pair of factors or 'alleles' (contrasting expressions of the same trait), as he named them. Now we know

that these are known as 'genes'. Gene is a segment of a nucleic acid called 'DNA' which is present in the nucleus of every cell. It controls the expression of a trait or character. In viruses RNA can also be controlling the expression of a character.

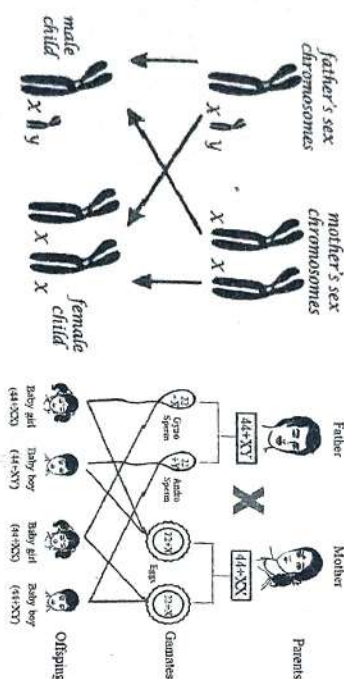
Sex determination in human beings

We inherited our traits from our parents.

Let us see how sex is determined in human beings. Each human cell contains 23 pairs (46) of chromosomes. Out of 23 pairs 22 pairs of chromosomes are autosomes.

Chromosomes whose number and morphology do not differ between males and females of a species are called autosomes.

The remaining pair is called allosomes or sex chromosomes. These are two types, one is 'X' and the other is 'Y'. These two chromosomes determine the sex of an individual. Females have two 'X' chromosomes in their cells (XX). Males have one 'X' and one 'Y' chromosomes in their cells (XY).



All the gametes (ova) produced by a woman have only X chromosomes. The gametes (sperm) produced by a man are of two types one with X chromosome and other Y chromosome. The sperm carries Y chromosome and fertilizes the ovum (X chromosome). Then the baby will have XY condition. So the baby will be a boy.

Evolution

Variations develop during reproduction in organisms. Sexual reproduction and errors in DNA copying leads to variations in offsprings in a population.

Acquired and Inherited Characters and Evolution

We discussed the idea that the germ cells of sexually reproducing population are formed in specialised reproductive tissue. If the weight of the beetles is reduced because of starvation that will not change the DNA of the germ cells. Therefore, low weight is not a trait that can be inherited by progeny of a starving beetle. Therefore even if some generations of beetles lose their weight because of starvation, that is not an example of evolution, since the change is not inherited over generations. Change in non reproductive tissues cannot be passed on to the DNA of the germ cells. Therefore the experiences of an individual during its lifetime cannot be passed on to its progeny, and cannot direct evolution.

Lamarckism

In the olden days people believed that all the organisms on the earth had not undergone any change. Jean Baptiste Lamarck was the first person to propose the theory of evolution. He thought that at some point of time in the history the size of giraffe was equal to that of deer. Due to shortage of food material on the ground and to reach the lower branches of trees giraffes started stretching their necks. Because of continuous stretching of neck, after several generations giraffes developed long necks. Such characters that are developed during the lifetime of an organism are called 'acquired characters'. Lamarck proposed that these acquired characters are passed on to its offsprings i.e. to next generation and proposed the theory of 'Inheritance of acquired characters'.

For example elongation of neck and forelimbs in giraffe.

But August Weismann, tested this theory by an experiment on rats. He removed tails of parental rats. He observed that their offsprings are normal with tails. He has done it again and again for twenty two generations but still offsprings are normal with tails. He proved that the bodily changes are not inherited. So they won't be passed to it's offsprings.

Darwinism

Charles Darwin proposed 'Natural selection' the famous 'theory of evolution'.

Charles Darwin (1809-1882) was born in England. He voyaged for five years, just when he was 22 years old. In the world survey ship HMS Beagle. He visited a number of places including Galapagos Islands. He keenly observed the flora and fauna of these places. He gathered a lot of information and evidences.

Darwin observed a small group of related birds which are exhibiting diversity in structure in the Galapagos islands. These birds are Finch birds.

He was influenced by the book 'Principles of geology' written by Sir Charles Lyell. He suggested that geological changes occurred in a uniform rate, Darwin did not agree to this idea. He felt that large changes occurred due to accumulation of small changes. Darwin was also influenced by the famous 'Malthus theory'. This was written in 'An essay on the principles of population'. Malthus observed that population grows in geometrical progression (1, 2, 4, 8,), where as food sources increases in arithmetic progression (1, 2, 3, 4, 5,).

Based on these ideas, Darwin proposed the theory of "Natural selection", which means that the nature only selects or decides which organism should survive or perish in nature. This is the meaning of survival of the fittest. The organisms with useful traits will survive. If traits are not useful to organisms then they are going to be perished or eliminated from its environment.

Alfred Russel Wallace also independently concluded that natural selection contributed to origin of new species. For example we have seen in the case of red beetles which were seen and eaten by crows. So, the population of red beetles gradually got eliminated or perished from its environment. But at the same time the beetles which are green in colour which are present on the green leaves were not noticed by crows. So the green beetles survived in the environment and their population has gradually increased. This is nothing but "Natural selection".

Variations which are useful to an individual are retained, while those which are not useful are lost. In a population when there is a struggle for the existence the 'fittest' will be survived. Nature favours only useful variations. Each species tend to produce large number of offsprings. They compete with each other for food, space, mating and other needs. In this struggle for existence, only the fittest can survive. This is called 'Survival of the fittest'. Over a long period of time this leads to the formation of new species.

Darwin's theory of evolution in a nutshell

- 1) Any group of population of an organism develops variations and all members of group are not identical.
 - 2) Variations are passed from parent to offspring through heredity.
 - 3) The natural selection over abundance of offspring leads to a constant struggle for their survival in any population.
 - 4) Individuals with variations that help them to survive and reproduce tend to live longer and have more offsprings than organisms with less useful features.
 - 5) The offsprings of survivors inherit the useful variations, and the same process happens with every new generation until the variation becomes a common feature.
 - 6) As the environment changes, the organism within the environment adapt and change to the new living conditions.
 - 7) Over a long period of time, each species of organism can accumulate so many changes that it becomes a new species, similar to but distinctly different from the original species. All species on the earth arise in this way.
 - 8) Evolution is a slow and continuous process.
- There are some limitations and objections to the Darwin theory. Many new theories like synthetic theory, mutation theory are put forward.

Speciation

How new species are evolved?

We have seen variations in a population of species, where the organism contain the traits that helped to adapt to the environment. These organisms are going to survive more efficiently. But in the same population the organism which contains the non beneficial traits may not be adapted in the environment. They are going to perish or eliminated slowly, like red and blue beetles in a population which we have discussed earlier in this chapter. These small changes within the species for example colour of beetles red and green is known as 'Micro evolution'.

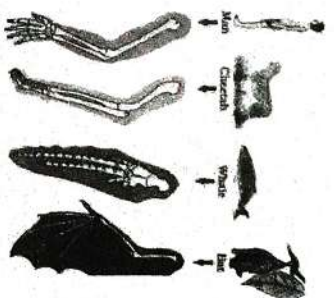
Now we are going to discuss how new species are formed. This is known as speciation which is also known as 'Macro evolution'.

Evidences of evolution

How does the evolution of organisms take place? Whatever scientists propose they require evidences or proofs. In the same way evolution of organisms requires evidences. Let us examine some of them.

Homologous and analogous organs

When we try to understand evolutionary relationships, we identify that some traits have common ancestors. These traits in different organisms would be similar because they are inherited from a common ancestor. You may be surprised to know that the internal structure of forelimb of a whale (swimmer) wing of a bat (flyer), leg of a cheetah (runner), claw of a mole (digger) and hand of a man (grasping), if we carefully observe the anatomy of all these animals, they show a common pattern in the arrangement of bones, even though their external form and functions are different. It indicates that all the vertebrates have evolved from a common ancestor and these organs are called homologous organs. This type of evolution is called divergent evolution.

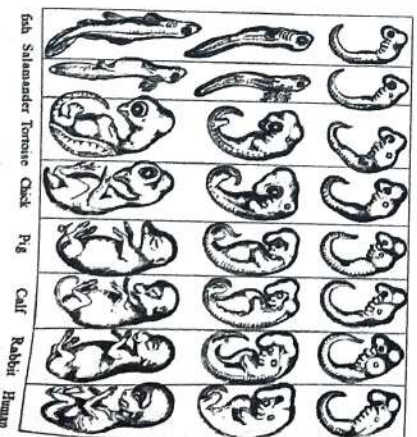


However, all similarities simply in organ shape are not necessarily to have a common ancestry. What would we think about the wings of birds and bats, for example (fig.). Birds and bats have wings, but squirrels and lizards do not. So birds and bats are more closely related to each other than squirrels or lizards.

When we observe, we find that the wings of bats have skin folds (patagium) stretched between elongated fingers. But the wings of birds have a feathery covering all along the arm. The designs of the two wings, their structure and components are different. They look similar because they have a common use for flying, but their origin is not common. This gives the 'analogous' characteristics (Traits). As the above mentioned organs which are structurally different but functionally similar are known as 'Analogous organs'. This type of evolution is called convergent evolution.

Evidences from embryology

Embryology is the study of the development of an organism from egg to adult stage. Tadpole of a frog resembles fish more than a frog. What does this indicates? Does it indicate that frogs have evolved from ancestors of fish? There are remarkable similarities in the embryos of different animals from fish to man. The resemblance is so close that at an early stage even an experienced embryologist would find it difficult to distinguish one embryo from the other. This strengthens the view of the existence of a common ancestor from which all these have evolved.



Evidences from fossils

We know some species which existed million years ago, but we may not find them now. They might be extinct and some of them may be found in the form of fossils. For example, we know Dinosaurs the biggest animal on land which were present long time ago but now they are extinct. The scientists got evidences of presence of Dinosaurs like animals in the form of fossils.

What are fossils?

Fossils are evidences of ancient life forms or ancient habitats which have been preserved by natural processes. Fossil evidence is typically preserved within the sediments deposited beneath water and land. They can be actual remains of once lived such as bones or shells or even traces of past event such as dinosaur foot print or ripple marks on a prehistoric shore. Usually when organisms die, their bodies will be decomposed and lost. Sometimes the body or some parts of the body do not decompose completely. For example if a dead insect get caught in mud, it will not decompose quickly and the mud will eventually harden and retain the impression of the body parts of insect. All such preserved traces of extinct organisms are called fossils.

Geologists can tell the age of a fossil. The study of fossil is called 'Palaeontology'. Palaeontologists determine the age of fossil by using carbon dating method. The breakdown of radioactive isotopes of certain elements such as Carbon, Uranium and Potassium take place at a known rate. So the age of rock or mineral containing isotopes can be calculated. A rare and magnificent fossil of the dinosaurs, ketosaurs belonging to the lower Jurassic age going back to about 160 million years were collected from Yamnapalli in Adilabad district of Telangana. This fossil has 14 metres length and 5 metres height. This fossil is preserved at BM Birla Science Centre in Hyderabad.

Human evolution

Diopithecus → Ramapithecus → Astrolipithecus → Homo habilis → Homo erectus → Homo Neanderthalensis → Cromagnan → Homo sapiens (modern man)

Human evolution is the evolutionary process leading up to the appearance of a modern human being. We the present human beings also have an evolutionary history like plants and other animals. Early man like forms appeared about 7 lakhs 50 thousand years ago. The first sure fossil of our own species of man the Homo sapiens, indicate that true man appeared on the earth 2 lakhs 50 thousand years ago.

Evolution of man through ages:

Homo habilis lived between 1.6 - 2.5 million years ago.

Homo erectus lived between 1 - 1.8 million years ago.

Homo sapiens neanderthalensis lived between 1,00,000 - 40,000 thousands years ago.

Homo sapiens (present man) appeared about 15,000 - 10,000 thousand years ago.

There is a great diversity in human forms and features across the planet.

So that for a long time, people used to talk about human 'races'. Skin colour used to be the commonest way of identifying the so called races. Some were called black, some white or brown. A major question debated for long time was, have these apparent groups evolved differently? Over recent years, the evidence has become very clear. The answer is that there is no biological basis to the notion of human races. All humans are a single species with a common ancestor.

Not only that, regardless of where we have lived for the past few thousand years, we all come from Africa. The earliest members of the human species, *Homo sapiens*, can be traced there. Our genetic footprints can be traced back to our African roots. A couple of hundred thousand years ago, some of our ancestors left Africa while others stayed on, while the residents spread across Africa. The migrants slowly spread across the planet—from Africa to west Asia, then to Central Asia, Eurasia, South Asia, East Asia. They travelled down the island of Indonesia and the Philippines to Australia, and they crossed the Bering land bridge to the Americas. They did not go in a single line, so were not travelling for the sake of travelling, obviously. They went forwards and backwards, with groups sometimes separating from each other, even moving in and out of Africa. Like all other species on planet, they had come into being as an accident of evolution, and were trying to live their lives the best they could.

Human being - a moving museum of vestigial organs

During the course of evolution some organs remain in the organisms. You have studied about appendix in the digestive system. In human beings it has no role to play in the process of digestion. But in herbivores like rabbit appendix plays an important role. Such type of organs which are not useful in animal are called vestigial organs.

Sometimes vestigial organs abruptly appear even in human beings. This phenomenon is called "ATAVISM". Eg : baby with tail.

There are nearly 180 vestigial organs in human beings. For example pinna, hair on skin, mammary glands in males, etc. That's why human being is said to be a moving museum of vestigial organs.

PRIORITY - 1**2 Marks Questions :**

1. What is meant by speciation ?
(AS,)(Q.No.27, P.No.85)
2. Define the term 'Genetic drift'.
(AS,)(Q.No.25(b), P.No.84)
3. Based on survival of fittest write few examples that you experience in your daily life.
(AS,)(Q.No.26, P.No.79)
4. What happens if there is no process of evolution ?
(AS,)(T.B, P.No. 189)
5. Why are traits acquired during the lifetime of an individual not inherited ?
(Q.No. 24, P.No. 84)
6. What are Darwin finches ?
(81)

- What would happen if the sex of the individual is determined by autosomes?
(AS₂)(TB, P.No. 188)
- Why do some traits of different organisms similar to each other?
(AS₁)(Q.No. 31, P.No. 85)
- What suggestions will you give to the farmer to develop desired varieties of mangoes?
(AS₂)(Q.No. 4, P.No. 70)
- If the theory of inheritance of acquired characters proposed by Lamarck was true, how will present world be? (Or) What are the consequences if acquired characters will be inherited? (Or) What happens if Lamarckism comes true?
(AS₂)(Q.No. 17, P.No. 75)

4 Marks Questions :

- Give an example of characteristics being used to determine how close two species are in evolutionary terms.
(AS₂)(Q.No. 2, P.No. 80)
- What variations have you generally observed in the species of a cow?
(AS₂)(Q.No. 9, P.No. 73)
- What are fossils? How are they formed? (AS₁) (Or) Write about the structures used by palaeontologists to study about fossils.
(AS₂)(Q.No. 1(a), P.No. 80)
- What differences did Mendel identified between parent and F₂ generations?
(AS₂)(Q.No. 12, P.No. 74)
- Why variations are important? How are variation useful for an organism or a population?
(AS₂)(TB, P.No. 178)
- How does the creation of variations in a species promote survival?
(AS₂)(Q.No. 20, P.No. 83)
- Explain analogous and homologous organs with examples. (AS₁) (Or) How are analogous organs different from homologous organs? (AS₂) (Or) Write a brief note on analogous organs.
(AS₂)(Q.No. 14, P.No. 75)
- What are the characters selected by Mendel for his experiments on pea plants?
(AS₂)(Q.No. 10, P.No. 74)
- Write a note on carbon dating method.
(AS₂)(Q.No. 21, P.No. 76)
- What is your understanding about survival of the fittest? Give some examples that you observe in your surroundings.
(AS₂)(Q.No. 26, P.No. 79)
- Define the following:
a) Alleles
b) Genotype
c) Phenotype
d) Gene
(AS₂)(Q.No. 7, P.No. 81)

6 Marks Questions :

- One experiment cut the tails of parent rats. What would be the traits in offsprings? Do daughter rats contain tails or not? Explain.
(AS₂)(Q.No. 3, P.No. 70)

- What is the law of independent assortment? Explain with an example. (AS₁) (Or) How do Mendel's experiments show that traits are inherited independently. (Independent assortment). (AS₂) (Or) Draw a checker board to show the law of independent assortment and explain the ratio. (AS₂) (Or) Explain Mendel's Dihybrid cross.
(AS₂)(Q.No. 6, P.No. 72)
- Explain monohybrid experiment with an example. Which law of inheritance can we understand. (Or) Explain the process to understand monohybrid cross of Mendel's experiment with a checker board. (AS₁) (Or) Explain the law of Dominance.
(AS₂)(Q.No. 5, P.No. 71)
- Explain the Darwin's theory of evolution 'Natural Selection', with an example.
(AS₂)(Q.No. 7, P.No. 73)
- How does sex determination takes place in humans? Explain with an example. (AS₁) (Or) Write a flowchart representing the sex determination of human beings. (AS₂) (Or) Male is responsible for sex determination of baby? Do you agree? If so explain your answer with a flowchart.
(AS₂)(Q.No. 36, P.No. 86)

PRIORITY - 2**2 Marks Questions :**

- In a forest there are two types of deers, in which one type of deer can run very fast. Whereas second type of deer can not run as fast as the first one. Lions. Tigers hunt deers for their food. Imagine which type of deers are going to survive in the forest and which type of deers population is going to be eliminated and why?
(AS₂)(Q.No. 1, P.No. 88)
- Why human beings are considered as the moving museum of vestigial organs.
(AS₂)(TB, P.No. 200)
- How embryology provides evidence for evolution?
(AS₂)(Q.No. 16, P.No. 83)
- What factors could lead to the rise of a new species?
(AS₂)(Q.No. 15, P.No. 82)
- What happens if a baby born with atavistic organs?
(AS₂)(Q.No. 13, P.No. 82)

4 Marks Questions :

- In what way has Mendel used the word "Traits". Explain with an example. (AS₁)
 - What are the functions of genes?
(Q.No. 11, P.No. 74)
 - Explain Mendel's Laws of inheritance.
(AS₂)(Q.No. 26, P.No. 84)
 - What are vestigial organs? Give examples.
(AS₂)(Q.No. 5, P.No. 71)
 - With the help of given information write your comment on evidences of evolution.
(AS₂)(Q.No. 14, P.No. 82)
- Mammals have four limbs as do birds, reptiles and amphibians. The basic structure of the limbs is similar though it has been modified to perform different functions.

6 Marks Questions :

- One student (researcher) wants to cross pure tall plant (TT) with pure dwarf plant (tt). What would be the F₁ and F₂ generations ? Explain. (AS₁)(Q.No. 2, P.No. 6)
- In a mango garden, a farmer saw one mango tree with full of mango fruits but with few pests. He also saw another mango tree without pests but with few mangoes. But the farmer wants the mango tree with full of mango fruits and pest free. Is it possible to create new mango tree which the farmer wants ? Can you explain how it is possible ? (AS₁)(Q.No. 4, P.No. 7)
- How do scientists utilise the information about fossils ? (AS₁)(Q.No. 15, P.No. 7)
- Why did Mendel select pea plants for his experiments ? (AS₁)(Q.No. 16, P.No. 7)
- What do you understand about the given terms in view of Mendel's experiments.
 - Heredity
 - Trait
 - Homozygous
 - Heterozygous
 (AS₁)(Q.No. 7, P.No. 8)
- "Archaeopteryx is considered as a connecting link between reptiles and birds". Why ? (Or) Archaeopteryx is considered as a connecting link between reptiles and birds. Justify the statement. (Or) (AS₁) What features of Archaeopteryx will make us to consider it as a connecting link between reptiles and birds? (AS₁)(Q.No. 18, P.No. 8)

PRIORITY - 3

Complete all the remaining question and answers of the lesson.

IMPORTANT DIAGRAMS

- Monohybrid
- Dihybrid
- Evolution of man through ages

PRACTICE SHEET

Choose the correct answer :

1. Variations in an organism refer to :

- differences in characters within very closely related groups
- differences in characters within the unrelated groups
- similarities in characters within very closely related groups
- similarities in characters within the unrelated groups

2. Find the incorrect statement about Mendel.

- He is considered as the father of genetics.
- He started working on how variations pass from one generation to other.
- He has worked on nearly 10,000 pea plants of 34 different varieties.
- He selected pea plants for his experiments as they have unisexual flowers.

3. The animals that have the parents of same breed or same variety are called :

- Impure breeds
- Mixed breeds
- Pure breeds
- Heterozygous breeds

- When true breeding tall pea plants are crossed with true breeding short pea plants, all of the _____ plants, and 3/4 of the _____ plants had tall stems.
 - F₁, F₂
 - G₁, G₁
 - Parental, F₁
 - F₁, parental
- A heritable feature that varies among individuals is called :
 - Trait
 - Character
 - Gene
 - Allele
- The study of fossils is called :
 - Paleontology
 - Embryology
 - Archeology
 - Geology
- In which of the following set of combinations we get only homozygous green plants ?
 - YY and YY
 - Yy and Yy
 - yy and yy
 - yY and yY
- According to the law of segregation the offspring receives :
 - its own pair of alleles for a trait one each from both parents
 - its own pair of alleles for a trait both from a single parent
 - a different pair of alleles for a trait each from both parents
 - a different pair of alleles for a trait both from a single parent
- The phenotypic ratio of dihybrid cross is 9 : 3 : 3 : 1, in this represents :
 - Green and Wrinkled seed
 - Round and Yellow seed
 - Round and Green seed
 - Yellow and Wrinkled seed
- Which of the following controls the expression or trait or a character ?
 - Gene
 - Loci
 - Cytoplasm
 - RNA
- The chromosomes whose number and morphology do not differ between males and females of a species are called :
 - Allosomes
 - Autosomes
 - Sex chromosomes
 - Acquired chromosomes
- Who among the following was the predecessor of Darwin's and developed the theory of acquired characters ?
 - Weismann
 - Mendel
 - Malthus
 - Lamarck
- Identify the true statements.
 - Characters are governed by genes.
 - Change in non reproductive tissues cannot be passed on to the DNA of germ cells.
 - Experiences of an individual during its lifetime can be passed on to its progeny.
 - I and III
 - I and II
 - II and III
 - I, II and III
 - Mutations
- Inheritance of variations was understood properly after the discovery of :
 - Genetics
 - Mitosis and Meiosis
 - Structure of DNA
 - Death of an organism
- Speciation takes place when variations occur with :
 - mood changes
 - death of an organism
 - changes due to accidents
 - geographical isolation

16. By studying analogous structures, we look for :
 A) similarities in appearance and function but differences in structure
 B) similarities in appearance but differences in functions
 C) similarities in organ structure
 D) similarities in cell make up
17. Which one of the following is the most direct and convincing evidence of evolution ?
 A) Vestigial organs B) Fossils C) Genetic code D) Hybridization
18. The number of chromosomes present in each human cell :
 A) 23 B) 23 pairs C) 22 pairs D) 46 pairs
19. Two nucleotide sequences found in two different species are almost exactly the same. This suggests that these species :
 A) contain identical DNA B) might have similar evolutionary histories
 C) have same number of mutations D) both are evolving into same species
20. Forelimbs of whale : Swimmer : : Claws of a mole : ?
 A) Flyer B) Digger C) Runner D) Grasper

9. OUR ENVIRONMENT

CHAPTER AT A GLANCE

The physical factors refer to abiotic factors (land, air, water, sunlight, etc.) and biological factors to biotic factors (flora and fauna). The place where organisms live is called biosphere. One organism cannot completely defy the balance to suit one's need. It would in some way or the other affect the balance in such a way that the survival of the organism affecting the damage would be at stake.

Food chains are interconnected and when we try to observe these connections among a number of food chains then it becomes a food web.

Food chain :

A food chain shows who eats what in a particular habitat. The arrows between each organism in the chain always point from the food to the feeder.

In an ecosystem the energy rich food passes from producers to consumers step wise, with respect to their (food) relationships.



Food chain shows how the energy is passed from one organism to another. At each transfer a large proportion (80 to 90 percent) of energy is dissipated as heat produced during the process of respiration and other ways. Thus in the above three steps of a food chain very little energy is still available for living organisms to use.

The main climatic influences which determine these ecosystems are rainfall, temperature and the availability of light from the sun. For instance, forests are usually associated with high rainfall, but the type is influenced by temperature and light; the same applies to deserts which occur in regions where rainfall is extremely low.

The term 'niche' denotes not only the animal's position in the food web and what it eats, but also its mode of life. Just as a habitat is the place where an animal lives, so a niche describes its occupation the way it 'goes about its business and earns its livings'.

Food web :

In nature food chains are usually not simple and linear. They are often interlinked, because at every stage or trophic level consumers have several alternative forms of food to choose from.

As a result, food chains become branched and interlinked at various stages leading to a web-like structure. This is called a food web.

Ecological pyramids :

Apart from the food chains pyramids are another type of representations which show flow of energy from one organism to another.

It was first introduced by a British ecologist Charles Elton in 1927. In the ecological pyramid the producers (first trophic level) are represented at the base; and other successive trophic levels (primary, secondary and tertiary consumers) are represented one above the other with top carnivores at the tip. There are three types of pyramids; pyramid of biomass, pyramid of number, pyramid of energy.

Pyramid of numbers :

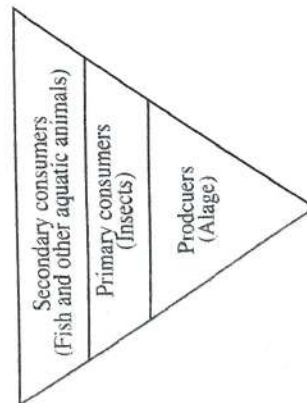
The number of organisms in a food chain can be represented graphically in a pyramid. Each bar represents the number of individuals at each trophic level in the food chain. At each link in a food chain, from the first-order consumers to the large carnivores, there is normally an increase in size, but decrease in number.

This relationship is best shown as a pyramid, which is upright.

Sometimes the pyramid of numbers does not look like a pyramid at all. This could happen if the producer is a large plant such as tree or if one of the organisms at any trophic level is very small. So keep one thing always in mind that whatever the situation, the producers still goes at the bottom of the pyramid.

Pyramid of Biomass :

Biomass is organic material of biological origin that is ultimately derived from the fixation of carbon dioxide by trapping solar energy during photosynthesis. This includes trees, shrubs, crops, grasses, algae, aquatic plants, agricultural and forest residues and all forms of human, animal and plant waste. Any type of plant or animal material that can be converted into energy is called biomass. When these materials are used for energy production. They are known as biofuels.



The pyramid of biomass represents the relationships that exist between the quantity of living matter (biomass) at different trophic levels.

In terrestrial ecosystems, the biomass progressively decreases from producers to top carnivores.

In an aquatic ecosystem, the pyramid of biomass is inverted. It is found that 10-20% of the biomass is transferred from one trophic level to the next in a food chain

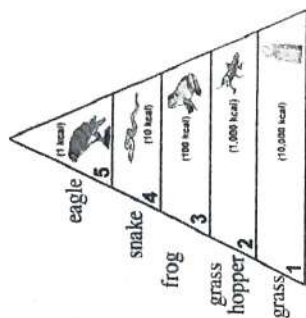
Pyramid of Energy :

Food is the source of energy for organisms that is used in the growth and rebuilding of the parts of the body; that are constantly wearing out. The food by its nature is the chemical energy and by its stored form, it is the potential energy. There are several mechanisms in organisms for continuous absorption of materials for the production of organic material, and for the release and conversion of organic material into inorganic form.

The food chains and food webs help in the transfer of the food and energy from the producers to different consumers. Animals obtain the minerals from the plant or animal food or both. Thus the mineral matter is constantly being removed from the earth to become a part of the plant, which may become a part of animal body. Plants absorb the minerals from the soil. They are absorbed into the plant along with water through roots. Energy enters the producers in the ecosystem from the sun in the form of solar energy or solar radiation. No other organisms except green plants and Photosynthetic bacteria due to the presence of chlorophyll are capable of absorbing solar energy and converting it into chemical energy.

From the producers, the chemical energy passes to the consumers from one trophic level to the next through food. At each trophic level, organisms use most of the food energy that they assimilate into their bodies to fulfil their metabolic requirements- performance of work, growth and reproduction. Because biological energy transformations are inefficient, a substantial proportion of metabolized food energy is lost, unused as heat.

Once the food is eaten, its energy follows a variety of pattern through the organisms. Not all the food can be fully digested and assimilated. Hair, feathers, insect exoskeletons, cartilage and bone in animal foods, cellulose and lignin in plant foods cannot be digested by most animals. These materials are either ejected by defecation or regurgitated in pellets of indigested remains. Assimilated energy (that is not lost through respiration or excretion) is available for the synthesis of new biomass through growth and reproduction. Organisms lose some biomass by death, disease or annual leaf-drop, where they enter the detritus pathways of the food chain i.e., after the death and decomposition of organisms the materials flow back into the environment. The remaining biomass is eventually consumed by herbivores or predators and its energy there by enters the next higher trophic level in the ecosystem.





The effects of human activities on ecosystem :

In the earlier classes we have studied about different types of pollution as a result of human interventions in ecosystem. In this section we will try to understand that when we cut forest to grow food crops, how this activity brings harmful changes in ecosystem and affects organisms of each trophic level.

Let us study a pond ecosystem to understand the components of environment, their interactions and effects of human interventions in the following story.

Story of Kolleru Lake :

Fresh water lakes provide the nutritional requirements of the world's poorest people. One such lake is Kolleru which is not an ordinary wetland. It is one of the largest fresh water lakes in India, existing between West Godavari and Krishna districts of the state of Andhra Pradesh. The catchment of the lake extends up to 6121 km². The lake Kolleru discharges its excess water in to the Bay of Bengal through the twisty channel called Upputeru, which is about 60 km long. Never the less, the Kolleru wetland receives huge quantity of nutrient rich sediments from the flood plains of these rivers.

In November 1999, the Government of India had declared the lake as Bird Sanctuary. This lake is hosting 193 species of birds and a variety of flora and fauna, including medicinal plants. It attracts migratory birds from northern Asia and Eastern Europe between the months of October and March and it is estimated as 20,00,000 birds per year. The lake was also an important habitat for an estimated 20 million residents. This largest sweet water lake has not only shrunk in size but faced great threat due to pollution in the last three decades as revealed by satellite pictures. The decrease in water area and muddy ground in the lake resulted in flooding problems in the lake area. Observe the data given in the following table.

Excessive nutrient addition, especially from anthropogenic sources, led to explosive weed growth. Ex: Eichhornia, Pistia (Eutrophication).

As a result, the water of the lake turned more alkaline in nature, turbid, nutrient rich, low in Dissolved Oxygen (DO) and high in Biological Oxygen Demand (BOD), which had great effect on aquatic animals.

Water borne diseases like diarrhoea, typhoid, amoebiasis and others are said to be common among the local inhabitants who are unaware of the state of pollution in the lake water.

The lake at present is prone to siltation, encroachment, blockages in canals. Due to aquaculture, 15 local varieties of fishes are endangered. As oxygen dissolution in water is decreasing, the number of fishes living in surface water is increased.

The Ministry of Environment and Forest (MoEF), Government of India constituted a committee "Operation Kolleru" to protect the lake. The objective of the programme is to

bringing back the ecological balance of Kolleru Lake which is a Gift of nature.

The process of entry of pollutants into a food chain is known as Bioaccumulation, whereas the tendency of pollutants to concentrate as they move from one trophic level to the next is known as Biomagnification.

Seasonal Bioaccumulation of heavy metals in fish (cyprinus carpio) of Edulabad Water Reservoir (EBWR)- effect on humans.

The aquatic biota is being contaminated with heavy metals due to industrialization and anthropogenic activities.

Recently, fish are considered to be the bioindicators of metal contamination in environmental monitoring because fish species strongly respond to stress conditions.

A study was undertaken to assess the enrichment of heavy metal such as Lead (Pb), Cadmium (Cd), Chromium (Cr), Manganese (Mn), Nickel (Ni) and Ferrum (Fe) contamination in Edulabad Water Reservoir (EBWR) which is located in Medchal district of Telangana state, highly polluted with industrial effluents.

Cyprinus carpio (common scale carp) is a cheap and high proteinaceous fish used as food for human beings, living in polluted EBWR was chosen for the study.

The metal concentrations in EBWR reservoir were found to be higher than Indian standard limits and exhibiting the following sequence, $Fe > Pb > Cr > Ni > Cd$.

The heavy metals could find their way into the human food chain, we analyzed bioaccumulation of these metals in the fish tissues. The bioaccumulation of these metals in fish tissues were of the following trend, $Cd > Cr > Fe > Ni > Pb$.

Higher bioaccumulation factors were found for Cd in liver, gill and kidney indicated the sensitivity of fish to this metal even at low concentrations.

Steps towards prevention :

If we think about the ways through which we can prevent ourselves and other living beings from the harmful effects of the use of toxic material as pesticides, then the instant reaction may be to Ban the pesticides. It is easy to say "Ban all pesticides" but the pests still have to be kept in check. After using pesticides then also we are having a significant amount of loss of food material because of pests. Now you can imagine if pesticides were totally banned what would happen to the diseases? Are they really controlling the pests that effect the crops we vitally need for our growing population?

Biological Principles

The long term solution to this problem is to find other effective methods of controlling pests which have far less harmful effects and are based on sound biological principles. Here are some of the important methods to be used.

Rotation of crops: Growing different crops on a particular piece of land in successive years reduces the occurrence of pests and damage to the crops from year to year in that area.